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1- OVERVIEW

Note

This procedure is for the RF/Pen 1 and 2 cabinets only.

RFT is an RF loopback test that quickly evaluates stability, magnitude and phase linearity, droop, pulse fidelity, and bandwidth of the RF transmit chain in a Signa system.

It is recommended that TPS Loopback Baseline Data Test is performed first. Run Body Sense Loop Baseline Data Test and Head Sense Loop Baseline Data Test next.

The receive line is not verified in this procedure.

For additional information, refer to Section 7-1 RFT Analysis or 8-1 RFT Overview.

Note

Disable the TNF (Transient Noise Filter) unit before running any RFT test. The TNF is located inside the rear door of the System Cabinet. Remember to enable the unit after RFT testing has completed

IMPORTANT!

RF/PEN 1 ONLY: With the APB (Analog Processor Board) in the system, low power stability and phase linearity (even with the correct entry in the config file) can significantly exceed the limits. When Body Dummy Load with EFB Enabled is performed, it is common to see low power magnitude stability numbers approaching 5 p-p, and low power phase close to 10 p-p. Magnitude linearity should pass with or without an APB in the system. Phase linearity with an APB in the system (EFB Enabled) can have numbers as high as 30, but usually drop to 12, or below, when bypassed. MR Engineering is currently investigating ways to reduce these numbers. Note that the numbers for Head Dummy Load (less RF drive) have phase linearity numbers about 40% lower than those in the Body mode.

2- REQUIRED TOOLS



EQUIPMENT DAMAGE POSSIBILITY! The Switchable Attenuator contains ferrous switch relays. Do not take the Switchable Attenuator into the magnet room; doing so can cause damage to the switch relays.

Refer to Tables 2-1, 2-2, 2-3, and 2-4 for the kits required and their contents.

TABLE 2-1
UNIVERSAL SST KIT 46-320383G1 - G7 CONTENTS

Item	Description	Part Number	Quantity
1	Cable, 5ft., RG58, with BNC Connectors	46-251710G4	5
2	Cable, 1ft., RG58, with BNC Connectors	46-282803G16	2
3	Cable, 0.5 ft., RG58, with BNC Connectors	46-251920G4	1
4	Cable, 5ft., RG188, with BNC Male/SMB Female	46-301549P5	1
5	DC Power Supply for Switchable Attenuator	46-320385P1	1
6	Switchable Attenuator	46-288458G1	1
7	1.0T RF Sense Loop (Only in G4 & G6 Kits)	46-321434G1	1
8	1.5T RF Sense Loop (Only in G1/G2/G3/G4/G5 Kits)	46-288446G1	1
9	Head Carriage Sense Loop "L" Bracket	2102499	1
10	Attenuator, Variable 0-10 in 1dB steps	46-255838P3	1
11	Attenuator, Variable 0-1 in 0.1dB steps	46-255838P7	1
12	Velcro Pads	46-307152P1	3
13	Case of Connectors	46-301042G1	1

TABLE 2-2
RF POWER MEASUREMENT KIT, 46-317724G1 OR G2 CONTENTS

Item	Description	Part Number	Quantity
1	Cable, N (M) to N (M)	46-317724P9	1
2	Cable, BNC (M) to SMB (F)	46-317724P13	1
3	Cable, BNC (M) to BNC (M)	46-317724P10	1
4	Attenuator, Variable 0-10 in 1dB steps	46-317724P8	1
5	Adapter, HN (M) to N (F)	46-221865P16	1
6	30dB, 200 Watt Attenuator (only in G2 kit)	46-317724P14	1
7	30dB Coupler	46-317724P4	1
8	40dB Coupler	46-317724P5	1
9	10 dB Inline Attenuator	46-317724P6 or P7	1

Note

The G2 version of this kit is the same as the G1 with the addition of a 30 dB, 200 Watt Dummy Load Attenuator.

TABLE 2-3
QTUNE KIT (46-287308G1/G2/G3) CONTENTS
(USED FOR RFT COIL TUNING CHECK ONLY)

Item	Description	Part Number	Quantity
1	Sense Coil Holder (for Head Coil) (G1 Kit only)	46-287006P1	1
2	Sense Coil Holder (for Head Coil) (G2 & G3 Kit only)	46-307899P1	1
3	1.0T RF Sense Loop (Only in G3 Kit)	46-321434G1	2
4	1.5T RF Sense Loop (Only in G1/G2/G3 Kits)	46-288446G1	2

TABLE 2-4
OTHER ITEMS NEEDED

Item	Description	Part Number	Quantity
1	30 dB, 200 Watt RF Attenuator	46-255837P10	1
2	Grafidy Holder Base Plate	46-271410G1	1
3	Attenuator, Variable 0-70 in 10db steps	46-255838P2	1

Note

A 30 dB, 200 Watt Attenuator (46-317724P14) from the RF Measurement Kit (46-317724G2) can be substituted for item 1. A second 30 dB Attenuator is needed for troubleshooting some Body RF Stability hardware problems.

3- BODY AND HEAD BASELINE LOOPBACK DATA TESTS

3-1 Initial Setup Requirements

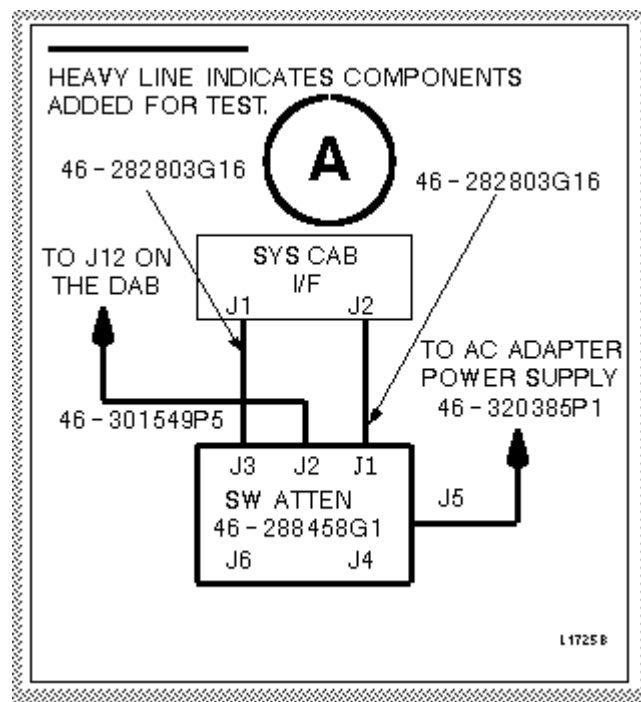


EQUIPMENT DAMAGE POSSIBILITY! The Switchable Attenuator contains ferrous switch relays. Do not take the attenuator into the magnet room; doing so can cause damage to the switch relays.

1. Install the Switchable Attenuator, 46-288458G1, on the rear of the Systems Cabinet as shown in Illustration 3-1.



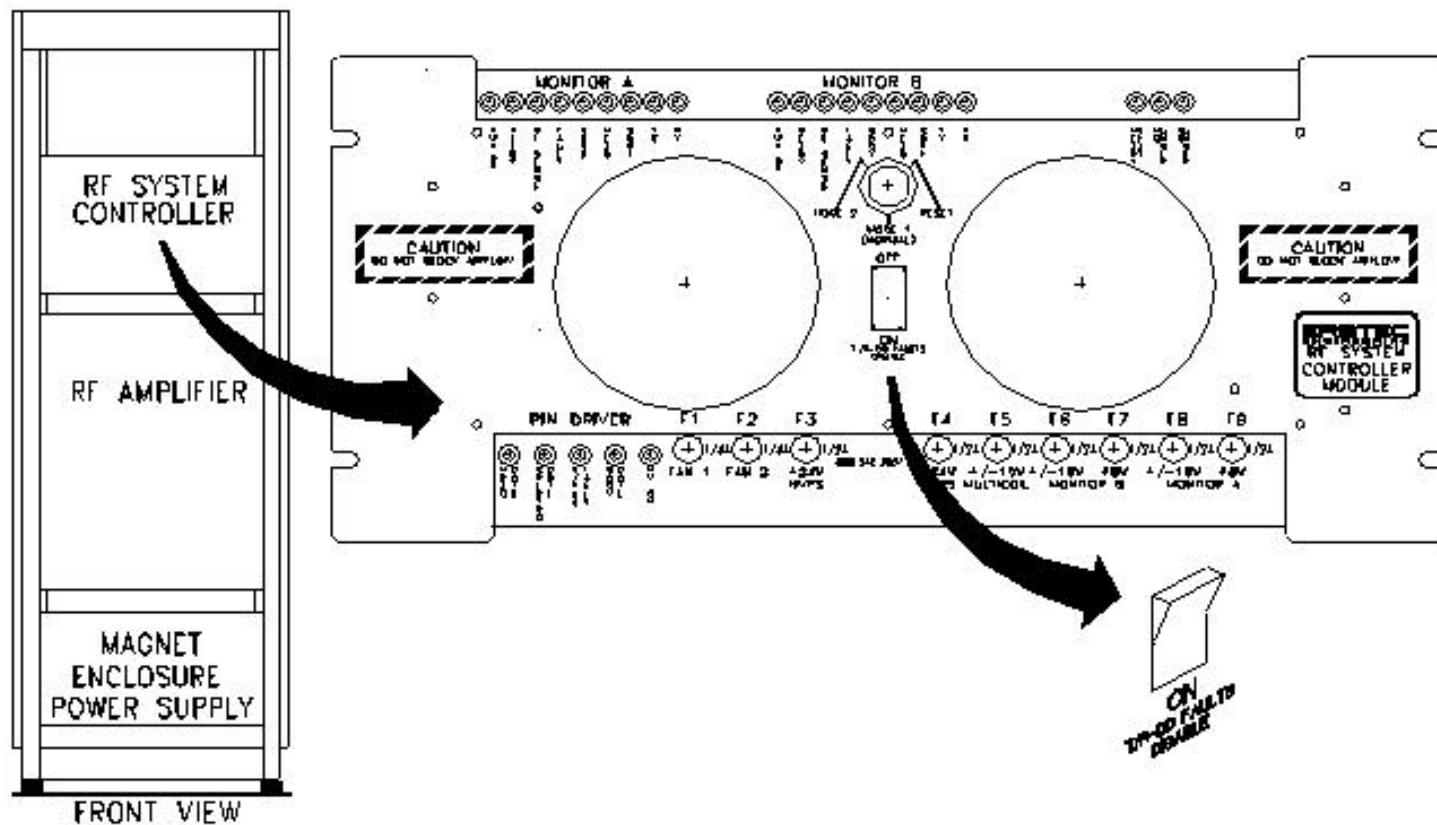
EQUIPMENT DAMAGE! CONNECT SWITCHABLE ATTENUATOR BEFORE CONNECTING SENSE LOOP TO PREVENT DAMAGE TO TPS RECEIVER BOARD. ALSO, LEAVE SWITCH S1 (PREAMP BIAS) ON SWITCHABLE ATTENUATOR IN "OUT" (DEFAULT) POSITION OR THE INTERNAL PROTECTIVE DC BLOCKER IS BYPASSED. (WITH S1 OUT, DS1 "BIAS ON" LED SHOULD BE OFF AFTER SWITCHABLE ATTENUATOR IS CONNECTED.



SWITCHABLE ATTENUATOR
ILLUSTRATION 3-1

2. On some Switchable Attenuators: Push in S1, then remove J5 power to restore the DS1 to OFF (DS1 latches).

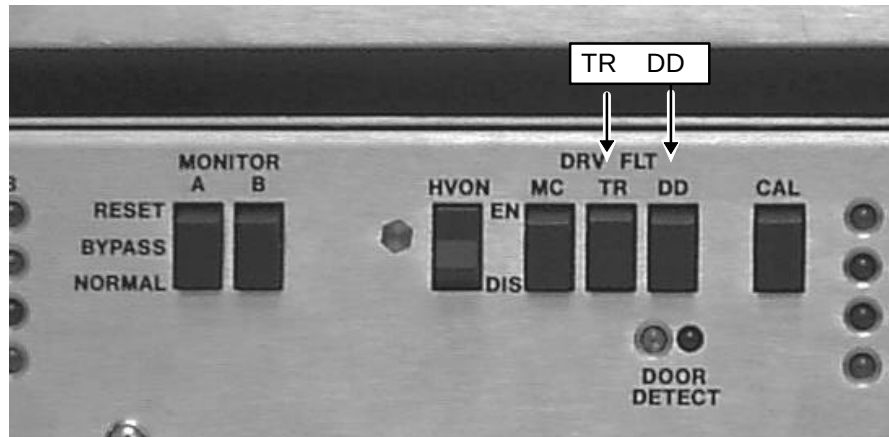
3. Disable the TNF (Transient Noise Filter) unit before running any RFT test. The TNF is located inside the rear door of the System Cabinet.
4. **RF/PEN 1:** see Illustration 3-2A. On the RFSC Module (MRIA15A3) front panel, place the T/R-DD Faults rocker switch to the ON (disable fault reporting) position. The RED portion of this switch is a visible indicator that the fault reporting is disabled.



T/R- DD FAULTS SWITCH TO ON POSITION
ILLUSTRATION 3-2A

5. **RF/PEN 2** — see Illustration 3-2B. On the front of the SSM place the DRV FLT switches as follows:

- TR switch to the DIS (disable faults) position.
- DD switch to the DIS (disable faults) position.



SSM FRONT PANEL TR AND DD FAULT SWITCHES DISABLED
ILLUSTRATION 3-2B

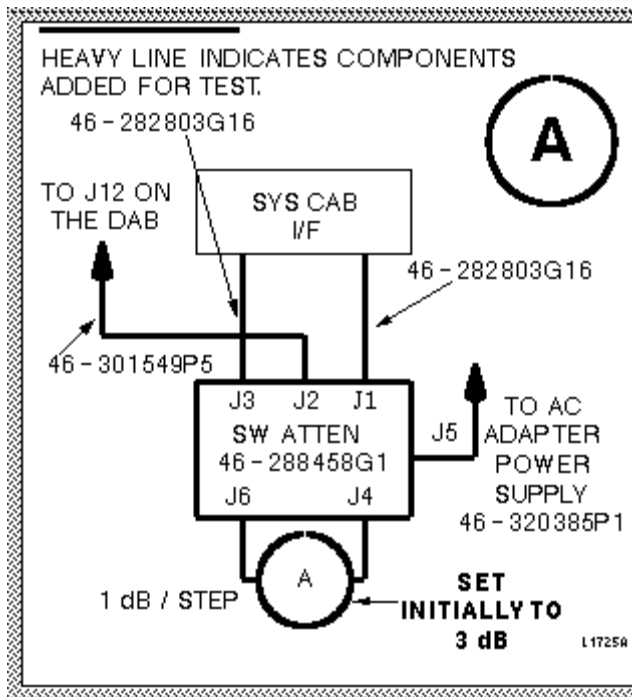
Note

Many sections contain alternate Stability T/S and Linearity T/S Tables.

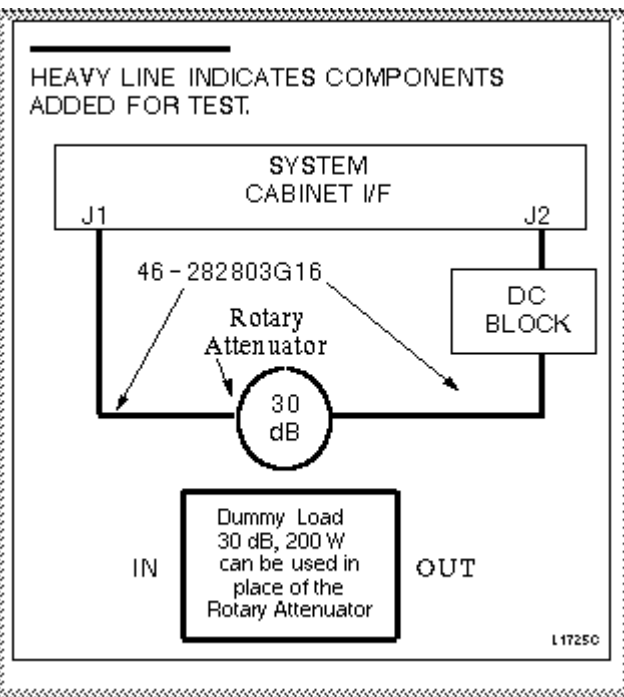
3-2 TPS Loopback Baseline Data Test

This test verifies the Exciter and Receiver hardware is good. TPS Loopback Baseline Data Test is performed in the Body mode. Results from this test are referenced automatically by subsequent RFT tests, therefore, this test is performed first. Control Variables for Stability and Linearity trouble-shooting are listed in Sub-Section 3-2-1, Table 3-2.

1. Prepare the system per sub-section 3-1 (TNF and SSM).
2. Install the Switchable Attenuator, 46-288458G1, on the rear of the Systems Cabinet as shown in Illustration 3-3A. **Or** install the 0 to 10 dB Variable Attenuator, 46-255838P3 and the DC Block, 46-301549P15, at the rear of the Systems Cabinet referring to Illustration 3-3B (Alternate Hardware Setup). This alternate test setup will perform the TPS Loopback test without the use of the Switchable Attenuator. The alternate hardware setup is useful if attempting to identify a hardware problem associated with the Switchable Attenuator.



TPS LOOPBACK SETUP
ILLUSTRATION 3-3A



ALTERNATE HADWARE SETUP FOR TPS LOOPBACK
ILLUSTRATION 3-3B

3. Prepare the system to scan in Body mode per Table 3-1.

TABLE 3-1
BODY SCAN PROTOCOL: TPS LOOPBACK BASELINE DATA TEST

—Note: This is the alternate proprietary procedure available for GE use, and to sites with a valid Advanced Service Package Limited License.

A. **[New Pt]**

Id: **geservice**<ENTER>

Name: **rft**

Weight (Lb.): **300**<ENTER>

Set Patient Protocols to **Service**.

At front enclosure, landmark in head area, press **LANDMARK**, then **MOVE TO SCAN**.

B. In the Patient Position Protocol field:

type **o.16.1**<ENTER>(o=Other, 16.1 =series) to load the body protocol

OR select **other** and select protocol **16** (RF test) and select series **1** (RFT Body).

C. In the Additional Parameters field:

[User CVs Screen].

CV Name: **EFB Bypass**, CV Value: **1**<ENTER>.

CV Name: **RF Amp Mode**, CV Value: **0**<ENTER>.

CV Name: **Sense Loop**, CV Value: **0**<ENTER>.

CV Name: **StabNoGrad**, CV Value: **1**<ENTER>.

CV Name: **Lin&Fidlty**, CV Value: **1**<ENTER>.

CV Name: **Stab/Grad**, CV Value: **4**<ENTER>.

CV Name: **Bandwidth**, CV Value: **1**<ENTER>.

CV Name: **BW Cal**, CV Value: **1**<ENTER>.

[Accept].

D. **[Save Series]**. If necessary --- **[Prepare to Scan]**.

E. **[Research Operations] [Setup Params]**.

R1 = 13

R2 = 15

TG = 50

Number of Frames: **4**<ENTER>

Window 1: Frame: **1**<ENTER>

+/- = +

Window 1: Frame: **0** <ENTER>

Window 2: Frame: **2** <ENTER>

+/- = +

Window 2: Frame: **0** <ENTER>

[Done].

F. **[Research Operations] [Download]**.

4. **[Manual Prescan].**

5. From the “Windows” Menu bar on the Manual Prescan window, select “**Two Windows**”.

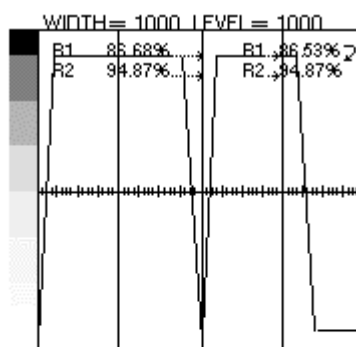
Plot Type = **Magnitude**

Plot Gain = **1**

Plot Type = **Magnitude**

Plot Gain = **1**

There are 4 Frames of rectangular pulses. Two of these pulses will be viewed, one in each window.



Note: This basic signal will be displayed during any of the tests on the following pages while in prescan.

Note: If the signal is low (~2%), check the AC Adapter Power Supply for power to the Switchable Attenuator.

L1726A

TPS LOOPBACK MANUAL PRESCAN SIGNAL DISPLAY
ILLUSTRATION 3-4

6. Set TG to 200.

IMPORTANT!

For TPS Loopback Set Up ONLY: See Illustration 3-4. The R2 Power Spectrum signal should be between 40% & 90%. This may require attenuator A to be set at 0 dB.

7. Select **[Done]**.

8. Select **[Scan]**. Scan time is approximately 5 minutes and 30 seconds.

9. When scan completes, view results by using the Report Manager tool. Record data for each test in Data Sheet 1.5T & Data Sheet 1.0T found in the appendix.

10. If not proceeding to next sub-section, perform System Restoration.

3-2-1 TPS Loopback Stability and Linearity Control Variables

- Control Variables for Stability and Linearity trouble-shooting are listed in Table 3-2:

TABLE 3-2
TPS LOOPBACK CONTROL VARIABLES

User CV Name	Baseline	Stability T/S	Linearity T/S
EFB Bypass	1	1	1
RFampMode	0	0	0
Sense Loop	0	0	0
StabNoGrad	1	1	0
Lin&Fidlty	1	0	1
Stab/Grad	4	4	0
Bandwidth	1	0	1
BW Cal	1	0	0

3-3 Body Sense Loop Baseline Data Test

3-3-1 Body Sense Loop Baseline Data Test — EFB Enabled

This test evaluates the health of the Body Transmit chain and the RF field in the bore. If this test passes the Body Transmit chain is good. TPS Loopback Baseline Data Test is performed first. Control Variables for Stability and Linearity trouble-shooting are listed in this Sub-Section.

1. Prepare the system per sub-section 3-1 (TNF and RFSC or SSM).
2. Prepare the system to scan in Body mode per Table 3-3.

TABLE 3-3

BODY SCAN PROTOCOL: BODY SENSE LOOP BASELINE — EFB ENABLED

Note: This is the alternate proprietary procedure available for GE use, and to sites with a valid Advanced Service Package Limited License.

A. **[New Pt]**

Id: **geservice**<ENTER>

Name: **rft**

Weight (Lb.): **300**<ENTER>

Set Patient Protocols to **Service**.

At front enclosure, landmark per Illustration, press **LANDMARK**, then **MOVE TO SCAN**.

B. In the Patient Position Protocol field:

type **o.16.1**<ENTER>(o=Other, 16.1 =series) to load the body protocol

OR select **other** and select protocol **16** (RF test) and select series **1** (RFT Body).

C. In the Additional Parameters field:

[User CVs Screen].

CV Name: **EFB Bypass**, CV Value: **0**<ENTER>.

CV Name: **RF Amp Mode**, CV Value: **4**<ENTER>.

CV Name: **Sense Loop**, CV Value: **1**<ENTER>.

CV Name: **StabNoGrad**, CV Value: **1**<ENTER>.

CV Name: **Lin&Fidlty**, CV Value: **1**<ENTER>.

CV Name: **Stab/Grad**, CV Value: **4**<ENTER>.

CV Name: **Bandwidth**, CV Value: **0**<ENTER>.

CV Name: **BW Cal**, CV Value: **0**<ENTER>.

[Accept].

D. **[Save Series]**. If necessary --- **[Prepare to Scan]**.

E. **[Research Operations] [Setup Params]**.

R1 = 13

R2 = 14

TG = 50

Number of Frames: **4**<ENTER>

Window 1: Frame: **1**<ENTER>

+/- = +

Window 1: Frame: **0** <ENTER>

Window 2: Frame: **2** <ENTER>

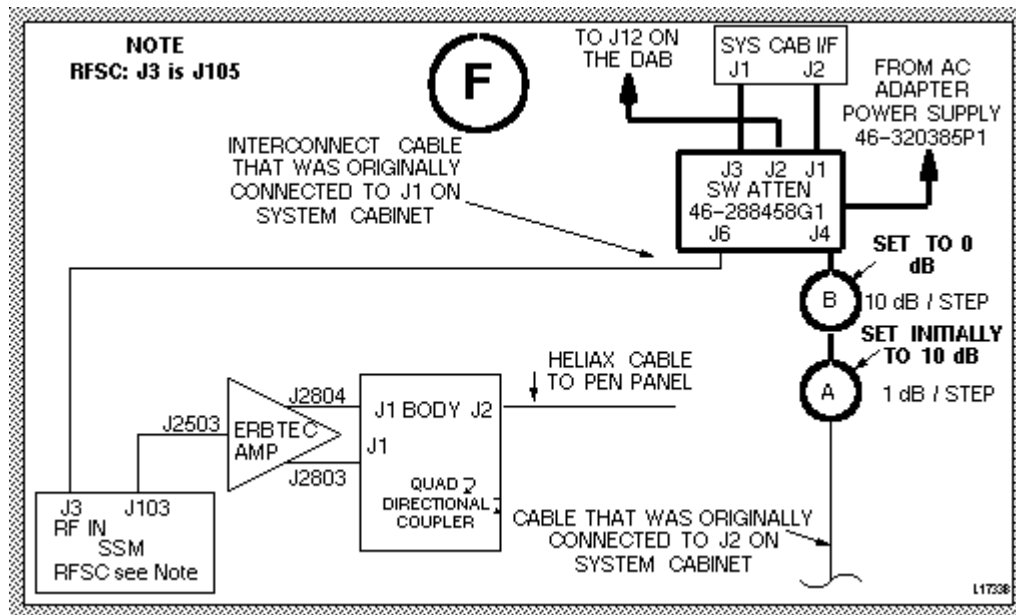
+/- = +

Window 2: Frame: **0** <ENTER>

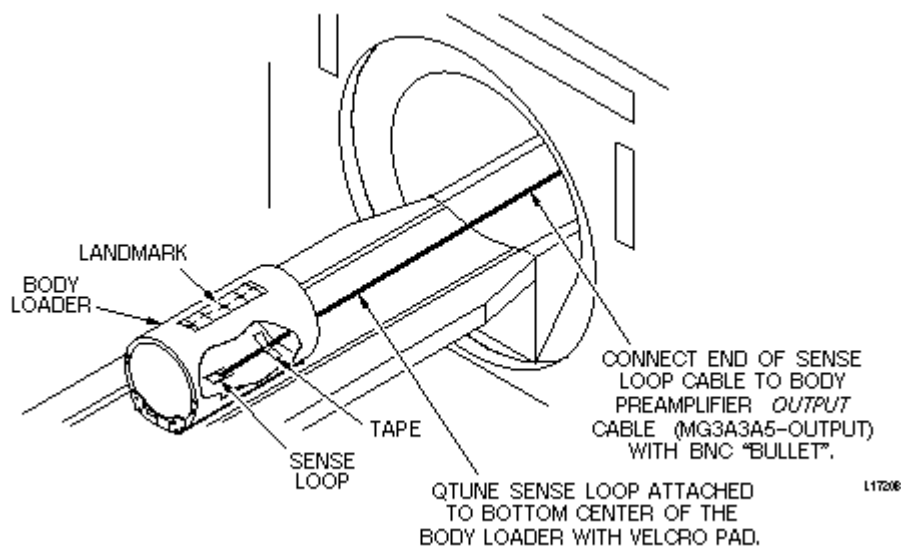
[Done].

F. **[Research Operations] [Download]**.

3. Refer to Illustrations 3-5 and 3-6 for the hardware setup.



BODY SENSE LOOP SET-UP — EFB ENABLED
ILLUSTRATION 3-5



BODY COIL HARDWARE SET-UP
ILLUSTRATION 3-6

4. **[Manual Prescan].**
5. From the "Windows" Menu bar on the Manual Prescan window, select **"Two Windows"**.
- Plot Type = **Magnitude**
Plot Gain = **1**
Plot Type = **Magnitude**
Plot Gain = **1**

6. Advance TG to 200, while adjusting variable attenuators A and B to prevent R1 and R2 overrange. Do NOT adjust R1 or R2.
7. With TG at 200, make final adjustments to variable attenuators A and B for R2 signal between 75 & 85%. Do NOT adjust R1 or R2.
8. Note the R2 signal value for the EFB Bypassed version.
9. **[Done]**.
10. **[Scan]**.
11. When scan completes, view results by using the Report Manager tool. Record data for each test in Data Sheet 1.5T & Data Sheet 1.0T found in the appendix.
12. Perform System Restoration.
13. Body Sense Loop — EFB Enabled Test Stability and Linearity Control Variables. Control Variables for Stability and Linearity trouble-shooting are listed in Table 3-4:

TABLE 3-4
BODY SENSE LOOP CONTROL VARIABLES — EFB ENABLED

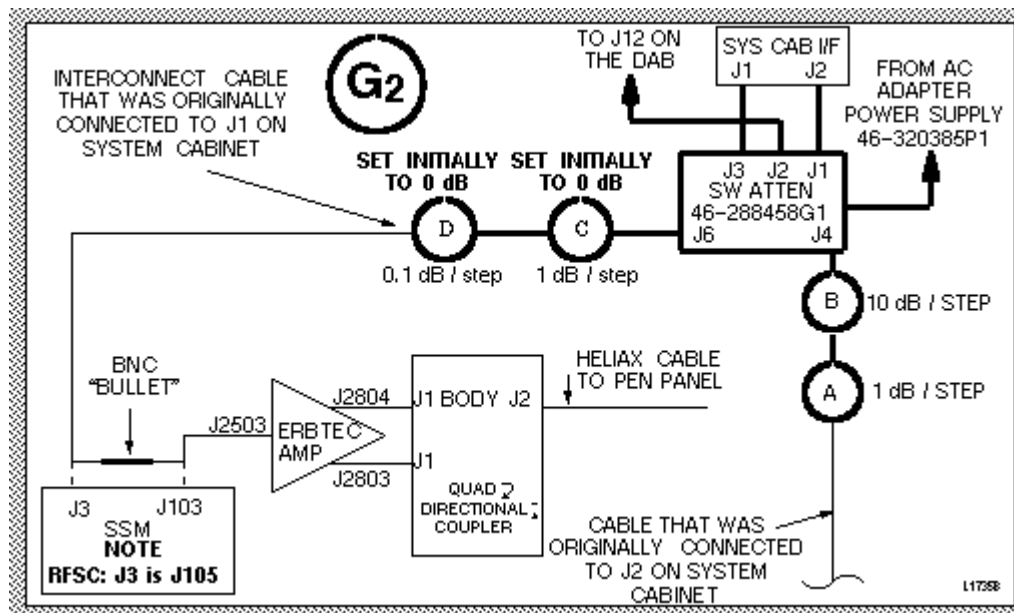
User CV Name	Baseline	Stability T/S	Linearity T/S
EFB Bypass	0	0	0
RFampMode	4	4	4
Sense Loop	1	1	1
StabNoGrad	1	1	0
Lin&Fidlty	1	0	1
Stab/Grad	4	4	0
Bandwidth	0	0	0
BW Cal	0	0	0

3-3-2 Body Sense Loop Baseline Data Test — EFB Bypassed

Note

This sub-section refers to the previous EFB Enabled sub-section which must be performed first.

1. Refer to Illustrations 3-6 and 3-7 for the hardware setup. Since scan will be performed with EFB bypassed, reconfigure as shown in Illustration 3-7.
 - a. Disconnect RFSC or SSM cables to bypass (RF IN and EFB RF Out). Join these two cables with a BNC "bullet".
 - b. Add variable attenuators C and D (from SST kit). Preset as indicated.



BODY SENSE LOOP SET-UP — EFB BYPASSED
ILLUSTRATION 3-7

2. Body Sense Loop — EFB Bypassed Baseline Test Control Variables should be entered using the same scan protocol listed in the previous sub-section. Control Variables for Stability and Linearity trouble-shooting are also listed in Table 3-5:

TABLE 3-5
BODY SENSE LOOP CONTROL VARIABLES — EFB BYPASSED

User CV Name	Baseline	Stability T/S	Linearity T/S
EFB Bypass	1	1	1
RFampMode	4	4	4
Sense Loop	1	1	1
StabNoGrad	1	1	0
Lin&Fidlty	1	0	1
Stab/Grad	4	4	0
Bandwidth	1	0	1
BW Cal	0	0	0

3. Refer to the R2 signal value noted in the previous EFB Enabled sub-section.
4. Resume manual prescan by selecting **[Manual Prescan]**.
5. At TG=200 adjust variable attenuators C and D to achieve the same R2 signal value that was recorded in step 8 (within + 1.0 point of the recorded value). (1.0 point is approximately 0.1dB).
6. Select **[Done]**.
7. Select **[Scan]**. When scan completes, view results by using the Report Manager tool. Record data for each test in the data sheets found in Appendix A.
8. After the baseline test is complete alternate shorter versions listed in Table 3-5 may be performed for Stability T/S or Linearity T/S.

3-4 Head Sense Loop Baseline Data Test

3-4-1 Head Sense Loop Baseline Data Test — EFB Enabled

This test evaluates the health of the Head Transmit chain and the RF field in the bore. If this test passes the Head Transmit chain is good. TPS Loopback Baseline Data Test is performed first. Control Variables for Stability and Linearity trouble-shooting are listed in this Sub-Section.

1. Prepare the system per sub-section 3-1 (TNF and RFSC or SSM).
2. Prepare the system to scan in Head mode per Table 3-6.

TABLE 3-6

HEAD SCAN PROTOCOL: HEAD SENSE LOOP BASELINE DATA TEST — EFB ENABLED

Note: This is the alternate proprietary procedure available for GE use, and to sites with a valid Advanced Service Package Limited License.

A. [New Pt]

Id: **geservice**<ENTER>

Name: **rft**

Weight (Lb.): **300**<ENTER>

Set Patient Protocols to **Service**.

At front enclosure, landmark per Illustration, press **LANDMARK**, then **MOVE TO SCAN**.

B. In the Patient Position Protocol field:

type **o.16.2**<ENTER>(o=Other, 16.2 =series) to load the head protocol

OR select **other** and select protocol **16** (RF test) and select series **2** (Head scan).

C. In the Additional Parameters field:

[User CVs Screen] .

CV Name: **EFB Bypass**, CV Value: **0**<ENTER>.

CV Name: **RFampMode**, CV Value: **2**<ENTER>.

CV Name: **Sense Loop**, CV Value: **1**<ENTER>.

CV Name: **StabNoGrad**, CV Value: **1**<ENTER>.

CV Name: **Lin&Fidlty**, CV Value: **1**<ENTER>.

CV Name: **Stab/Grad**, CV Value: **4**<ENTER>.

CV Name: **Bandwidth**, CV Value: **0**<ENTER>.

CV Name: **BW Cal**, CV Value: **0**<ENTER>.

[Accept].

D. [Save Series]. If necessary --- [Prepare to Scan].

E. [Research Operations] [Setup Params].

R1 = 13

R2 = 14

TG = 50

Number of Frames: **4**<ENTER>

Window 1: Frame: **1**<ENTER>

+/- = +

Window 1: Frame: **0** <ENTER>

Window 2: Frame: **2** <ENTER>

+/- = +

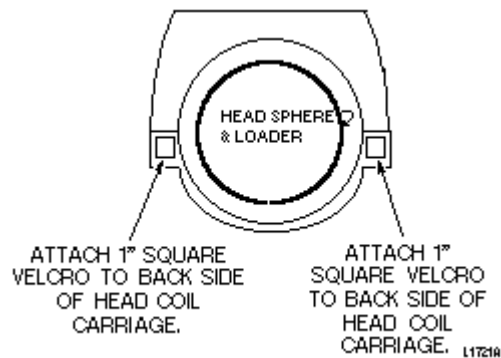
Window 2: Frame: **0** <ENTER>

[Done].

F. [Research Operations] [Download].

3. See Illustration 3-8, 3-9 and 3-10 for the head setup.

A. Attach Velcro as shown in Illustration 3-8.

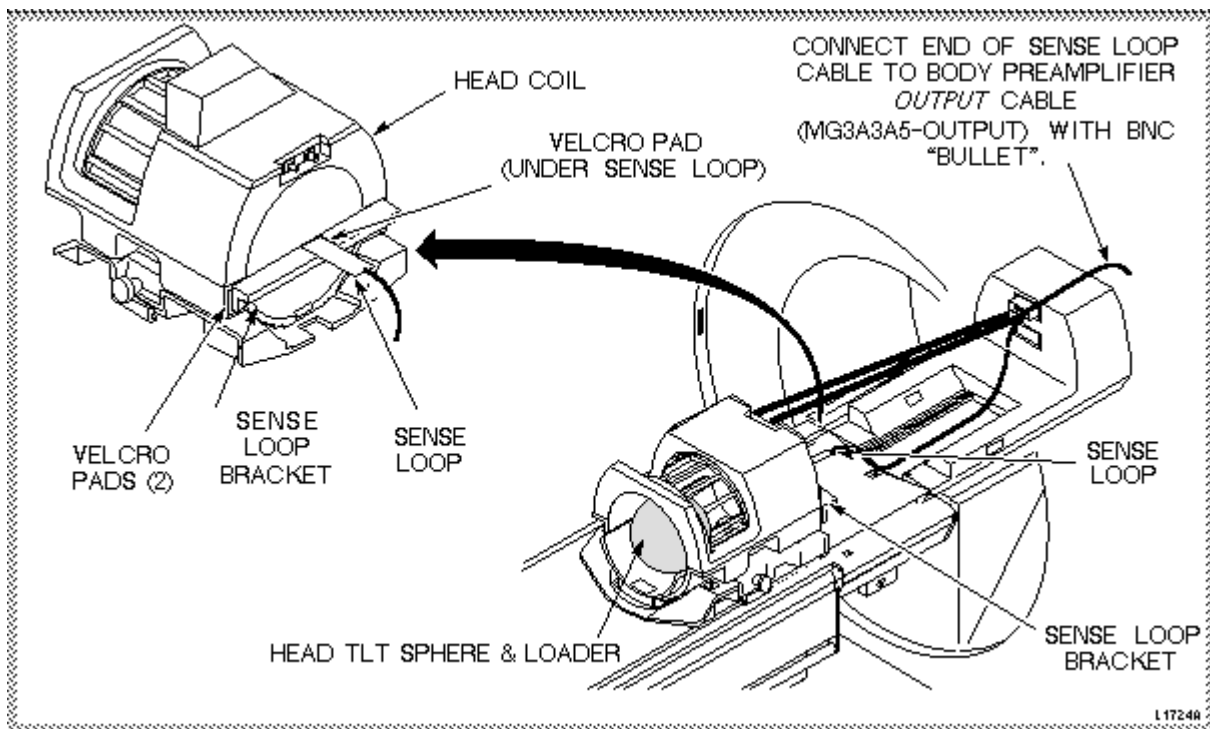


VELCRO ATTACHEMENT
ILLUSTRATION 3-8

B. Position sense loop per Illustration 3-9. DO NOT use tuning ring. Position the sense loop on the sense loop bracket as shown.

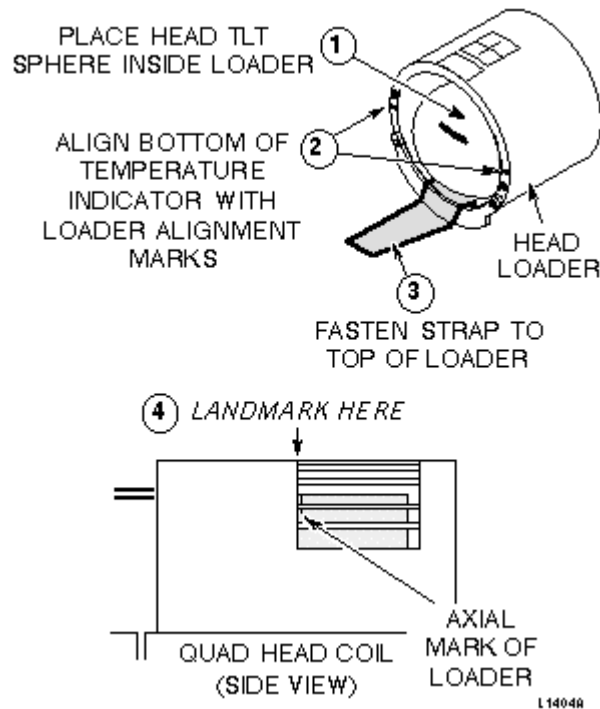
WARNING!

FOR THIS TEST CONFIGURATION, DO NOT PLACE THE SENSE LOOP IN THE HEAD COIL. THE HEAD COIL ASSEMBLY MAY BE DAMAGED.



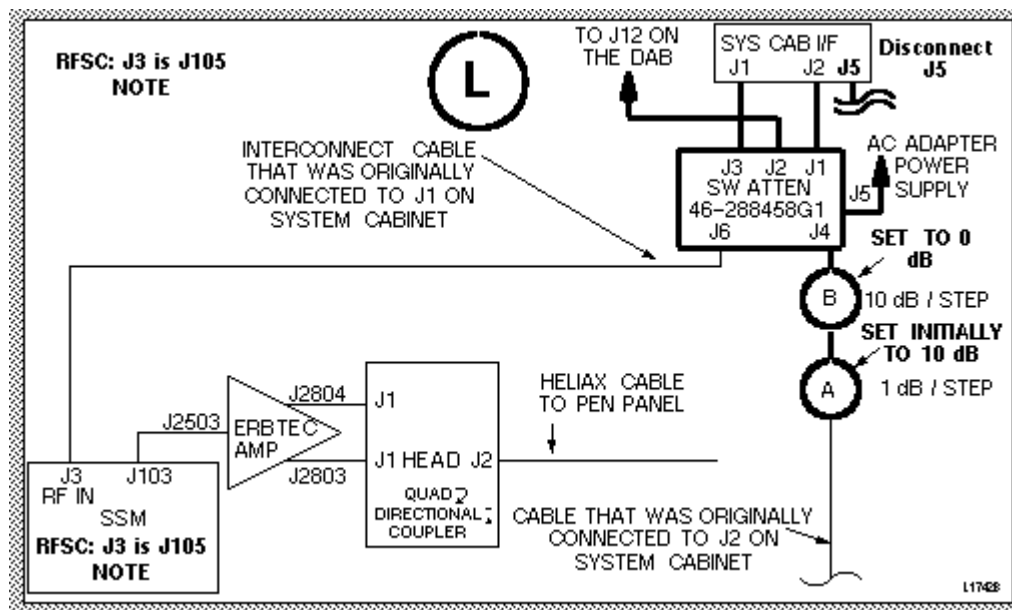
POSITIONING SENSE LOOP AT HEAD COIL
ILLUSTRATION 3-9

- C. Position and landmark Head TLT Sphere in the Loader per Illustration 3-10. DO NOT use tuning ring.



POSITIONING HEAD TLT SPHERE INSIDE THE HEAD LOADER
ILLUSTRATION 3-10

4. See Illustration 3-11 for the hardware setup.



HEAD SENSE LOOP SETUP — EFB ENABLED
ILLUSTRATION 3-11

5. Disconnect the Head Receive line at the System Cabinet I/F Panel J5 to prevent crosstalk issues.
6. **[Manual Prescan]**.
7. From the “Windows” Menu bar on the Manual Prescan window, select “**Two Windows**”.
 Plot Type = **Magnitude**
 Plot Gain = **1**
 Plot Type = **Magnitude**
 Plot Gain = **1**
7. Advance TG from 50 to 200, while adjusting variable attenuators A and B to prevent R1 and R2 over-range. Do NOT adjust R1 or R2.
8. With TG at 200, make final adjustments to variable attenuators A and B for R2 signal between 40 & 85%. Do NOT adjust R1 or R2.
9. Note the R2 signal value for the EFB Bypassed version.
10. Select **[Done]**.
11. Select **[Scan]**.
12. When scan completes, view results by using the Report Manager tool. Record data for each test in Data Sheet 1.5T & Data Sheet 1.0T found in Appendix A.
13. If not proceeding to next sub-section, perform System Restoration.
14. Head Sense Loop — EFB Enabled Test Stability and Linearity Control Variables. Control Variables for Stability and Linearity trouble-shooting are listed in Table 3-7:

TABLE 3-7
HEAD SENSE LOOP CONTROL VARIABLES — EFB ENABLED

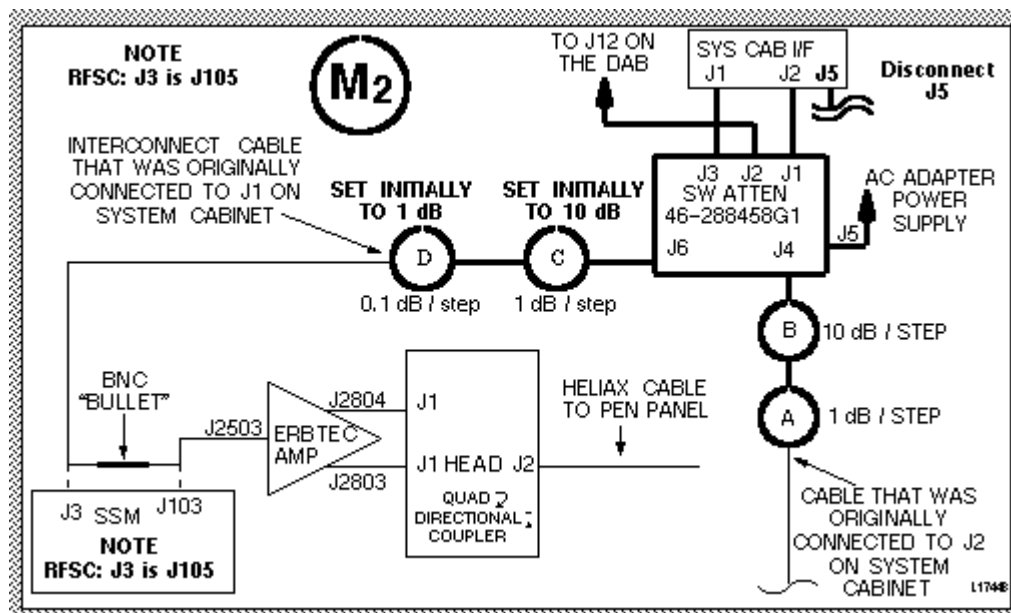
User CV Name	Baseline	Stability T/S	Linearity T/S
EFB Bypass	0	0	0
RFampMode	2	2	2
Sense Loop	1	1	1
StabNoGrad	1	1	0
Lin&Fidlty	1	0	1
Stab/Grad	4	4	0
Bandwidth	0	0	0
BW Cal	0	0	0

3-4-2 Head Sense Loop Baseline Data Test — EFB Bypassed

Note

This sub-section refers to the previous EFB Enabled sub-section which must be performed first.

1. Refer to Illustrations 3-8, 3-9, 3-10, and 3-12 and for the hardware setup. Since scan will be performed with EFB bypassed, reconfigure as shown in Illustration 3-12.
 - a. Disconnect RFSC or SSM cables to bypass (RF IN and EFB RF Out). Join these two cables with a BNC "bullet".
 - b. Add variable attenuators C and D (from SST kit). Preset as indicated.



HEAD SENSE LOOP SET-UP — EFB BYPASSED
ILLUSTRATION 3-12

2. Head Sense Loop — EFB Bypassed Baseline Test Control Variables should be entered using the same scan protocol listed in the previous sub-section. Control Variables for Stability and Linearity trouble-shooting are also listed in Table 3-8:

TABLE 3-8
HEAD SENSE LOOP CONTROL VARIABLES — EFB BYPASSED

User CV Name	Baseline	Stability T/S	Linearity T/S
EFB Bypass	1	1	1
RFampMode	2	2	2
Sense Loop	1	1	1
StabNoGrad	1	1	0
Lin&Fidlty	1	0	1
Stab/Grad	4	4	0
Bandwidth	1	0	1
BW Cal	0	0	0

3. Refer to the R2 signal value noted in the previous EFB Enabled sub-section.
4. Resume manual prescan by selecting **[Manual Prescan]**.
5. At TG=200 adjust variable attenuators C and D to achieve the same R2 signal value that was recorded in step 8 (within + 1.0 point of the recorded value). (1.0 point is approximately 0.1dB).
6. Select **[Done]**.
7. Select **[Scan]**. When scan completes, view results by using the Report Manager tool. Record data for each test in the data sheets found in Appendix A.
8. After the baseline test is complete alternate shorter versions listed in Table 3-8 may be performed for Stability T/S or Linearity T/S.

3-5 Body Coil Load Baseline Data Test

3-5-1 Body Coil Load Baseline Data Test — EFB Enabled

This test evaluates the health of the Body Transmit chain out to the Body Coil with the RF Amplifier loaded by the coil, however, it does not evaluate the RF field in the bore. TPS Loopback Baseline Data Test is performed first. Control Variables for Stability and Linearity trouble-shooting are listed in this Sub-Section.

1. Prepare the system per sub-section 3-1 (TNF and RFSC or SSM).
2. Prepare the system to scan in Body mode per Table 3-9.

TABLE 3-9

BODY SCAN PROTOCOL: BODY COIL LOAD BASELINE DATA TEST — EFB ENABLED

Note: This is the alternate proprietary procedure available for GE use, and to sites with a valid Advanced Service Package Limited License.

A. [New Pt]

Id: **geservice**<ENTER>

Name: **rft**

Weight (Lb.): **300**<ENTER>

Set Patient Protocols to **Service**.

At front enclosure, landmark in head area, press **LANDMARK**, then **MOVE TO SCAN**.

B. In the Patient Position Protocol field:

type **o.16.1**<ENTER>(o=Other, 16.1 =series) to load the body protocol

OR select **other** and select protocol **16** (RF Test) and select series **1** (RFT Body).

C. In the Additional Parameters field:

[User CVs Screen].

CV Name: **EFB Bypass**, CV Value: **0**<ENTER>.

CV Name: **RFampMode**, CV Value: **4**<ENTER>.

CV Name: **Sense Loop**, CV Value: **0**<ENTER>.

CV Name: **StabNoGrad**, CV Value: **1**<ENTER>.

CV Name: **Lin&Fidlty**, CV Value: **1**<ENTER>.

CV Name: **Stab/Grad**, CV Value: **4**<ENTER>.

CV Name: **Bandwidth**, CV Value: **0**<ENTER>.

CV Name: **BW Cal**, CV Value: **0**<ENTER>.

[Accept].

D. [Save Series]. If necessary --- [Prepare to Scan].

E. [Research Operations] [Setup Params].

R1 = 13

R2 = 14

TG = 50

Number of Frames: **4**<ENTER>

Window 1: Frame: **1**<ENTER>

+/- = +

Window 1: Frame: **0** <ENTER>

Window 2: Frame: **2** <ENTER>

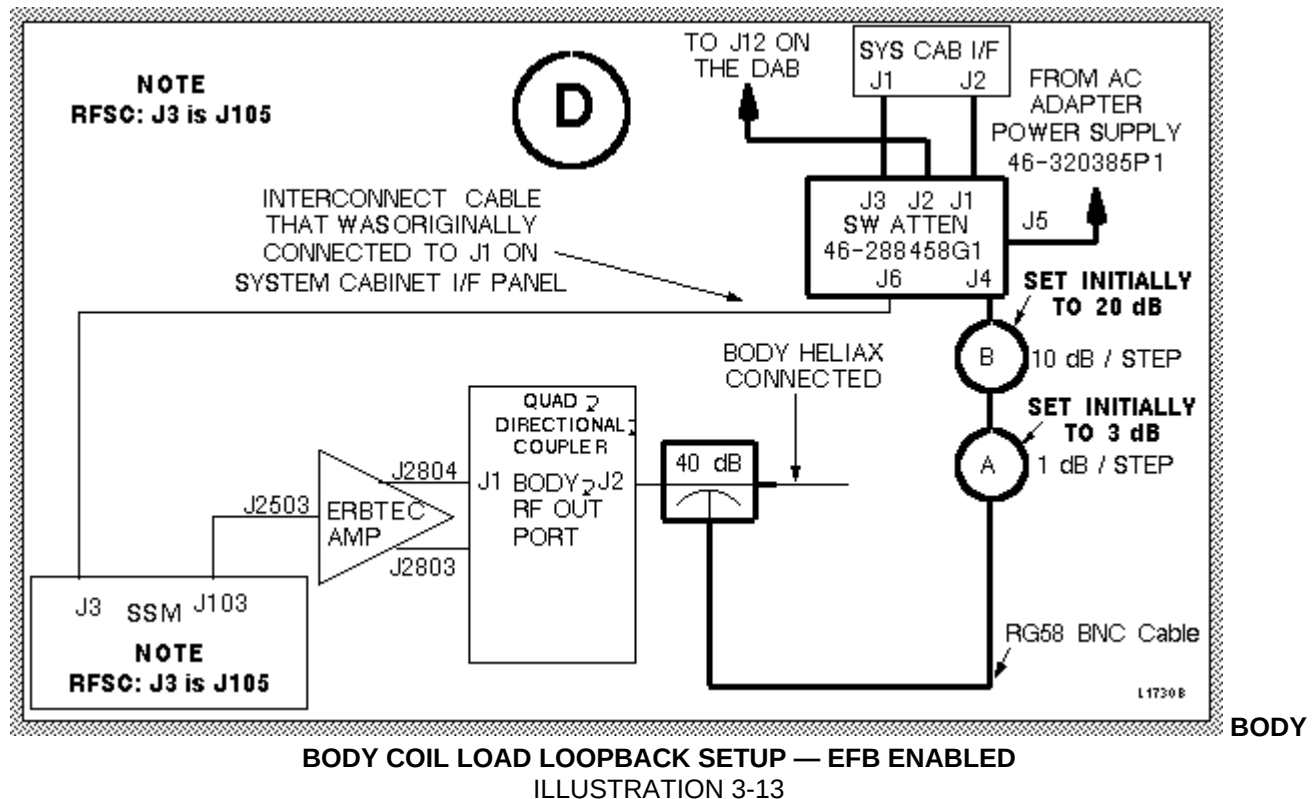
+/- = +

Window 2: Frame: **0** <ENTER>

[Done].

F. [Research Operations] [Download].

- See Illustration 3-13 for the hardware setup.



- [Manual Prescan].**
- From the “Windows” Menu bar on the Manual Prescan window, select “**Two Windows**”.
Plot Type = **Magnitude**
Plot Gain = **1**
Plot Type = **Magnitude**
Plot Gain = **1**
- Advance TG from 50 to 200, while adjusting variable attenuators A and B to prevent R1 and R2 over-range. Do NOT adjust R1 or R2.
- With TG at 200, make final adjustments to variable attenuators A and B for R2 signal between 75 & 85%. Do NOT adjust R1 or R2.
- Note the R2 signal value for the EFB Bypassed version.
- Select **[Done]**.
- Select **[Scan]**.
- When scan completes, view results by using the Report Manager tool. Record data for each test in Data Sheet 1.5T & Data Sheet 1.0T found in Appendix A.
- After the baseline test is complete alternate shorter versions listed in Table 3-10 may be performed for Stability T/S or Linearity T/S.

13. If not proceeding to next sub-section, perform System Restoration.

14. Body Coil Load — EFB Enabled Test Stability and Linearity Control Variables. Control Variables for Stability and Linearity trouble-shooting are listed in Table 3-10:

TABLE 3-10
BODY COIL LOAD CONTROL VARIABLES — EFB ENABLED

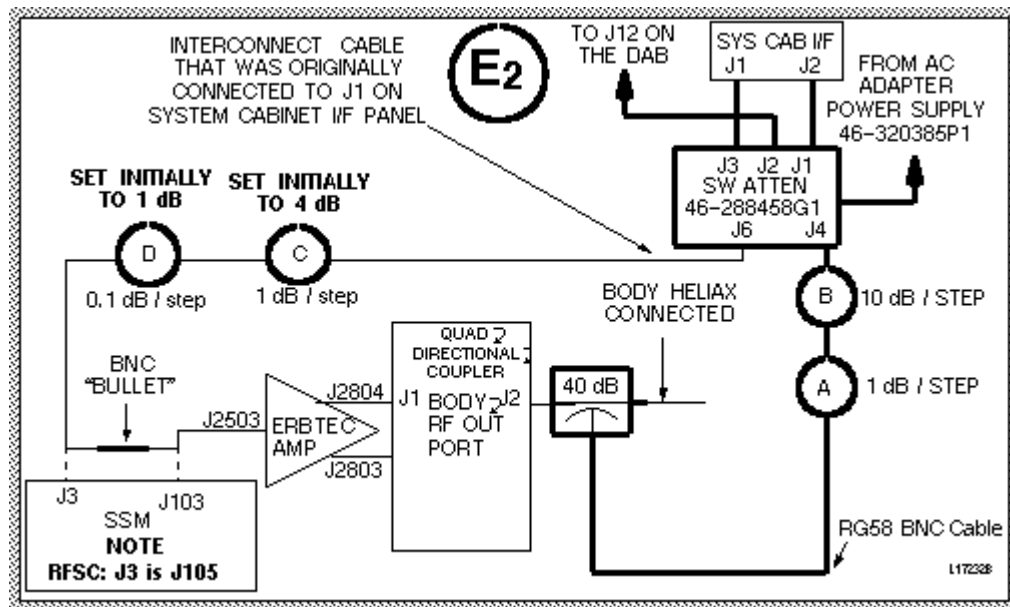
CV Name	Baseline	Stability T/S	Linearity T/S
EFB Bypass	0	0	0
RFampMode	4	4	4
SenseLoop	0	0	0
StabNoGrad	1	1	0
Lin&Fidlty	1	0	1
StabGrad	4	4	0
Bandwidth	0	0	0
BW Cal	0	0	0

3-5-2 Body Coil Load Baseline Data Test — EFB Bypassed

Note

This sub-section refers to the previous EFB Enabled sub-section which must be performed first.

1. Refer to Illustration 3-14 for the hardware setup. Scan will be performed with EFB bypassed, reconfigure as shown in Illustration 3-14.
 - a. Disconnect RFSC or SSM cables to bypass (RF IN and EFB RF Out). Join these two cables with a BNC "bullet".
 - b. Add variable attenuators C and D (from SST kit). Preset as indicated.



BODY COIL LOAD SET-UP — EFB BYPASSED
ILLUSTRATION 3-14

2. Body Coil Load — EFB Bypassed Baseline Test Control Variables should be entered using the same scan protocol listed in the previous sub-section. Control Variables for Stability and Linearity trouble-shooting are also listed in Table 3-11:

TABLE 3-11
BODY COIL LOAD CONTROL VARIABLES — EFB BYPASSED

User CV Name	Baseline	Stability T/S	Linearity T/S
EFB Bypass	1	1	1
RFampMode	4	4	4
Sense Loop	0	0	0
StabNoGrad	1	1	0
Lin&Fidlty	1	0	1
Stab/Grad	4	4	0
Bandwidth	1	0	1
BW Cal	1	0	0

3. Refer to the R2 signal value noted in the previous EFB Enabled sub-section.
4. Resume manual prescan by selecting **[Manual Prescan]**.
5. At TG=200 adjust variable attenuators C and D to achieve the same R2 signal value that was recorded in step 8 (within + 1.0 point of the recorded value). (1.0 point is approximately 0.1dB).
6. Select **[Done]**.
7. Select **[Scan]**. When scan completes, view results by using the Report Manager tool. Record data for each test in the data sheets found in Appendix A.
8. After the baseline test is complete alternate shorter versions listed in Table 3-11 may be performed for Stability T/S or Linearity T/S.

3-6 Body Dummy Load Baseline Data Test

3-6-1 Body Dummy Load Baseline Data Test — EFB Enabled

This test evaluates the Body Transmit chain up to the Body RF Out at the RF/PDU cabinet with the RF Amplifier loaded by a 50 ohm Dummy Load. Control Variables for Stability and Linearity trouble-shooting are listed in Sub-Section 3-6-1.

1. Prepare the system per sub-section 3-1 (TNF and RFSC or SSM).
2. Prepare the system to scan in Body mode per Table 3-12.

TABLE 3-12
BODY SCAN PROTOCOL: BODY DUMMY LOAD BASELINE — EFB ENABLED

Note: This is the alternate proprietary procedure available for GE use, and to sites with a valid Advanced Service Package Limited License.

A. **[New Pt]**

Id: **geservice**<ENTER>

Name: **rft**

Weight (Lb.): **300**<ENTER>

Set Patient Protocols to **Service**.

At front enclosure, landmark in head area, press **LANDMARK**, then **MOVE TO SCAN**.

B. In the Patient Position Protocol field:

type **o.16.1**<ENTER>(o=Other, 16.1 =series) to load the body protocol

OR select **other** and select protocol **16** (RF test) and select series **1** (RFT Body).

C. In the Additional Parameters field:

[User CVs Screen]

CV Name: **EFB Bypass**, CV Value: **0**<ENTER>.

CV Name: **RFampMode**, CV Value: **3**<ENTER>.

CV Name: **Sense Loop**, CV Value: **0**<ENTER>.

CV Name: **StabNoGrad**, CV Value: **1**<ENTER>.

CV Name: **Lin&Fidlty**, CV Value: **1**<ENTER>.

CV Name: **Stab/Grad**, CV Value: **4**<ENTER>.

CV Name: **Bandwidth**, CV Value: **0**<ENTER>.

CV Name: **BW Cal**, CV Value: **0**<ENTER>.

[Accept].

D. **[Save Series].** If necessary --- **[Prepare to Scan].**

E. **[Research Operations] [Setup Params].**

R1 = 13

R2 = 14

TG = 50

Number of Frames: **4**<ENTER>

Window 1: Frame: **1**<ENTER>

+/- = +

Window 1: Frame: **0** <ENTER>

Window 2: Frame: **2** <ENTER>

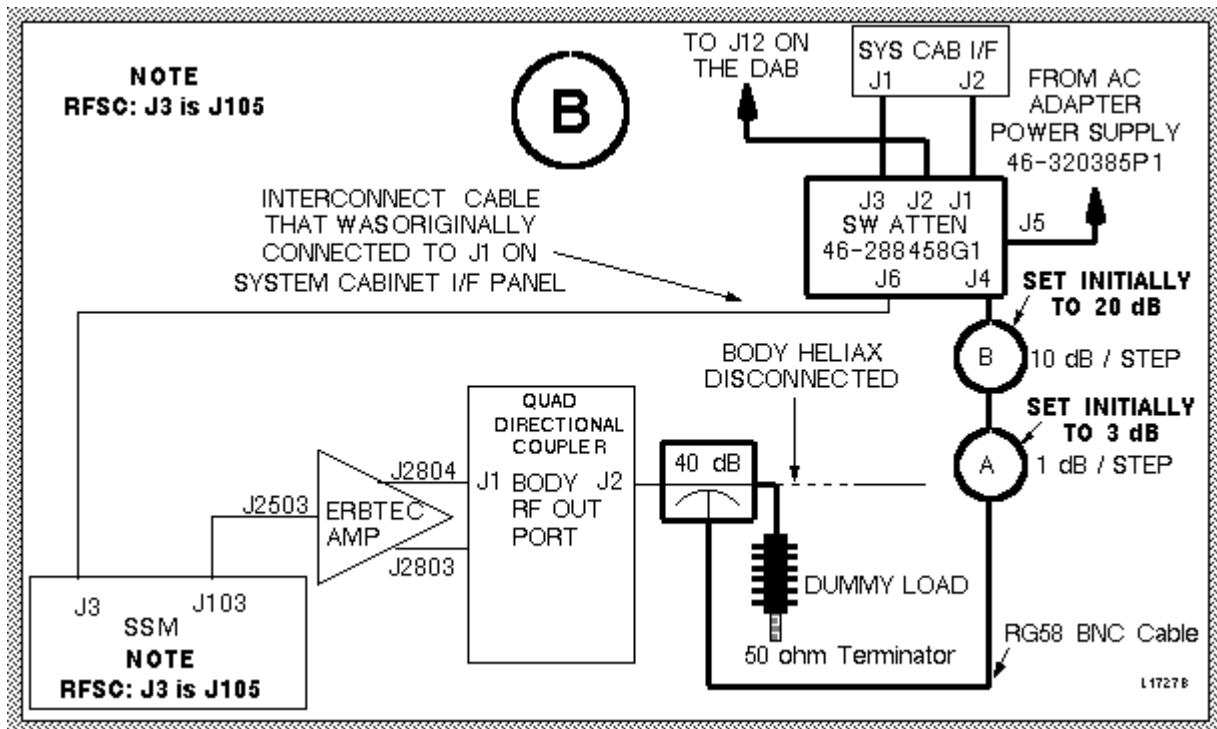
+/- = +

Window 2: Frame: **0** <ENTER>

[Done].

F. **[Research Operations] [Download].**

- See Illustration 3-15 for the hardware setup.



BODY DUMMY LOAD LOOPBACK SETUP — EFB ENABLED
ILLUSTRATION 3-15

- [Manual Prescan].**
- From the “Windows” Menu bar on the Manual Prescan window, select “**Two Windows**”.
Plot Type = **Magnitude**
Plot Gain = **1**
Plot Type = **Magnitude**
Plot Gain = **1**
- Advance TG from 50 to 200, while adjusting variable attenuators A and B to prevent R1 and R2 overrange. Do NOT adjust R1 or R2.
- With TG at 200, make final adjustments to variable attenuators A and B for R2 signal between 75 & 85%. Do NOT adjust R1 or R2.
- Note the R2 signal value for the EFB Bypassed version.
- [Done].**
- [Scan].**
- When scan completes, view results by using the Report Manager tool. Record data for each test in Data Sheet 1.5T & Data Sheet 1.0T found in the appendix.
- After the baseline test is complete alternate shorter versions listed in Table 3-13 may be performed for Stability T/S or Linearity T/S.

13. If not proceeding to the next sub-section perform System Restoration.

14. Body Dummy Load — EFB Enabled Test Stability and Linearity Control Variables. Control Variables for Stability and Linearity trouble-shooting are listed in Table 3-13:

TABLE 3-13
BODY DUMMY LOAD CONTROL VARIABLES — EFB ENABLED

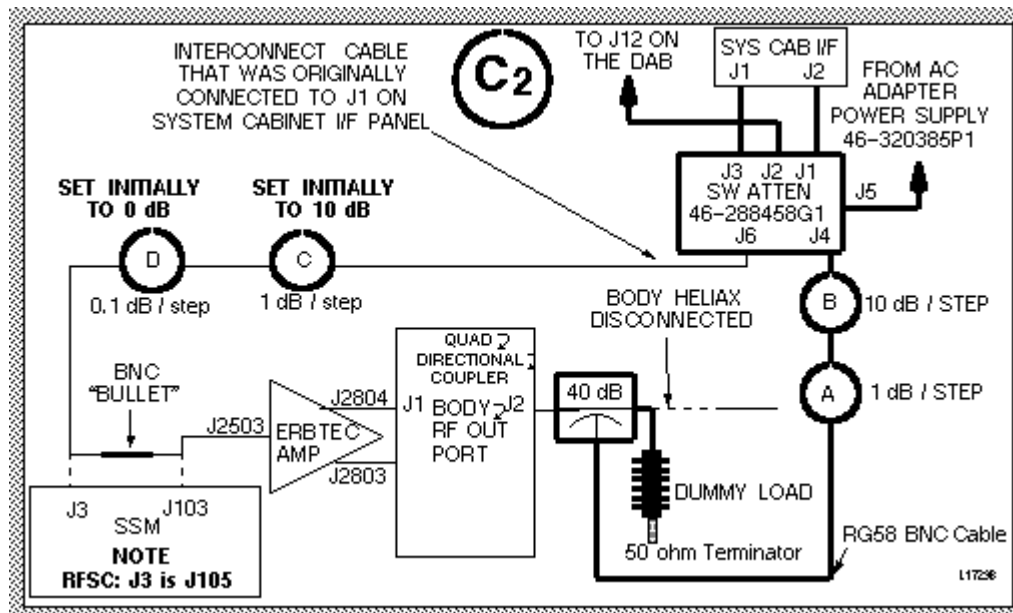
User CV Name	Baseline	Stability T/S	Linearity T/S
EFB Bypass	0	0	0
RFampMode	3	3	3
Sense Loop	0	0	0
StabNoGrad	1	1	0
Lin&Fidlty	1	0	1
Stab/Grad	4	4	0
Bandwidth	0	0	0
BW Cal	0	0	0

3-6-2 Body Dummy Load Baseline Data Test — EFB Bypassed

Note

This sub-section refers to the previous EFB Enabled sub-section which must be performed first.

1. Refer to Illustration 3-16 for the hardware setup. Scan will be performed with EFB bypassed, reconfigure as shown in Illustration 3-16.
 - a. Disconnect RFSC or SSM cables to bypass (RF IN and EFB RF Out). Join these two cables with a BNC "bullet".
 - b. Add variable attenuators C and D (from SST kit). Preset as indicated.



BODY DUMMY LOAD SET-UP — EFB BYPASSED
ILLUSTRATION 3-16

2. Body Dummy Load — EFB Bypassed Baseline Test Control Variables should be entered using the same scan protocol listed in the previous sub-section. Control Variables for Stability and Linearity trouble-shooting are also listed in Table 3-14:

TABLE 3-14
BODY DUMMY LOAD CONTROL VARIABLES — EFB BYPASSED

User CV Name	Baseline	Stability T/S	Linearity T/S
EFB Bypass	1	1	1
RFampMode	3	3	3
Sense Loop	0	0	0
StabNoGrad	1	1	0
Lin&Fidlty	1	0	1
Stab/Grad	4	4	0
Bandwidth	1	0	1
BW Cal	0	0	0

3. Refer to the R2 signal value noted in the previous EFB Enabled sub-section.
4. Resume manual prescan by selecting **[Manual Prescan]**.
5. At TG=200 adjust variable attenuators C and D to achieve the same R2 signal value that was recorded in step 8 (within + 1.0 point of the recorded value). (1.0 point is approximately 0.1dB).
6. Select **[Done]**.
7. Select **[Scan]**. When scan completes, view results by using the Report Manager tool. Record data for each test in the data sheets found in Appendix A.
8. After the baseline test is complete alternate shorter versions listed in Table 3-14 may be performed for Stability T/S or Linearity T/S.

3-7 Head Dummy Load Baseline Data Test

3-7-1 Head Dummy Load Baseline Data Test — EFB Enabled

This test evaluates the Head Transmit chain up to the Head RF Out at the RF/PDU cabinet with the RF Amplifier loaded by a 50 ohm Dummy Load. Control Variables for Stability and Linearity trouble-shooting are listed in Sub-Section 3-7-1.

1. Prepare the system per sub-section 3-1 (TNF and RFSC or SSM).
2. Prepare the system to scan in Head mode per Table 3-15.

TABLE 3-15

HEAD SCAN PROTOCOL: HEAD DUMMY LOAD BASELINE — EFB ENABLED

Note: This is the alternate proprietary procedure available for GE use, and to sites with a valid Advanced Service Package Limited License.

A. **[New Pt]**

Id: **geservice**<ENTER>

Name: **rft**

Weight (Lb.): **300**<ENTER>

Set Patient Protocols to **Service**.

At front enclosure, landmark per Illustration, press **LANDMARK**, then **MOVE TO SCAN**.

B. In the Patient Position Protocol field:

type **o.16.2**<ENTER>(o=Other, 16.2 =series) to load the head protocol

OR select **other** and select protocol **16** (RF test) and select series **2** (Head scan).

C. In the Additional Parameters field:

[User CVs Screen] .

CV Name: **EFB Bypass**, CV Value: **0**<ENTER>.

CV Name: **RFampMode**, CV Value: **1**<ENTER>.

CV Name: **Sense Loop**, CV Value: **0**<ENTER>.

CV Name: **StabNoGrad**, CV Value: **1**<ENTER>.

CV Name: **Lin&Fidlty**, CV Value: **1**<ENTER>.

CV Name: **Stab/Grad**, CV Value: **4**<ENTER>.

CV Name: **Bandwidth**, CV Value: **0**<ENTER>.

CV Name: **BW Cal**, CV Value: **0**<ENTER>.

[Accept].

D. **[Save Series]**. If necessary --- **[Prepare to Scan]**.

E. **[Research Operations]** **[Setup Params]**.

R1 = 13

R2 = 14

TG = 50

Number of Frames: **4**<ENTER>

Window 1: Frame: **1**<ENTER>

+/- = +

Window 1: Frame: **0** <ENTER>

Window 2: Frame: **2** <ENTER>

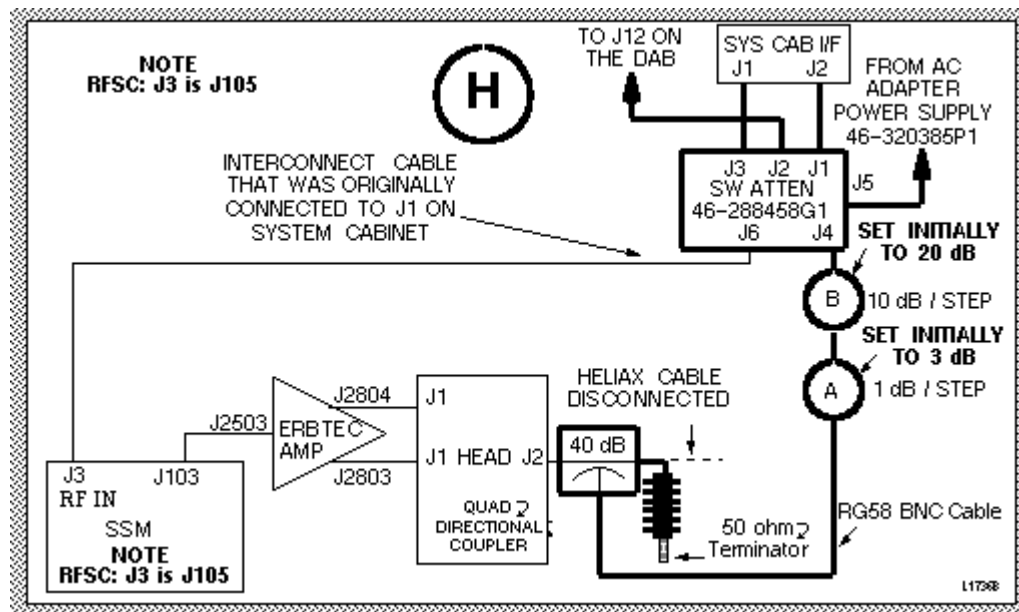
+/- = +

Window 2: Frame: **0** <ENTER>

[Done].

F. **[Research Operations]** **[Download]**.

- See Illustration 3-17 for the hardware setup. Use RF Power Measurement Kit 40 dB Coupler. Connect 50 ohm Terminator to Dummy Load Output.



HEAD MODE, DUMMY LOAD LOOPBACK SETUP - EFB ENABLED
ILLUSTRATION 3-17

- Start **[Manual Prescan]**.
- From the "Windows" Menu bar on the Manual Prescan window, select **"Two Windows"**.
Plot Type = **Magnitude**
Plot Gain = **1**
Plot Type = **Magnitude**
Plot Gain = **1**
- Advance TG from 50 to 200, while adjusting variable attenuators A and B to prevent R1 and R2 overrange. Do NOT adjust R1 or R2.
- With TG at 200, make final adjustments to variable attenuators A and B for R2 signal between 75 & 85%. Do NOT adjust R1 or R2.
- Note the R2 signal value for the EFB Bypassed version.
- Select **[Done]**.
- Select **[Scan]**. When scan completes, view results by using the Report Manager tool. Record data for each test in Data Sheet 1.5T & Data Sheet 1.0T found in the Appendix A.
- After the baseline test is complete alternate shorter versions listed in Table 3-13 may be performed for Stability T/S or Linearity T/S.
- If not proceeding to the next sub-section perform System Restoration.

13. Head Dummy Load — EFB Enabled Test Stability and Linearity Control Variables. Control Variables for Stability and Linearity trouble-shooting are listed in Table 3-16:

TABLE 3-16
HEAD DUMMY LOAD CONTROL VARIABLES — EFB ENABLED

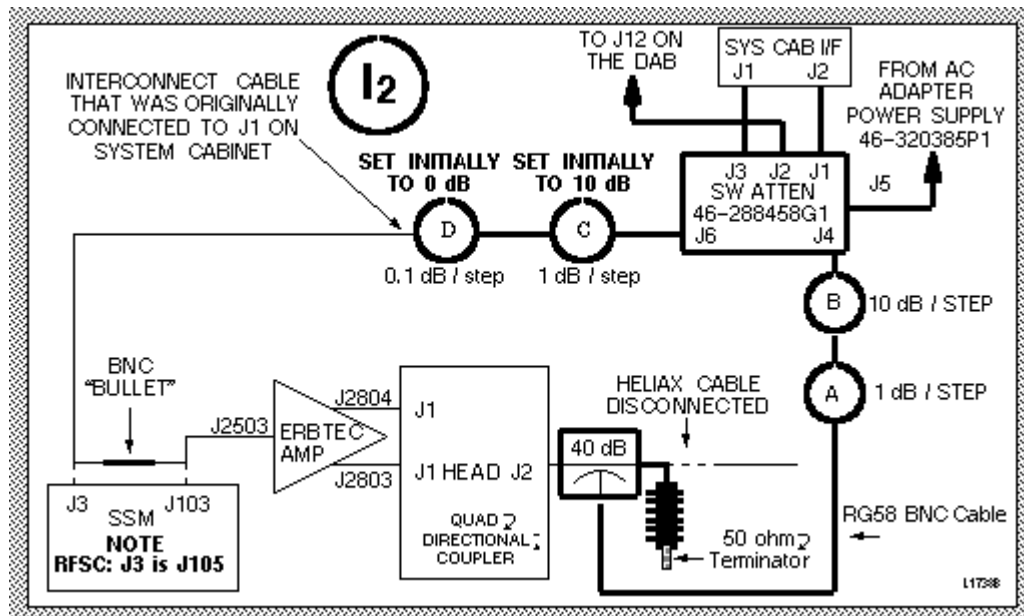
User CV Name	Baseline	Stability T/S	Linearity T/S
EFB Bypass	0	0	0
RFampMode	1	1	1
Sense Loop	0	0	0
StabNoGrad	1	1	0
Lin&Fidlty	1	0	1
Stab/Grad	4	4	0
Bandwidth	0	0	0
BW Cal	0	0	0

3-7-2 Head Dummy Load Baseline Data Test — EFB Bypassed

Note

This sub-section refers to the previous EFB Enabled sub-section which must be performed first.

1. Refer to Illustration 3-18 for the hardware setup. Scan will be performed with EFB bypassed, reconfigure as shown in Illustration 3-18.
 - a. Disconnect RFSC or SSM cables to bypass (RF IN and EFB RF Out). Join these two cables with a BNC "bullet".
 - b. Add variable attenuators C and D (from SST kit). Preset as indicated.



HEAD MODE DUMMY LOAD LOOPBACK SETUP - EFB BYPASSED
ILLUSTRATION 3-18

2. Head Dummy Load — EFB Bypassed Baseline Test Control Variables should be entered using the same scan protocol listed in the previous sub-section. Control Variables for Stability and Linearity trouble-shooting are also listed in Table 3-17:

TABLE 3-17
HEAD DUMMY LOAD CONTROL VARIABLES — EFB BYPASSED

User CV Name	Baseline	Stability T/S	Linearity T/S
EFB Bypass	1	1	1
RFampMode	1	1	1
Sense Loop	0	0	0
StabNoGrad	1	1	0
Lin&Fidlty	1	0	1
Stab/Grad	4	4	0
Bandwidth	1	0	1
BW Cal	0	0	0

3. Refer to the R2 signal value noted in the previous EFB Enabled sub-section.
4. Resume manual prescan by selecting **[Manual Prescan]**.
5. At TG=200 adjust variable attenuators C and D to achieve the same R2 signal value that was recorded in step 8 (within + 1.0 point of the recorded value). (1.0 point is approximately 0.1dB).
6. Select **[Done]**.
7. Select **[Scan]**. When scan completes, view results by using the Report Manager tool. Record data for each test in the data sheets found in Appendix A.
8. After the baseline test is complete alternate shorter versions listed in Table 3-17 may be performed for Stability T/S or Linearity T/S.

3-8 Head Coil Load Baseline Data Test

3-8-1 Head Coil Load Baseline Data Test — EFB Enabled

This test evaluates the health of the Head Transmit chain out to the Head Coil with the RF Amplifier loaded by the coil, however, it does not evaluate the RF field in the bore. TPS Loopback Baseline Data Test is performed first. Control Variables for Stability and Linearity trouble-shooting are listed in this Sub-Section.

1. Prepare the system per sub-section 3-1 (TNF and RFSC or SSM).
2. Prepare the system to scan in Head mode per Table 3-18.

TABLE 3-18

HEAD SCAN PROTOCOL: HEAD COIL LOAD BASELINE — EFB ENABLED

Note: This is the alternate proprietary procedure available for GE use, and to sites with a valid Advanced Service Package Limited License.

A. **[New Pt]**

Id: **geservice**<ENTER>

Name: **rft**

Weight (Lb.): **300**<ENTER>

Set Patient Protocols to **Service**.

At front enclosure, landmark per Illustration, press **LANDMARK**, then **MOVE TO SCAN**.

B. In the Patient Position Protocol field:

type **o.16.2**<ENTER>(o=Other, 16.2 =series) to load the head protocol

OR select **other** and select protocol **16** (RF test) and select series **2** (Head scan).

C. In the Additional Parameters field:

[User CVs Screen] .

CV Name: **EFB Bypass**, CV Value: **0**<ENTER>.

CV Name: **RFampMode**, CV Value: **2**<ENTER>.

CV Name: **Sense Loop**, CV Value: **0**<ENTER>.

CV Name: **StabNoGrad**, CV Value: **1**<ENTER>.

CV Name: **Lin&Fidlty**, CV Value: **1**<ENTER>.

CV Name: **Stab/Grad**, CV Value: **4**<ENTER>.

CV Name: **Bandwidth**, CV Value: **0**<ENTER>.

CV Name: **BW Cal**, CV Value: **0**<ENTER>.

[Accept].

D. **[Save Series]**. If necessary --- **[Prepare to Scan]**.

E. **[Research Operations]** **[Setup Params]**.

R1 = 13

R2 = 14

TG = 50

Number of Frames: **4**<ENTER>

Window 1: Frame: **1**<ENTER>

+/- = +

Window 1: Frame: **0** <ENTER>

Window 2: Frame: **2** <ENTER>

+/- = +

Window 2: Frame: **0** <ENTER>

[Done].

F. **[Research Operations]** **[Download]**.

13. Head Coil Load — EFB Enabled Test Stability and Linearity Control Variables. Control Variables for Stability and Linearity trouble-shooting are listed in Table 3-19:

TABLE 3-19
HEAD COIL LOAD CONTROL VARIABLES — EFB ENABLED

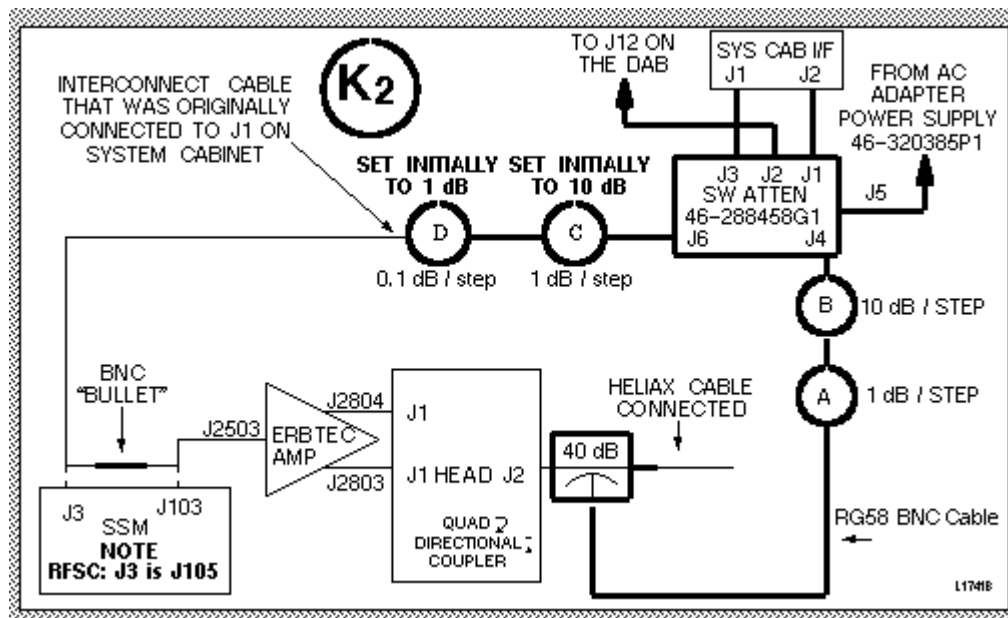
User CV Name	Baseline	Stability T/S	Linearity T/S
EFB Bypass	0	0	0
RFampMode	2	2	2
Sense Loop	0	0	0
StabNoGrad	1	1	0
Lin&Fidlty	1	0	1
Stab/Grad	4	4	0
Bandwidth	0	0	0
BW Cal	0	0	0

3-8-1 Head Coil Load Baseline Data Test — EFB Bypassed

Note

This sub-section refers to the previous EFB Enabled sub-section which must be performed first.

1. Refer to Illustration 3-20 for the hardware setup. Scan will be performed with EFB bypassed, reconfigure as shown in Illustration 3-20.
 - a. Disconnect RFSC or SSM cables to bypass (RF IN and EFB RF Out). Join these two cables with a BNC "bullet".
 - b. Add variable attenuators C and D (from SST kit). Preset as indicated.



HEAD MODE COIL LOAD LOOPBACK SETUP - EFB BYPASSED
ILLUSTRATION 5-27

2. Head Coil Load — EFB Bypassed Baseline Test Control Variables should be entered using the same scan protocol listed in the previous sub-section. Control Variables for Stability and Linearity trouble-shooting are also listed in Table 3-20:

TABLE 3-20
HEAD COIL LOAD CONTROL VARIABLES — EFB BYPASSED

User CV Name	Baseline	Stability T/S	Linearity T/S
EFB Bypass	1	1	1
RFampMode	2	2	2
Sense Loop	0	0	0
StabNoGrad	1	1	0
Lin&Fidlty	1	0	1
Stab/Grad	4	4	0
Bandwidth	1	0	1
BW Cal	1	0	0

3. Refer to the R2 signal value noted in the previous EFB Enabled sub-section.
4. Resume manual prescan by selecting **[Manual Prescan]**.
5. At TG=200 adjust variable attenuators C and D to achieve the same R2 signal value that was recorded in step 8 (within + 1.0 point of the recorded value). (1.0 point is approximately 0.1dB).
6. Select **[Done]**.
7. Select **[Scan]**. When scan completes, view results by using the Report Manager tool. Record data for each test in the data sheets found in Appendix A.
8. After the baseline test is complete alternate shorter versions listed in Table 3-20 may be performed for Stability T/S or Linearity T/S.

4- COIL TUNING SENSE LOOP TESTS

The Coil Tuning Sense Loop Tests evaluate the coils (body or head) resonant frequency and quality factor ($q=cf/bw$). Perform TPS Loopback and Coil Tune Preamplifier Check, then perform Body Coil Tune Sense Loop Check on all planes.

4-1 Body Coil Tuning Sense Loop Test

1. Prepare the system per sub-section 3-1 (TNF and RFSC or SSM).
2. Prepare the system to scan in Body mode per Table 4-1.

TABLE 4-1
BODY SCAN PROTOCOL: BODY COIL TUNING SENSE LOOP TEST

Note: This is the alternate proprietary procedure available for GE use, and to sites with a valid Advanced Service Package Limited License.

A. **[New Pt]**

Id: **geservice**<ENTER>

Name: **rft**

Weight (Lb.): **300**<ENTER>

Set Patient Protocols to **Service**.

At front enclosure, landmark in head area, press **LANDMARK**, then **MOVE TO SCAN**.

B. In the Patient Position Protocol field:

type **o.16.1**<ENTER>(o=Other, 16.1 =series) to load the body protocol

OR select **other** and select protocol **16** (RF Test) and select series **1** (RFT Body).

C. In the Additional Parameters field:

[User CVs Screen].

CV Name: **EFB Bypass**, CV Value: **1**<ENTER>.

CV Name: **RFampMode**, CV Value: **7**<ENTER>.

CV Name: **Sense Loop**, CV Value: **1**<ENTER>.

CV Name: **StabNoGrad**, CV Value: **0**<ENTER>.

CV Name: **Lin&Fidlty**, CV Value: **0**<ENTER>.

CV Name: **Stab/Grad**, CV Value: **0**<ENTER>.

CV Name: **Bandwidth**, CV Value: **1**<ENTER>.

CV Name: **BW Cal**, CV Value: **0**<ENTER>.

[Accept].

D. **[Save Series]**. If necessary --- **[Prepare to Scan]**.

E. **[Research Operations]** **[Setup Params]**.

R1 = 13

R2 = 14

TG = 50

Number of Frames: **4**<ENTER>

Window 1: Frame: **1**<ENTER>

+/- = +

Window 1: Frame: **0** <ENTER>

Window 2: Frame: **2** <ENTER>

+/- = +

Window 2: Frame: **0** <ENTER>

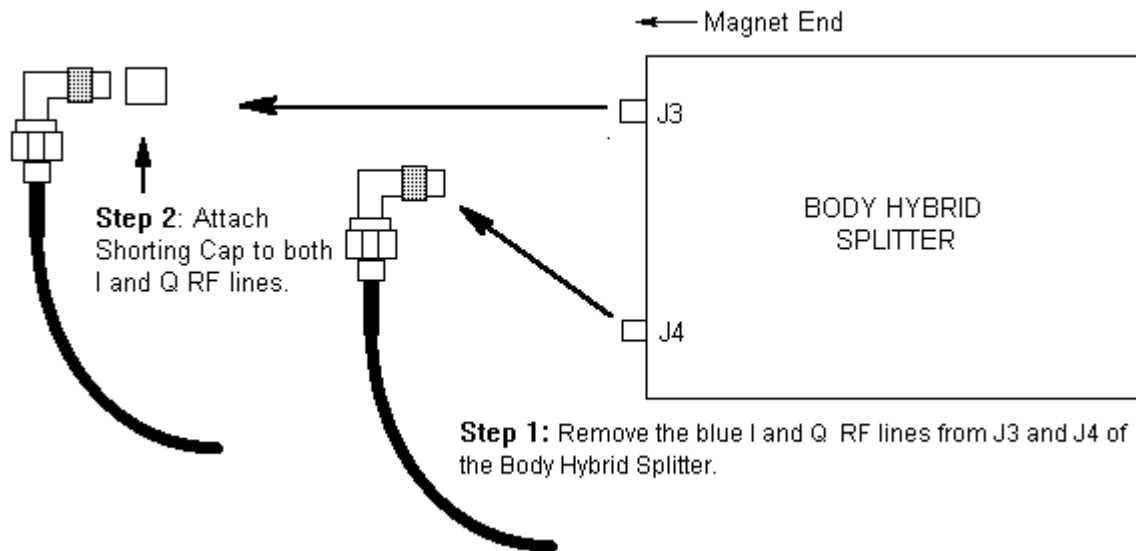
[Done].

F. **[Research Operations]** **[Download]**.

- For an accurate body coil tuning reading, the Body Hybrid Splitter must be electrically isolated from the Body Coil. Remove the blue I and Q RF lines from J3 and J4 of the Body Hybrid Splitter.
- Once the I and Q lines have been removed, attach an N Shorting Cap (46-265916P1) to each line. Refer to Illustration 4-1.

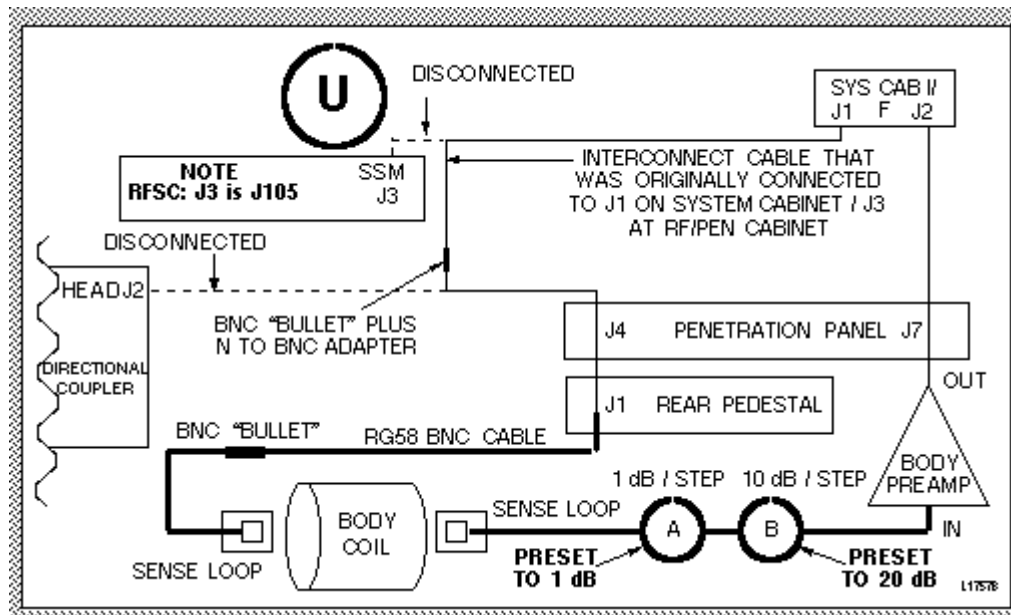
Note

The N Shorting cap is supplied with the Coil Tune Extension kit (46-265917G1).



TERMINATING I & Q RF LINES WITH SHORTING CAPS
ILLUSTRATION 4-1

- Connect system as shown in Illustration 4-2. Refer to Illustration 4-3 and Table 4-2 for Sense Loop placement.



BODY COIL TUNE SENSE LOOP CHECK
ILLUSTRATION 4-2

- A. Position the Sense coils per Illustration 4-3 using sticky backed velcro.
- B. RF Coil type identified by part number (not description) on the sticker at rear of the Coil Assembly, see Table 4-2 for help:

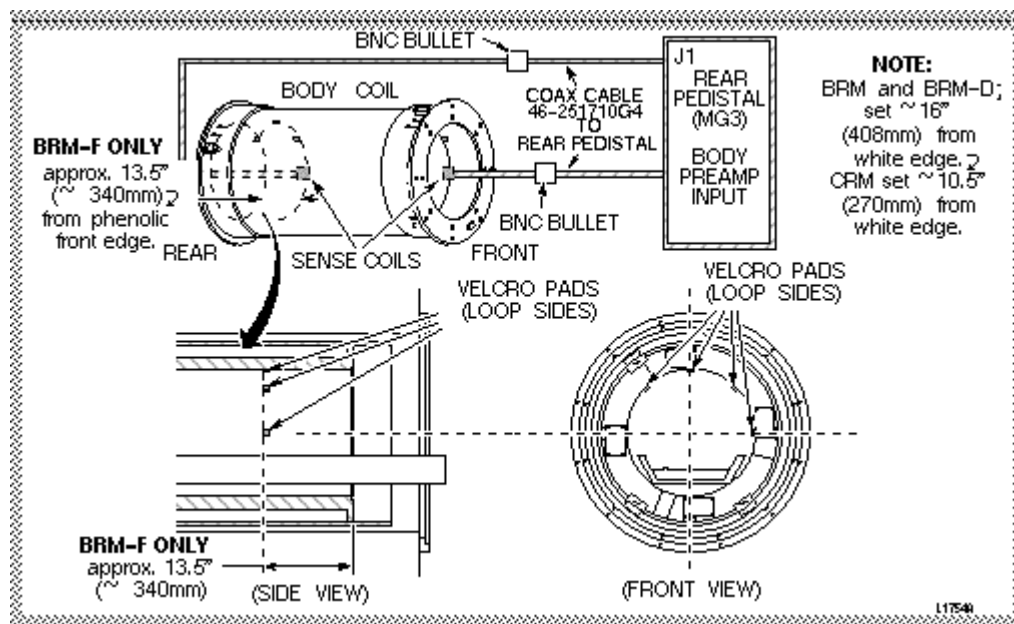
TABLE 4-2
SENSE COIL POSITIONING

RF Coil Type	Part Number	Measure from:	Measured Value
BRM or BRM/D	46-328500G1 46-328500G2	Front edge of white RF Coil Tube (Cone/RF Coil Tube edge).	approximately 16 inches (approximately 408 mm)
CRM	2186651	Front edge of white RF Coil Tube (Cone/RF Coil Tube edge).	approximately 10.5 inches (approximately 270 mm)
BRM/F	2218111 2218111-2	Front edge of white RF Coil Tube (Cone/RF Coil Tube edge).	approximately 0.9 inches (approximately 32.2 mm)

Note

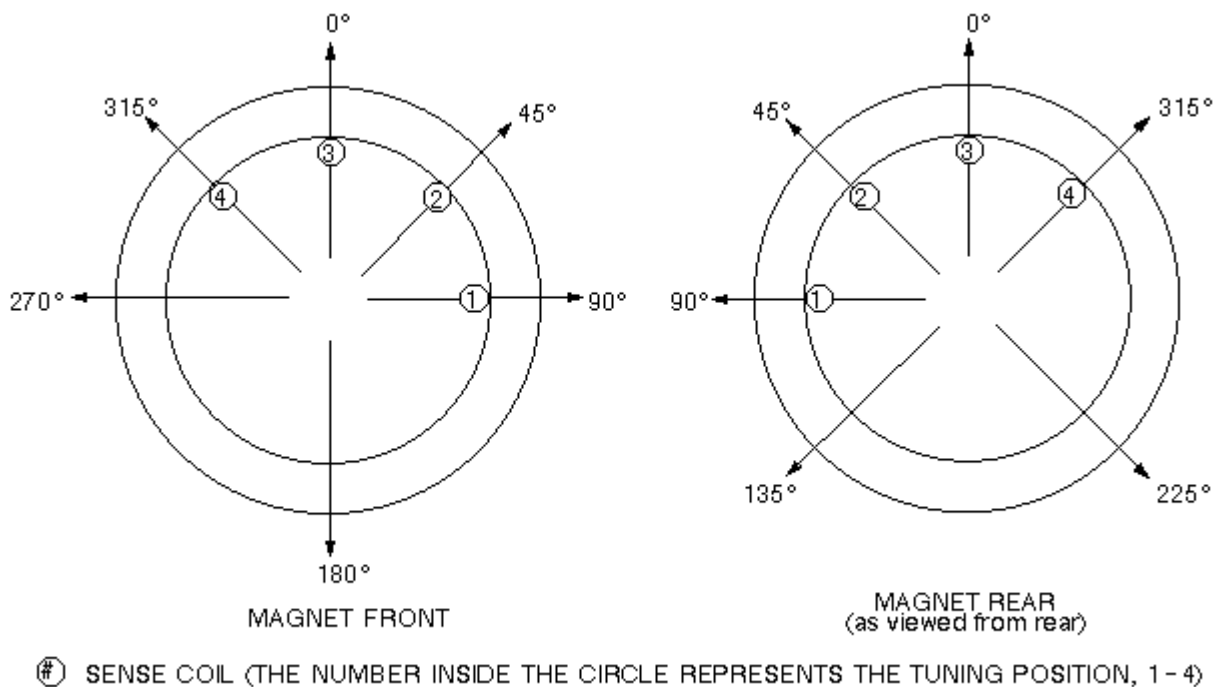
When routing the cables for the Sense Coils, try to lay them as straight as possible and without crossing them. Crossed cables will affect data collection due to crosstalk.

The sense coils must be positioned correctly or tuning problems will occur. Placing the sense coils too far into the Body Coil will cause over-coupling (creating 2 false peaks). Placing the sense coils too far out of the Body Coil will cause a decrease in frequency and sensitivity. Ensure that the sense coil foil run is placed the proper distance in per Table 4-3, Sense Coil Positioning).



SENSE COIL POSITIONING
ILLUSTRATION 4-3

4. Start **[Manual Prescan]**.
5. From the "Windows" Menu bar on the Manual Prescan window, select **"Two Windows"**.
Plot Type = **Magnitude**
Plot Gain = **1**
Plot Type = **Magnitude**
Plot Gain = **1**
6. Advance TG from 50 to 200, while adjusting variable attenuators A and B to prevent R1 and R2 overrange. Do NOT adjust R1 or R2.
7. With TG at 200, make final adjustments to variable attenuators A and B for R2 signal between 75 & 85%. Do NOT adjust R1 or R2.
8. Select **[Done]**.
9. Select **[Scan]**. When scan completes, view results by using the Report Manager tool. Record the "Center Frequency" value for the sense coil plane. (This value will later be used in calculating Peak Frequency).
10. Perform a new Body Coil Tune Sense Loop on all remaining planes. Refer to Illustration 4-4 for plane references (positions 2 through 4).



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BODY SENSE COIL PLACEMENT
ILLUSTRATION 4-4

11. After obtaining Body Coil Tune Check Scans from all planes (0, 45, 90, and 315 degrees), perform the following analysis:

- A. Select **Report Tool** under **[Utilities]** on the Service Desktop. Select the **File** menu, then select **Open** from the pull-down sub-menu. Choose the pertinent file from the *Load Data File* pop up window. The most recent files appear at the end of the file directory.
- B. **Center Frequency:** Record the center frequency obtained at each plane. Subtract the lowest frequency value from the highest frequency. The “mode delta” difference must be less than or equal to 150,000 Hz (1.5T) or 100,000 Hz (1.0T).

Example:

- degree plane: body coil tune check scan: 42.731 MHz
- degree plane: body coil tune check scan: 42.683 MHz
- $42.731\text{MHz} - 42.683\text{ MHz} = 48,000\text{ Hz}$ or 48 KHz. These frequency values are within spec.

12. If the bandwidth is not correct, then perform “Preamplifier Bandwidth Check” and troubleshoot the problem.

4-2 Head Coil Tuning Sense Loop Test

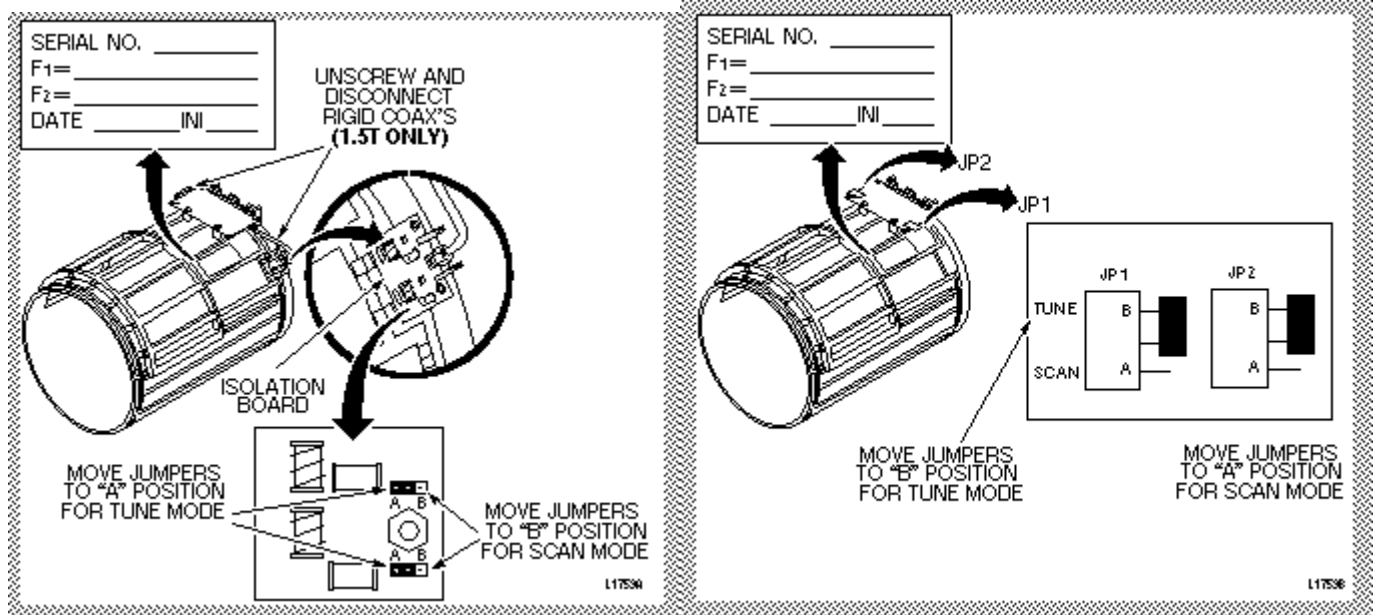
1. Prepare the system per sub-section 3-1 (TNF and RFSC or SSM).
2. Setup the initial hardware using the following steps:
 - A. Disconnect the 2 (two) white BNC cables and the Head Coil Quick Disconnect Adapter from the Head Coil.
 - B. Locate and move both JP1 and JP2 jumpers to the Tune position (specific to the coil design). Refer to Illustrations 4-5A and 4-5B.

Note

The 1.5T Isolation Boards and Hybrid Assembly are not removed, however, the semi-rigid coax's are removed.

WARNING!

FOR THIS TEST CONFIGURATION, DO NOT LEAVE THE HEAD COIL CONNECTED TO THE CARRIAGE COVER ASSEMBLY. THE HEAD T/R ASSEMBLY MAY BE DAMAGED. REMOVE THE BNC CABLES AND HEAD COIL QUICK DISCONNECT ADAPTER BEFORE INSTALLING THE HEAD COIL IN THE BORE.



3. Prepare the system to scan in Head mode per Table 4-3.

TABLE 4-3
HEAD SCAN PROTOCOL: HEAD COIL TUNE SENSE LOOP CHECK TEST

Note: This is the alternate proprietary procedure available for GE use, and to sites with a valid Advanced Service Package Limited License.

A. **[New Pt]**

Id: **geservice**<ENTER>

Name: **rft**

Weight (Lb.): **300**<ENTER>

Set Patient Protocols to **Service**.

At front enclosure, , landmark in head area, press **LANDMARK**, then **MOVE TO SCAN**.

B. In the Patient Position Protocol field:

type **o.16.2**<ENTER>(o=Other, 16.2 =series) to load the head protocol

OR select **other** and select protocol **16** (RF test) and select series **2** (Head scan).

C. In the Additional Parameters field:

[User CVs Screen]

CV Name: **EFB Bypass**, CV Value: **1**<ENTER>.

CV Name: **RF Amp Mode**, CV Value: **7**<ENTER>.

CV Name: **Sense Loop**, CV Value: **1**<ENTER>.

CV Name: **StabNoGrad**, CV Value: **0**<ENTER>.

CV Name: **Lin&Fidlty**, CV Value: **0**<ENTER>.

CV Name: **Stab/Grad**, CV Value: **0**<ENTER>.

CV Name: **Bandwidth**, CV Value: **1**<ENTER>.

CV Name: **BW Cal**, CV Value: **0**<ENTER>.

[Accept].

D. **[Save Series]**. If necessary --- **[Prepare to Scan]**.

E. **[Research Operations]** **[Setup Params]**.

R1 = 13

R2 = 14

TG = 50

Number of Frames: **4**<ENTER>

Window 1: Frame: **1**<ENTER>

+/- = +

Window 1: Frame: **0** <ENTER>

Window 2: Frame: **2** <ENTER>

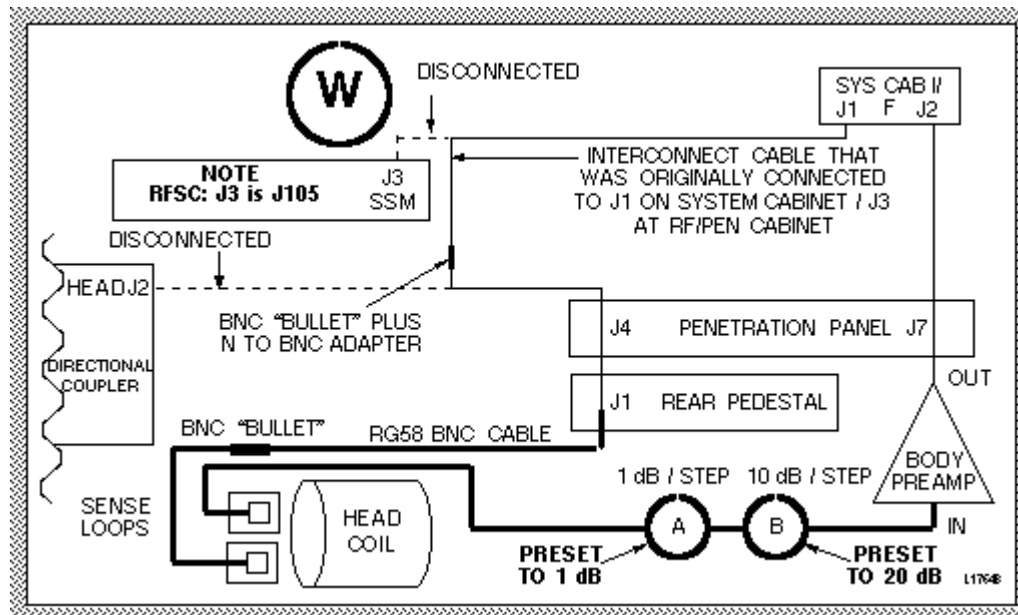
+/- = +

Window 2: Frame: **0** <ENTER>

[Done].

F. **[Research Operations]** **[Download]**.

4. Setup system per Illustration 4-6 and Illustration 4-7.

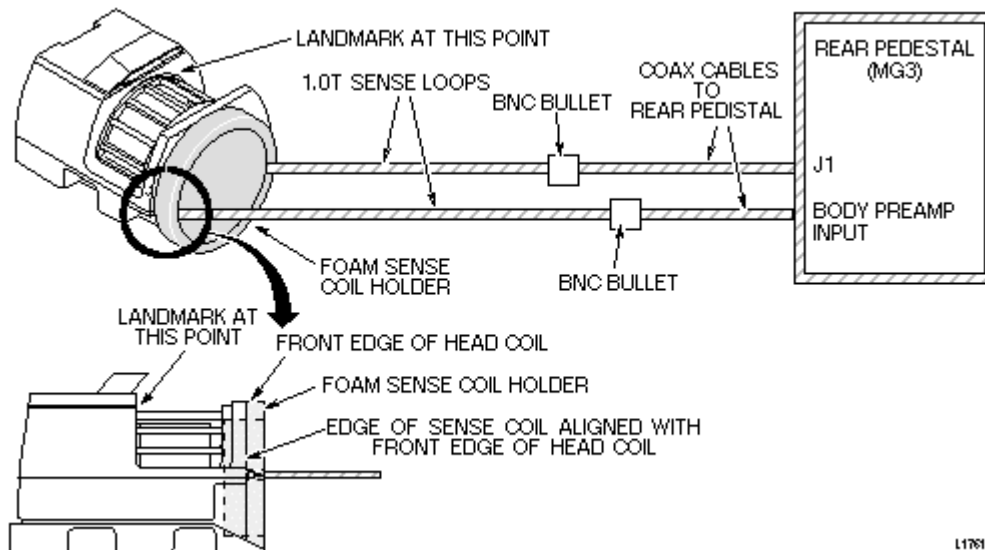


HEAD COIL TUNE SENSE LOOP CHECK
ILLUSTRATION 4-6

5. Position the Sense coils per Illustration 4-7:
- Place Head foam piece into front of head coil.
 - Place Sense Coils into foam positioner until the edge of the sense coil is aligned with the front edge of the Head Coil.
 - Landmark per Illustration 4-7.

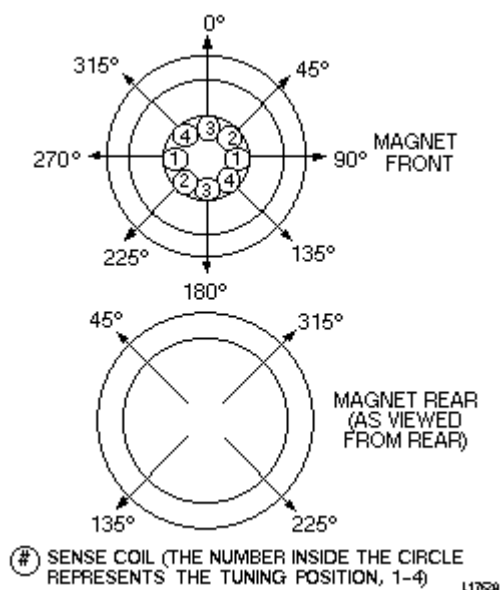
Note

When routing the cables for the Sense Coils, try to lay them as straight as possible and without crossing them. Crossed cables will affect data collection due to crosstalk.



HEAD COIL SENSE COIL POSITIONING
ILLUSTRATION 4-7

6. Start **[Manual Prescan]**.
7. From the "Windows" Menu bar on the Manual Prescan window, select **"Two Windows"**.
Plot Type = **Magnitude**
Plot Gain = **1**
Plot Type = **Magnitude**
Plot Gain = **1**
8. Advance TG from 50 to 200, while adjusting variable attenuators A and B to prevent R1 and R2 overrange. Do NOT adjust R1 or R2.
9. With TG at 200, make final adjustments to variable attenuators A and B for R2 signal between 75 & 85%. Do NOT adjust R1 or R2.
10. Select **[Done]**.
11. Select **[Scan]**.
12. For scan #1, place sense coils at position 1. For subsequent scans, place sense coils at next location. Refer to Illustration 4-8.



SENSE COIL POSITIONING / PLANE REFERENCES
ILLUSTRATION 4-8

13. Perform a new Head Coil Tune Sense Loop Test scan on all remaining planes (positions 2 through 4). Refer to Illustration 4-8 for plane / position references.

14. After obtaining Head Coil Tune Check Scans from all planes (0, 45, 90, and 315 degrees), view results by using the Report Manager tool. Record data for each test in Data Sheet 1.5T & Data Sheet 1.0T found in the appendix.

- A. Select **Report Tool** under **[Utilities]** on the Service Desktop. Select the **File** menu, then select **Open** from the pull-down sub-menu. Choose the pertinent file from the *Load Data File* pop up window. The most recent files appear at the end of the file directory.
- B. **Center Frequency:** Record the center frequency obtained at each plane. Subtract the lowest frequency value from the highest frequency. The “mode delta” difference must be less than or equal to 150,000 Hz (1.5T) or 100,000 Hz (1.0T).

Example:

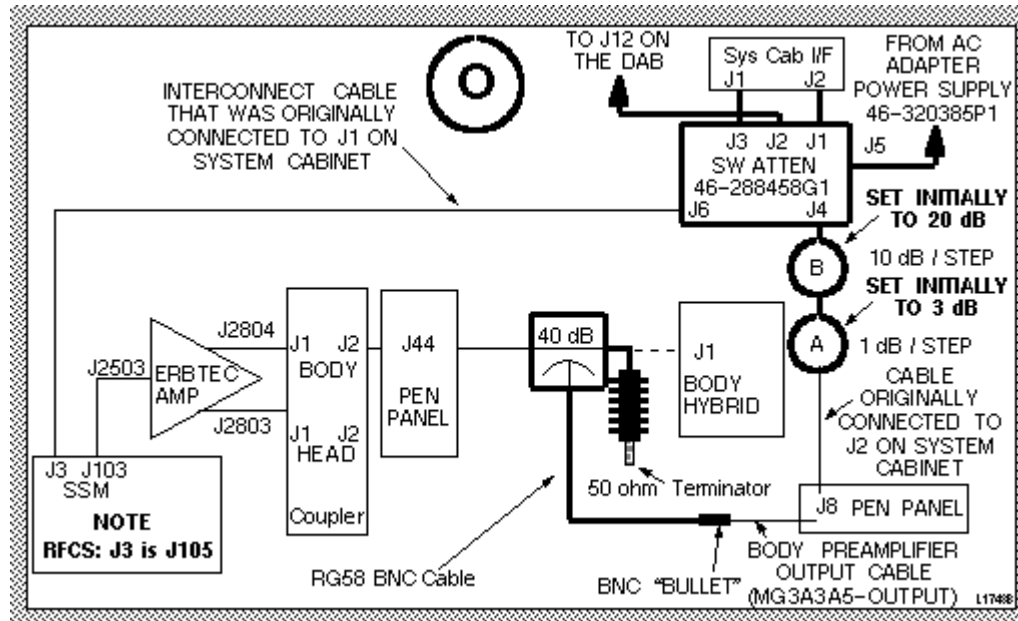
- degree plane: body coil tune check scan: 42.731 MHz
- degree plane: body coil tune check scan: 42.683 MHz
- $42.731\text{MHz} - 42.683\text{ MHz} = 48,000\text{ Hz}$ or 48 kHz. These frequency values are within spec.

15. If the bandwidth is not correct, then perform “Pre-amplifier Bandwidth Check” and troubleshoot the problem.

5- BODY AND HEAD TROUBLESHOOTING TESTS

5-1 Body Dummy Load (Body Heliax Output)

1. Prepare the system per sub-section 3-1 (TNF and RFSC or SSM).
2. Refer to Illustration 5-1 for hardware setup:



BODY MODE DUMMY LOAD LOOPBACK (BODY HELIAX OUTPUT)
ILLUSTRATION 5-1

3. Prepare the system to scan in Body mode per Table 5-1.

TABLE 5-1

BODY SCAN PROTOCOL: BODY DUMMY LOAD (BODY HELIAX OUTPUT) TEST

Note: This is the alternate proprietary procedure available for GE use, and to sites with a valid Advanced Service Package Limited License.

A. **[New Pt]**

Id: **geservice**<ENTER>

Name: **rft**

Weight (Lb.): **300**<ENTER>

Set Patient Protocols to **Service**.

At front enclosure, landmark in head area, press **LANDMARK**, then **MOVE TO SCAN**.

B. In the Patient Position Protocol field:

type **o.16.1**<ENTER>(o=Other, 16.1 =series) to load the body protocol

OR select **other** and select protocol **16** (RF Test) and select series **1** (RFT Body).

C. In the Additional Parameters field:

[User CVs Screen] (see Table 5-2 for CV Values):

CV Name: **EFB Bypass**, CV Value: <ENTER>.

CV Name: **RFampMode**, CV Value: <ENTER>.

CV Name: **Sense Loop**, CV Value: <ENTER>.

CV Name: **StabNoGrad**, CV Value: <ENTER>.

CV Name: **Lin&Fidlty**, CV Value: <ENTER>.

CV Name: **Stab/Grad**, CV Value: <ENTER>.

CV Name: **Bandwidth**, CV Value: <ENTER>.

CV Name: **BW Cal**, CV Value: <ENTER>.

[Accept].

D. **[Save Series]**. If necessary --- **[Prepare to Scan]**.

E. **[Research Operations]** **[Setup Params]**.

R1 = 13

R2 = 14

TG = 50

Number of Frames: **4**<ENTER>

Window 1: Frame: **1**<ENTER>

+/- = +

Window 1: Frame: **0** <ENTER>

Window 2: Frame: **2** <ENTER>

+/- = +

Window 2: Frame: **0** <ENTER>

[Done].

F. **[Research Operations]** **[Download]**.

4. Control Variables for Stability and Linearity trouble-shooting are listed in Table 5-2 for **[User CV's]** screen select (refer to Appendix B for CV descriptions):

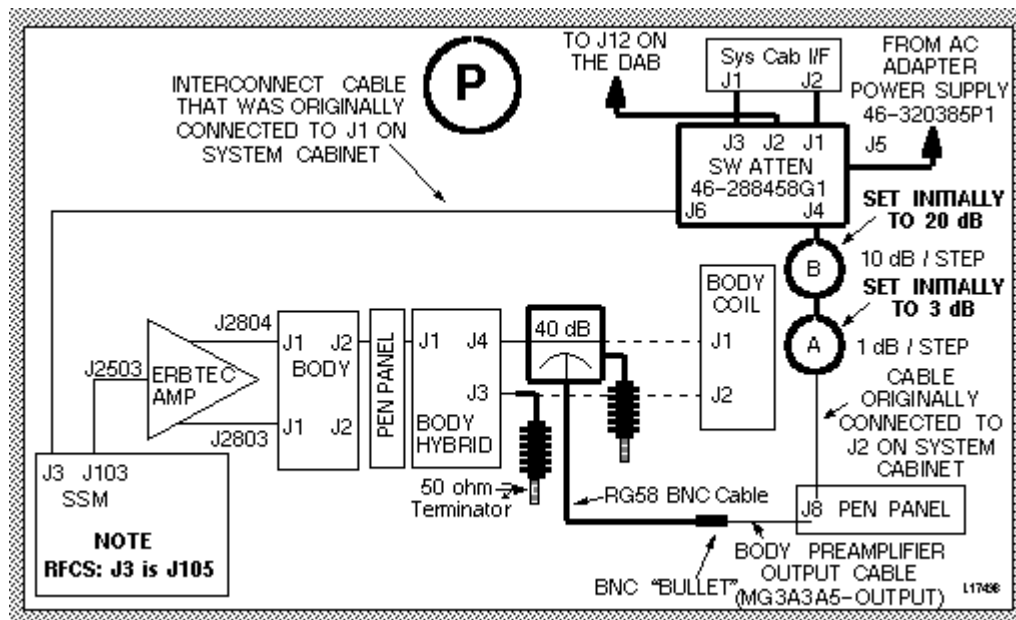
TABLE 5-2
BODY DUMMY LOAD (BODY HELIAX OUTPUT) CONTROL VARIABLES

User CV Name	Stability T/S	Linearity T/S
EFB Bypass	0	0
RFampMode	3	3
Sense Loop	0	0
StabNoGrad	1	0
Lin&Fidly	0	1
Stab/Grad	4	0
Bandwidth	0	0
BW Cal	0	0

5. Start **[Manual Prescan]**.
6. From the “Windows” Menu bar on the Manual Prescan window, select “**Two Windows**”.
- Plot Type = **Magnitude**
Plot Gain = **1**
Plot Type = **Magnitude**
Plot Gain = **1**
7. Advance TG from 50 to 200, while adjusting variable attenuators A and B to prevent R1 and R2 overrange. Do NOT adjust R1 or R2.
8. With TG at 200, make final adjustments to variable attenuators A and B for R2 signal between 75 & 85%. Do NOT adjust R1 or R2.
9. Select **[Done]**.
10. Select **[Scan]**. When scan completes, view results by using the Report Manager tool (Release 8.2) or the system PC FTP tool (Release 8.1). Record data for each test in Data Sheet 1.5T & Data Sheet 1.0T found in the appendix.

5-2 Body Dummy Load (Body T/R Switch & Hybrid Output)

1. Prepare the system per sub-section 3-1 (TNF and RFSC or SSM).
2. Refer to Illustration 5-2 for hardware setup.



BODY MODE DUMMY LOAD LOOPBACK (BODY TIR & HYBRID OUTPUT) SETUP

3. Prepare the system to scan in Body mode per Table 5-3.

TABLE 5-3

BODY SCAN PROTOCOL: BODY DUMMY LOAD (BODY T/R SWITCH & HYBRID OUTPUT) TEST

Note: This is the alternate proprietary procedure available for GE use, and to sites with a valid Advanced Service Package Limited License.

A. **[New Pt]**

Id: **geservice**<ENTER>

Name: **rft**

Weight (Lb.): **300**<ENTER>

Set Patient Protocols to **Service**.

At front enclosure, landmark in head area, press **LANDMARK**, then **MOVE TO SCAN**.

B. In the Patient Position Protocol field:

type **o.16.1**<ENTER>(o=Other, 16.1 =series) to load the body protocol

OR select **other** and select protocol **16** (RF Test) and select series **1** (RFT Body).

C. In the Additional Parameters field:

[User CVs Screen] (see Table 5-2 for CV Values):

CV Name: **EFB Bypass**, CV Value: <ENTER>.

CV Name: **RF Amp Mode**, CV Value: <ENTER>.

CV Name: **Sense Loop**, CV Value: <ENTER>.

CV Name: **StabNoGrad**, CV Value: <ENTER>.

CV Name: **Lin&Fidlty**, CV Value: <ENTER>.

CV Name: **Stab/Grad**, CV Value: <ENTER>.

CV Name: **Bandwidth**, CV Value: <ENTER>.

CV Name: **BW Cal**, CV Value: <ENTER>.

[Accept].

D. **[Save Series]**. If necessary --- **[Prepare to Scan]**.

E. **[Research Operations]** **[Setup Params]**.

R1 = 13

R2 = 14

TG = 50

Number of Frames: **4**<ENTER>

Window 1: Frame: **1**<ENTER>

+/- = +

Window 1: Frame: **0** <ENTER>

Window 2: Frame: **2** <ENTER>

+/- = +

Window 2: Frame: **0** <ENTER>

[Done].

F. **[Research Operations]** **[Download]**.

4. Control Variables for Stability and Linearity trouble-shooting are listed in Table 5-4 for **[User CV's]** screen select (refer to Appendix B for CV descriptions):

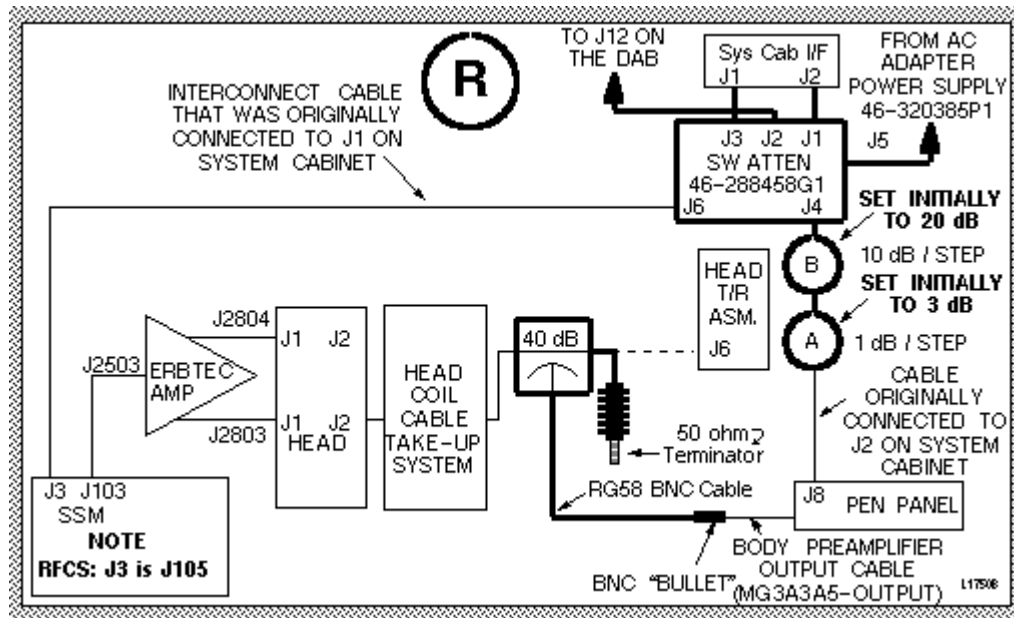
TABLE 5-4
BODY DUMMY LOAD (BODY T/R SWITCH & HYBRID OUTPUT) CONTROL VARIABLES

User CV Name	Stability T/S	Linearity T/S
EFB Bypass	0	0
RFampMode	3	3
Sense Loop	0	0
StabNoGrad	1	0
Lin&Fidly	0	1
Stab/Grad	4	0
Bandwidth	0	0
BW Cal	0	0

5. Start **[Manual Prescan]**.
6. From the “Windows” Menu bar on the Manual Prescan window, select “**Two Windows**”.
- Plot Type = **Magnitude**
Plot Gain = **1**
Plot Type = **Magnitude**
Plot Gain = **1**
7. Advance TG from 50 to 200, while adjusting variable attenuators A and B to prevent R1 and R2 overrange. Do NOT adjust R1 or R2.
8. With TG at 200, make final adjustments to variable attenuators A and B for R2 signal between 75 & 85%. Do NOT adjust R1 or R2.
9. Select **[Done]**.
10. Select **[Scan]**. When scan completes, view results by using the Report Manager tool (Release 8.2) or the system PC FTP tool (Release 8.1). Record data for each test in Data Sheet 1.5T & Data Sheet 1.0T found in the appendix.

5-3 Head Dummy Load (Head Heliax Output)

1. Prepare the system per sub-section 3-1 (TNF and RFSC or SSM).
2. Refer to Illustration 5-3 for hardware setup.



HEAD MODE DUMMY LOAD LOOPBACK (HEAD HELIAX OUTPUT)
ILLUSTRATION 5-3

3. Prepare the system to scan in Head mode per Table 5-5.

TABLE 5-5

HEAD SCAN PROTOCOL: HEAD DUMMY LOAD (HEAD HELIAX OUTPUT) TEST

Note: This is the alternate proprietary procedure available for GE use, and to sites with a valid Advanced Service Package Limited License.

A. **[New Pt]**

Id: **geservice**<ENTER>

Name: **rft**

Weight (Lb.): **300**<ENTER>

Set Patient Protocols to **Service**.

At front enclosure, landmark in head area, press **LANDMARK**, then **MOVE TO SCAN**.

B. In the Patient Position Protocol field:

type **o.16.2**<ENTER>(o=Other, 16.2 =series) to load the body protocol

OR select **other** and select protocol **16** (RF Test) and select series **2** (RFT Head).

C. In the Additional Parameters field:

[User CVs Screen] (see Table 5-2 for CV Values):

CV Name: **EFB Bypass**, CV Value: <ENTER>.

CV Name: **RFampMode**, CV Value: <ENTER>.

CV Name: **Sense Loop**, CV Value: <ENTER>.

CV Name: **StabNoGrad**, CV Value: <ENTER>.

CV Name: **Lin&Fidlty**, CV Value: <ENTER>.

CV Name: **Stab/Grad**, CV Value: <ENTER>.

CV Name: **Bandwidth**, CV Value: <ENTER>.

CV Name: **BW Cal**, CV Value: <ENTER>.

[Accept].

D. **[Save Series]**. If necessary --- **[Prepare to Scan]**.

E. **[Research Operations]** **[Setup Params]**.

R1 = 13

R2 = 14

TG = 50

Number of Frames: **4**<ENTER>

Window 1: Frame: **1**<ENTER>

+/- = +

Window 1: Frame: **0** <ENTER>

Window 2: Frame: **2** <ENTER>

+/- = +

Window 2: Frame: **0** <ENTER>

[Done].

F. **[Research Operations]** **[Download]**.

4. Control Variables for Stability and Linearity trouble-shooting are listed in Table 5-6 for **[User CV's]** screen select (refer to Appendix B for CV descriptions):

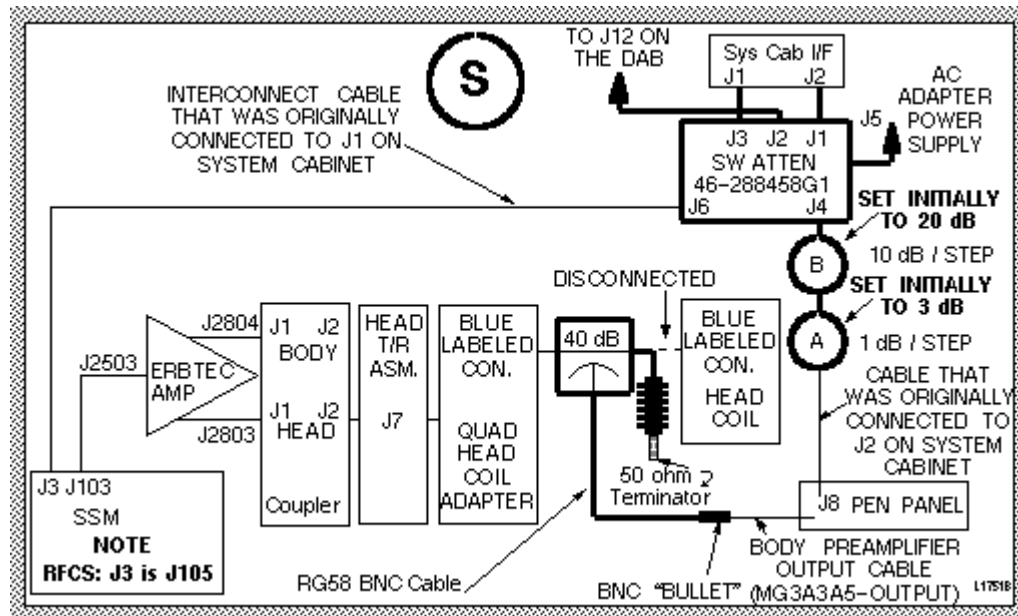
TABLE 5-6
HEAD DUMMY LOAD (HEAD HELIAX OUTPUT) CONTROL VARIABLES

User CV Name	Stability T/S	Linearity T/S
EFB Bypass	0	0
RFampMode	1	1
Sense Loop	0	0
StabNoGrad	1	0
Lin&Fidly	0	1
Stab/Grad	4	0
Bandwidth	0	0
BW Cal	0	0

5. Start **[Manual Prescan]**.
6. From the "Windows" Menu bar on the Manual Prescan window, select **"Two Windows"**.
 Plot Type = **Magnitude**
 Plot Gain = **1**
 Plot Type = **Magnitude**
 Plot Gain = **1**
7. Advance TG from 50 to 200, while adjusting variable attenuators A and B to prevent R1 and R2 overrange. Do NOT adjust R1 or R2.
8. With TG at 200, make final adjustments to variable attenuators A and B for R2 signal between 75 & 85%. Do NOT adjust R1 or R2.
9. Select **[Done]**.
10. Select **[Scan]**. When scan completes, view results by using the Report Manager tool (Release 8.2) or the system PC FTP tool (Release 8.1). Record data for each test in Data Sheet 1.5T & Data Sheet 1.0T found in the appendix.

5-4 Head Dummy Load (Head T/R Output) (Head Heliax Output)

1. Prepare the system per sub-section 3-1 (TNF and RFSC or SSM).
2. Refer to Illustration 5-4 for the hardware setup:



HEAD MODE DUMMY LOAD LOOPBACK (HEAD T/R) SETUP
ILLUSTRATION 5-4

3. Prepare the system to scan in Head mode per Table 5-7.

TABLE 5-7

HEAD SCAN PROTOCOL: HEAD DUMMY LOAD (HEAD T/R OUTPUT) (HEAD HELIAX OUTPUT) TEST

Note: This is the alternate proprietary procedure available for GE use, and to sites with a valid Advanced Service Package Limited License.

A. **[New Pt]**

Id: **geservice**<ENTER>

Name: **rft**

Weight (Lb.): **300**<ENTER>

Set Patient Protocols to **Service**.

At front enclosure, landmark in head area, press **LANDMARK**, then **MOVE TO SCAN**.

B. In the Patient Position Protocol field:

type **o.16.2**<ENTER>(o=Other, 16.2 =series) to load the body protocol

OR select **other** and select protocol **16** (RF Test) and select series **2** (RFT Head).

C. In the Additional Parameters field:

[User CVs Screen] (see Table 5-2 for CV Values):

CV Name: **EFB Bypass**, CV Value: <ENTER>.

CV Name: **RFampMode**, CV Value: <ENTER>.

CV Name: **Sense Loop**, CV Value: <ENTER>.

CV Name: **StabNoGrad**, CV Value: <ENTER>.

CV Name: **Lin&Fidlty**, CV Value: <ENTER>.

CV Name: **Stab/Grad**, CV Value: <ENTER>.

CV Name: **Bandwidth**, CV Value: <ENTER>.

CV Name: **BW Cal**, CV Value: <ENTER>.

[Accept].

D. **[Save Series]**. If necessary --- **[Prepare to Scan]**.

E. **[Research Operations]** **[Setup Params]**.

R1 = 13

R2 = 14

TG = 50

Number of Frames: **4**<ENTER>

Window 1: Frame: **1**<ENTER>

+/- = +

Window 1: Frame: **0** <ENTER>

Window 2: Frame: **2** <ENTER>

+/- = +

Window 2: Frame: **0** <ENTER>

[Done].

F. **[Research Operations]** **[Download]**.

4. Control Variables for Stability and Linearity trouble-shooting are listed in Table 5-8 for **[User CV's]** screen select (refer to Appendix B for CV descriptions):

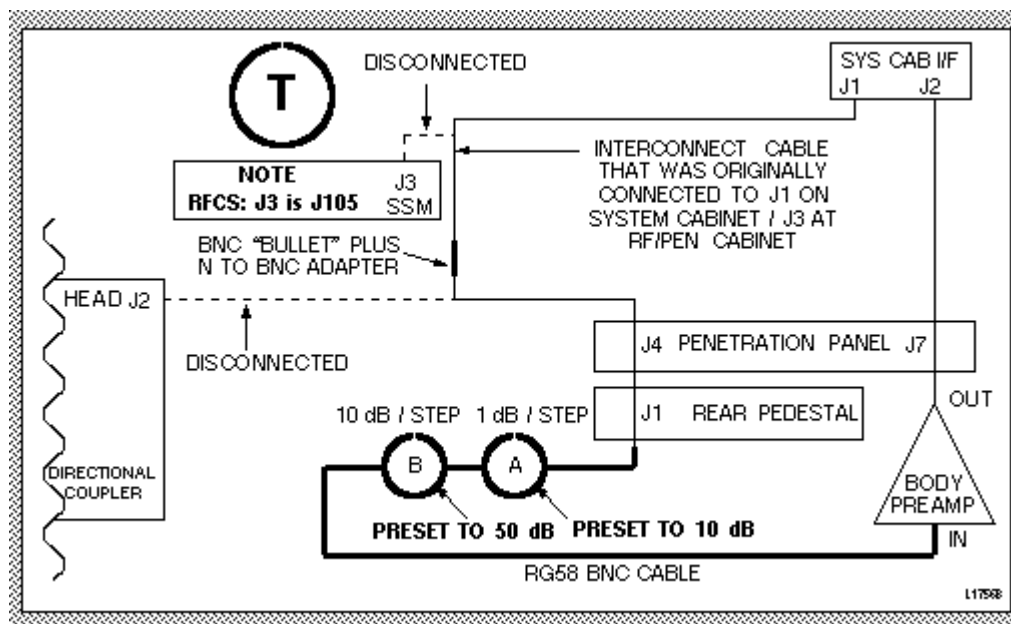
TABLE 5-8
HEAD DUMMY LOAD (HEAD T/R OUTPUT) (HEAD HELIAX OUTPUT) CONTROL VARIABLES

User CV Name	Stability T/S	Linearity T/S
EFB Bypass	0	0
RFampMode	1	1
Sense Loop	0	0
StabNoGrad	1	0
Lin&Fidly	0	1
Stab/Grad	4	0
Bandwidth	0	0
BW Cal	0	0

5. Start **[Manual Prescan]**.
6. From the “Windows” Menu bar on the Manual Prescan window, select “**Two Windows**”.
- Plot Type = **Magnitude**
Plot Gain = **1**
Plot Type = **Magnitude**
Plot Gain = **1**
7. Advance TG from 50 to 200, while adjusting variable attenuators A and B to prevent R1 and R2 overrange. Do NOT adjust R1 or R2.
8. With TG at 200, make final adjustments to variable attenuators A and B for R2 signal between 75 & 85%. Do NOT adjust R1 or R2.
9. Select **[Done]**.
10. Select **[Scan]**. When scan completes, view results by using the Report Manager tool (Release 8.2) or the system PC FTP tool (Release 8.1). Record data for each test in Data Sheet 1.5T & Data Sheet 1.0T found in the appendix.

5-5 Coil Tune Preamplifier Check

1. Prepare the system per sub-section 3-1 (TNF and RFSC or SSM).
2. Refer to Illustration 5-5 for the hardware setup:



COIL TUNE CHECK PREAMPLIFIER LOOP BACK MANUAL PRESCAN SETUP
ILLUSTRATION 5-5

3. Prepare the system to scan in Body mode per Table 5-9.

TABLE 5-9
BODY SCAN PROTOCOL: BODY COIL TUNE PREAMPLIFIER CHECK TEST

Note: This is the alternate proprietary procedure available for GE use, and to sites with a valid Advanced Service Package Limited License.

A. **[New Pt]**

Id: **geservice**<ENTER>

Name: **rft**

Weight (Lb.): **300**<ENTER>

Set Patient Protocols to **Service**.

At front enclosure, landmark in head area, press **LANDMARK**, then **MOVE TO SCAN**.

B. In the Patient Position Protocol field:

type **o.16.1**<ENTER>(o=Other, 16.1 =series) to load the body protocol

OR select **other** and select protocol **16** (RF Test) and select series **1** (RFT Body).

C. In the Additional Parameters field:

[User CVs Screen] (see Table 5-2 for CV Values):

CV Name: **EFB Bypass**, CV Value: **1**<ENTER>.

CV Name: **RFampMode**, CV Value: **7**<ENTER>.

CV Name: **Sense Loop**, CV Value: **1**<ENTER>.

CV Name: **StabNoGrad**, CV Value: **0**<ENTER>.

CV Name: **Lin&Fidlty**, CV Value: **0**<ENTER>.

CV Name: **Stab/Grad**, CV Value: **0**<ENTER>.

CV Name: **Bandwidth**, CV Value: **1**<ENTER>.

CV Name: **BW Cal**, CV Value: **1**<ENTER>.

[Accept].

D. **[Save Series]**. If necessary --- **[Prepare to Scan]**.

E. **[Research Operations]** **[Setup Params]**.

R1 = 13

R2 = 14

TG = 50

Number of Frames: **4**<ENTER>

Window 1: Frame: **1**<ENTER>

+/- = +

Window 1: Frame: **0** <ENTER>

Window 2: Frame: **2** <ENTER>

+/- = +

Window 2: Frame: **0** <ENTER>

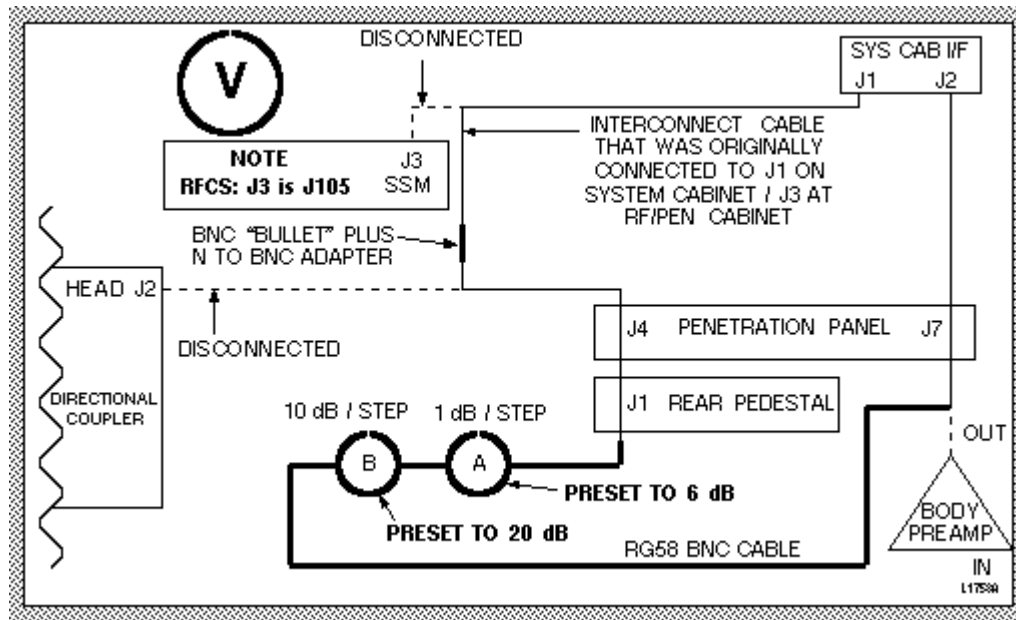
[Done].

F. **[Research Operations]** **[Download]**.

4. Start **[Manual Prescan]**.
5. From the “Windows” Menu bar on the Manual Prescan window, select “**Two Windows**”.
Plot Type = **Magnitude**
Plot Gain = **1**
Plot Type = **Magnitude**
Plot Gain = **1**
6. Advance TG from 50 to 200, while adjusting variable attenuators A and B to prevent R1 and R2 over range. Do NOT adjust R1 or R2.
7. With TG at 200, make final adjustments to variable attenuators A and B for R2 signal between 75 & 85%. Do NOT adjust R1 or R2.
8. Select **[Done]**.
9. Select **[Scan]**. The scan data is used by RFT Analysis for frequency compensation with hardware configurations (Body Coil Tune Sense Loop Check) and (Head Coil Tune Sense Loop Check). (Viewing RFT results for this scan not necessary.)

5-6 Preamplifier Bandwidth Check

1. Prepare the system per sub-section 3-1 (TNF and RFSC or SSM).
2. Refer to Illustration 5-6 for hardware setup.



PREAMPLIFIER BANDWIDTH CHECK TPS LOOPBACK MANUAL PRESCAN
ILLUSTRATION 5-6

3. Prepare the system to scan in Body mode per Table 5-10.

TABLE 5-10

BODY SCAN PROTOCOL: BODY COIL PREAMPLIFIER BANDWIDTH CHECK TEST

Note: This is the alternate proprietary procedure available for GE use, and to sites with a valid Advanced Service Package Limited License.

A. **[New Pt]**

Id: **geservice**<ENTER>

Name: **rft**

Weight (Lb.): **300**<ENTER>

Set Patient Protocols to **Service**.

At front enclosure, landmark in head area, press **LANDMARK**, then **MOVE TO SCAN**.

B. In the Patient Position Protocol field:

type **o.16.1**<ENTER>(o=Other, 16.1 =series) to load the body protocol

OR select **other** and select protocol **16** (RF Test) and select series **1** (RFT Body).

C. In the Additional Parameters field:

[User CVs Screen] (see Table 5-2 for CV Values):

CV Name: **EFB Bypass**, CV Value: **1**<ENTER>.

CV Name: **RF Amp Mode**, CV Value: **7**<ENTER>.

CV Name: **Sense Loop**, CV Value: **0**<ENTER>.

CV Name: **StabNoGrad**, CV Value: **0**<ENTER>.

CV Name: **Lin&Fidlty**, CV Value: **0**<ENTER>.

CV Name: **Stab/Grad**, CV Value: **0**<ENTER>.

CV Name: **Bandwidth**, CV Value: **1**<ENTER>.

CV Name: **BW Cal**, CV Value: **0**<ENTER>.

[Accept].

D. **[Save Series]**. If necessary --- **[Prepare to Scan]**.

E. **[Research Operations]** **[Setup Params]**.

R1 = 13

R2 = 14

TG = 50

Number of Frames: **4**<ENTER>

Window 1: Frame: **1**<ENTER>

+/- = +

Window 1: Frame: **0** <ENTER>

Window 2: Frame: **2** <ENTER>

+/- = +

Window 2: Frame: **0** <ENTER>

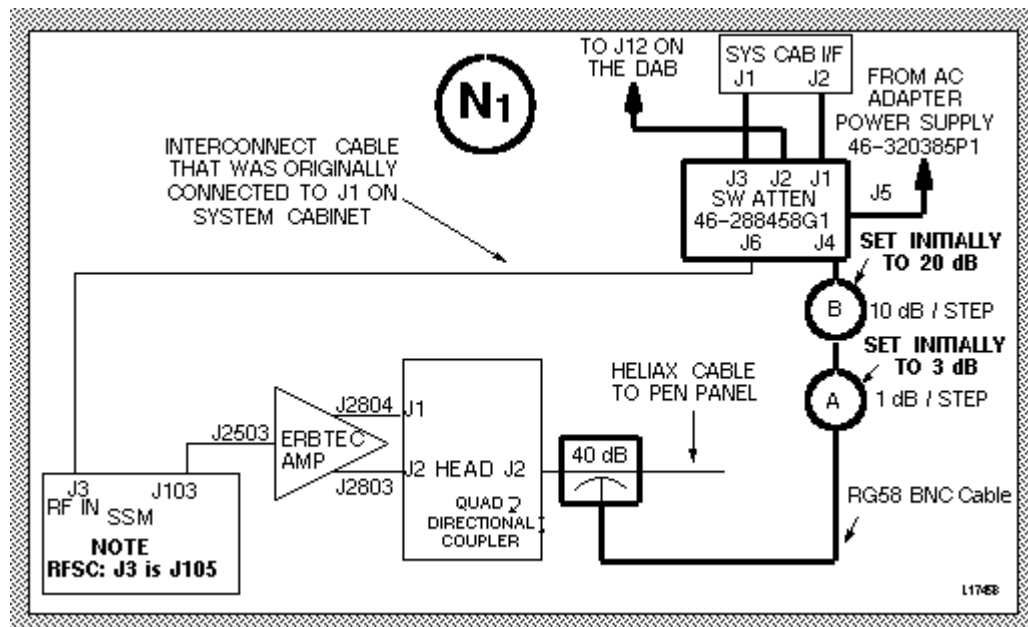
[Done].

F. **[Research Operations]** **[Download]**.

4. Start **[Manual Prescan]**.
5. From the “Windows” Menu bar on the Manual Prescan window, select “**Two Windows**”.
Plot Type = **Magnitude**
Plot Gain = **1**
Plot Type = **Magnitude**
Plot Gain = **1**
6. Advance TG from 50 to 200, while adjusting variable attenuators A and B to prevent R1 and R2 overrange. Do NOT adjust R1 or R2.
7. With TG at 200, make final adjustments to variable attenuators A and B for R2 signal between 75 & 85%. Do NOT adjust R1 or R2.
8. Select **[Done]**.
9. Select **[Scan]**. When scan completes, view results by using the Report Manager tool. Isolate/troubleshoot the bandwidth problem based on the results of this test.

5-7 Head, SS Driver — EFB Bypassed

1. Prepare the system per sub-section 3-1 (TNF and RFSC or SSM).
2. Refer to Illustration 5-7 for the initial hardware setup. The RF Power Measurement Kit 30 dB Coupler can be used in place of the 40 dB Coupler for the 3 (three) hardware setups throughout this sub-section (N1, N2, and N3) in place of the 40 dB Coupler.



HEAD MODE SS DRIVER SETUP - INITIAL
ILLUSTRATION 5-7

3. Prepare the system to scan in Head mode per Table 5-11.

TABLE 5-11
HEAD SCAN PROTOCOL: HEAD, SS DRIVER — EFB BYPASSED TEST

Note: This is the alternate proprietary procedure available for GE use, and to sites with a valid Advanced Service Package Limited License.

A. **[New Pt]**

Id: **geservice**<ENTER>

Name: **rft**

Weight (Lb.): **300**<ENTER>

Set Patient Protocols to **Service**.

At front enclosure, landmark in head area, press **LANDMARK**, then **MOVE TO SCAN**.

B. In the Patient Position Protocol field:

type **o.16.2**<ENTER>(o=Other, 16.1 =series) to load the body protocol

OR select **other** and select protocol **16** (RF Test) and select series **2** (Head scan).

C. In the Additional Parameters field:

[User CVs Screen] (see Table 5-2 for CV Values):

CV Name: **EFB Bypass**, CV Value: **1**<ENTER>.

CV Name: **RFampMode**, CV Value: **6**<ENTER>.

CV Name: **Sense Loop**, CV Value: **0**<ENTER>.

CV Name: **StabNoGrad**, CV Value: **1**<ENTER>.

CV Name: **Lin&Fidlty**, CV Value: **1**<ENTER>.

CV Name: **Stab/Grad**, CV Value: **4**<ENTER>.

CV Name: **Bandwidth**, CV Value: **0**<ENTER>.

CV Name: **BW Cal**, CV Value: **0**<ENTER>.

[Accept].

D. **[Save Series]**. If necessary --- **[Prepare to Scan]**.

E. **[Research Operations]** **[Setup Params]**.

R1 = 13

R2 = 14

TG = 50

Number of Frames: **4**<ENTER>

Window 1: Frame: **1**<ENTER>

+/- = +

Window 1: Frame: **0** <ENTER>

Window 2: Frame: **2** <ENTER>

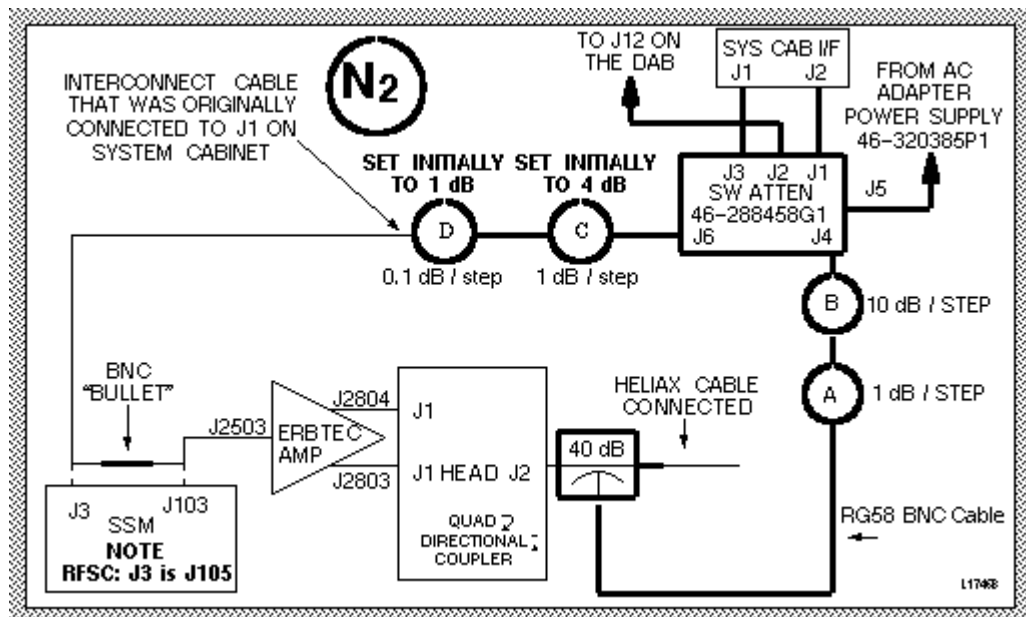
+/- = +

Window 2: Frame: **0** <ENTER>

[Done].

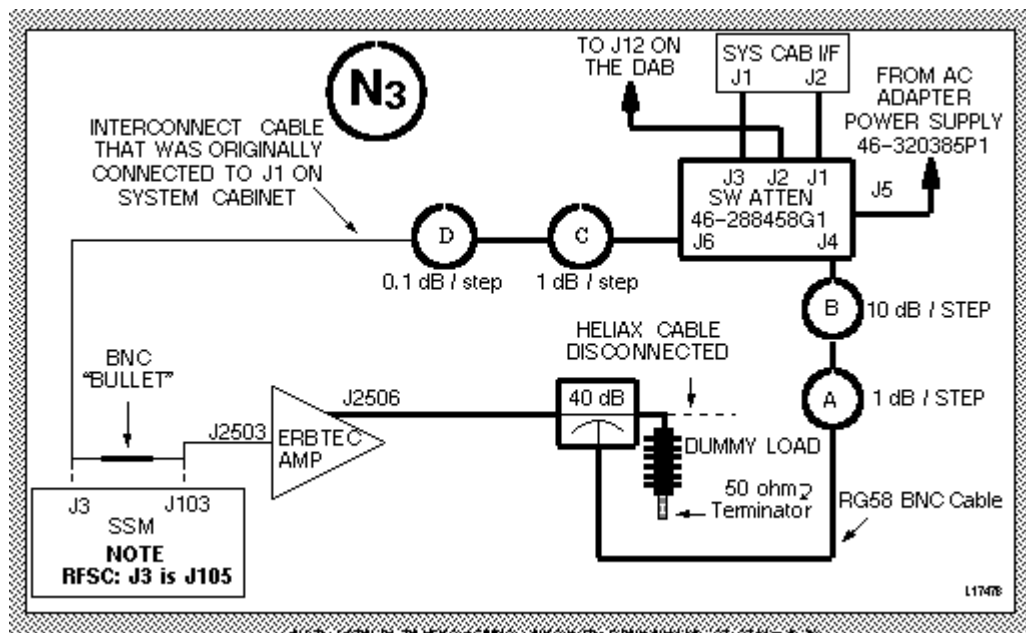
F. **[Research Operations]** **[Download]**.

4. Start [Manual Prescan].
5. From the "Windows" Menu bar on the Manual Prescan window, select **"Two Windows"**.
 Plot Type = **Magnitude**
 Plot Gain = **1**
 Plot Type = **Magnitude**
 Plot Gain = **1**
6. Advance TG from 50 to 200, while adjusting variable attenuators A and B to prevent R1 and R2 overrange. Do NOT adjust R1 or R2.
7. With TG at 200, make final adjustments to variable attenuators A and B for R2 signal between 75 & 85%. Do NOT adjust R1 or R2.
8. Note the R2 signal value.
9. Select **[Done]**.
10. Since scan will be performed with EFB bypassed, reconfigure as shown in Illustration 5-8.
 - A. Disconnect SSM cables at J103 and J3. Join these two cables with a BNC "bullet".
 - B. Add variable attenuators C and D (from SST kit). Preset as indicated.
 - C. Use RF Power Measurement Kit 40 dB Coupler. Connect 50 ohm Terminator to Dummy Load Output.



HEAD MODE SS DRIVER SETUP — INTERMEDIATE
ILLUSTRATION 5-8

11. Resume **[Manual Prescan]**.
12. Adjust variable attenuators C and D to achieve the same R2 signal value that was recorded in step 7 (within + 1.0 point of the recorded value of N1). (1.0 point is approximately 0.1dB).
13. Select **[Done]**.
14. Reconfigure as shown in Illustration 5-9 and perform pre-scan calibration. Scan will be performed using a Coupler to pick-off the SS driver output with EFB Bypassed. Use RF Power Measurement Kit 40 dB Coupler. Connect 50 ohm Terminator to Dummy Load Output.



HEAD MODE SS DRIVER SETUP — FINAL
ILLUSTRATION 5-9

15. Resume **[Manual Prescan]**.
16. Make final adjustments to variable attenuators A and B for R2 signal at previously noted value (between 75% and 85%). Do NOT adjust R1, R2, TG, or variable attenuators C and D.
17. Select **[Done]**.

18. Select **[Scan]**. When scan completes, view results by using the Report Manager tool. Record data for each test in the data sheets found in Appendix A.

19. After the baseline test is complete alternate shorter versions listed below may be performed for Stability T/S or Linearity T/S. See the **[User CV's]** screen selections below (or refer to Appendix B for CV descriptions):

	Baseline	Stability T/S	Linearity T/S
EFB Bypass	1	1	1
RFampMode	6	6	6
SenseLoop	0	0	0
StabNoGrad	1	1	0
Lin&Fidlty	1	0	1
StabGrad	4	4	0
Bandwidth	0	0	0
BW Cal	0	0	0

6- RESTORATION CHECKLIST

1. Remove Sense Coils and foam from inside of Body Coil or Head Coil.
2. At the System Cabinet:
 - A. Remove Switchable Attenuator and associated test hardware.
 - B. Enable the TNF (Transient Noise Filter) unit in the rear of the System Cabinet.
 - C. Re-connect System Cabinet cables to original configuration.
 - D. Reconnect I/F Panel J5 coaxial cable.
3. At the RF Cabinet:
 - A. Remove associated test hardware.
 - B. Re-connect RF Cabinet cables to original configuration.
 - C. Place the RFSC to Disable faults (no red) or SSM back to the enable faults for TR and DD.
4. At the rear pedestal:
 - A. Remove associated test hardware.
 - B. Re-connect Body Preamplifier to original configuration.

CAUTION

Equipment damage possibility. Application of RF energy to TR Switches with Jumpers in TUNE position can cause permanent damage or destruction of components. The Jumpers must be placed back to the scan (normal) position. This JP1 and JP2 position differs for 1.5T and 1.0T Head Coils. Refer to the Head Coil Tuning Sense Loop Test procedure for specific details.

5. **For 1.5T Head Coil Tune Only:** Reinstall rigid coax's and install JP1 and JP2 jumpers on Isolation Boards in the B (scan) position.
6. **For 1.0T Head Coil Tune Only:** Install JP1 and JP2 jumpers in the A (scan) position.
7. **For Body Coil Tune Only:** Remove N shorting caps from I and Q lines and reconnect I and Q lines to the Body Hybrid Splitter.
8. When finished with RFT, delete all unneeded RFT studies.
9. Perform at least one head or body scan to verify proper system operation.
10. Remove all remaining test hardware, restore the system back to normal configuration.

7- RFT RESULTS

7-1 RFT Analysis

RFT data analysis begins immediately after the scan is completed. The following is displayed on the display plasma while the data analysis is occurring:

Note

Only those tests which were selected during scan prescription will be processed.

Running RF Transmit Test Analysis

```
*****
*                                     RFT Analysis Tool                                     *
*****
```

```
RF Stability:Slice 1, echo 1
RF Stability:Slice 2, echo 1
RF Stability:Slice 3, echo 1
RF Stability:Slice 4, echo 1
RF Linearity:Slice 1, echo 2
RF Linearity:Slice 2, echo 2
RF Droop:Slice 2, echo 1
RF Fidelity:Slice 2, echo 3
RF Fidelity:Slice 2, echo 4
RF Bandwidth:Slice 5, echo 1
```

Note: RF Bandwidth appears only
when Bandwidth = 1.

```
***** RFT Analysis Completed *****
```

Press any key to quit-->Enter any key to quit.

7-2 Report Program

RFT results (both text and graphics) can be viewed using the Report program from the Signa Tools menu. Refer to Table 7-1. This section describes how to initiate the Report program. Remaining sections provide examples of all reported parameters for RFT results. Record data for each test in Data Sheet 1.5T or Data Sheet 1.0T found in Appendix A.

Note

In Release 8.2 (and later), the RFT data sheet can automatically be generated. Refer to the Report Manager procedure.

TABLE 7-1
RESULT FILE ORGANIZATION

Group	Slice	Echo	Title	Plots
1	1	1	Low Power Magnitude Stability - No Gradients	1 Sample deviation all (max) 2 Sample deviation all (min) 3 View means 4 Min sample deviation 5 Max sample deviation
2	1	1	Low Power Phase Stability - No Gradients	1 Sample deviation all (max) 2 Sample deviation all (min) 3 View means 4 Min sample deviation 5 Max sample deviation
3	2	1	High Power Magnitude Stability - No Gradients	1 Sample deviation all (max) 2 Sample deviation all (min) 3 View means 4 Min sample deviation 5 Max sample deviation
4	2	1	High Power Phase Stability - No Gradients	1 Sample deviation all (max) 2 Sample deviation all (min) 3 View means 4 Min sample deviation 5 Max sample deviation
5	3	1	Low Power Magnitude Stability - With Gradients	1 Sample deviation all (max) 2 Sample deviation all (min) 3 View means 4 Min sample deviation 5 Max sample deviation

TABLE 7-1 - CONTINUED
RESULT FILE ORGANIZATION

6	3	1	Low Power Phase Stability - With Gradients	1 Sample deviation all (max) 2 Sample deviation all (min) 3 View means 4 Min sample deviation 5 Max sample deviation
7	4	1	High Power Magnitude Stability - With Gradients	1 Sample deviation all (max) 2 Sample deviation all (min) 3 View means 4 Min sample deviation 5 Max sample deviation
8	4	1	High Power Phase Stability - With Gradients	1 Sample deviation all (max) 2 Sample deviation all (min) 3 View means 4 Min sample deviation 5 Max sample deviation
9	1 & 2	2	Linearity -61 to 0 dB	1 Magnitude Error 2 Phase Error
10	2	1	Magnitude Droop	1 Raw Droop 2 Early Droop 3 Average Droop 4 Late Droop
11	2	3	Sinc 2 Pulse Fidelity	1 Half Scale Odd Views 2 Full Scale Odd Views 3 Half Scale Even Views 4 Full Scale Even Views 5 Ideal Slice

TABLE 7-1 - CONTINUED
RESULT FILE ORGANIZATION

12	2	4	Sinc 8 Pulse Fidelity	1 Half Scale Odd Views 2 Full Scale Odd Views 3 Half Scale Even Views 4 Full Scale Even Views 5 Ideal Slice
13	5	1	Bandwidth	1 Raw Magnitude vs. Frequency 2 Corrected Magnitude vs. Frequency
14	5	1	Coil Tune Check	1 Raw Magnitude vs. Frequency 2 Corrected Magnitude vs. Frequency

7-3 RFT Header Data

Refer to Table 7-2 for definition of legal value entries for all RFT Report header parameters:

TABLE 7-2
RFT HEADER PARAMETERS

Parameter	Legal Values	Source
Sitename	Site Name	Raw Header
Usn	Unique System Number (GE Cares Issued)	Config File
Srvconfig	Date/Time Srv Config File Last Changed	Srv Config File ("sysconfig")
Receiver	# Of Tps Rcvr Bds (4-4); # Channels/Bd (1) Starting Rec., Ending Rec, Port Enable	Srv Config File
Xmtrfcoil	Body, Head	Raw Header
Rcvrfcoil	Body, Head, <Surface Coil Name>	Raw Header
Freq	Magnet Frequency	Raw Header
Time	YY/MM/DD HH:MM:SS	Raw Header
Baserun	Base Run Number Of Scan	Raw Header
Configcode	xxxxxxxxxx	Raw Header
Softrev	Software Revision	"mrswrev" Script
Rcvcoilgain	Calibration Value (Typ. 4-10)	Config File
R1	4-7	Raw Header
R2	4-30	Raw Header
TG	0-200	Raw Header
C1	Exam Description	Raw Header
C2	Series Description	Raw Header

Refer to the following for a sample header:

```
17JE1325.RFT/0 Header Info RDF/GRP Revision: 1.0 /41
=====
SITENAME      = BAY 12
USN           = B15B0C0
MLN           = 9999
SRVCONFIG     = YY/MM/DD HH/MM/SS
EXCITER       = 000
RECEIVER      = S0/E0/PE0
XMTRFCOIL     = BODY
RCVRFCOIL     = BODY
FREQ          = 42663520 Hz
TIME          = 91/7/27 07:40:00
BASERUN       = 44032
CONFIGCODE    = _0201240000
SOFTREV       = 5.3
NUCLIDE       = 000
HEADERCODE    = 0x000000000
RCVCOILGAIN   = 6.090 R1 = 7 R2 = 14 TG = 200
----- Exam description -----
NO COMMENT
----- Series description -----
Body,Ax,2D,Spin Echo,None
-----
```

7-4 RFT Sample Test Outputs

Refer to the following sections for sample RFT data for each test mode.

7-4-1 Low Power Magnitude (No Gradients) Stability Data

```
xxxxxxx.RFT/1 LO PWR MAGNITUDE STABILITY - NO GRADIENTS
=====
All values below are percent of peak pulse amplitude.
-----
max-min(pp)   + stdev   - stdev   + area   - area
      0.20      0.01      0.01     14.46    -14.99
-----
max sample column 157      p-p      stdev
min sample column  32      0.16      0.02
view means           0.07      0.01
-----
```

7-4-2 Low Power Phase (No Gradients) Stability Data

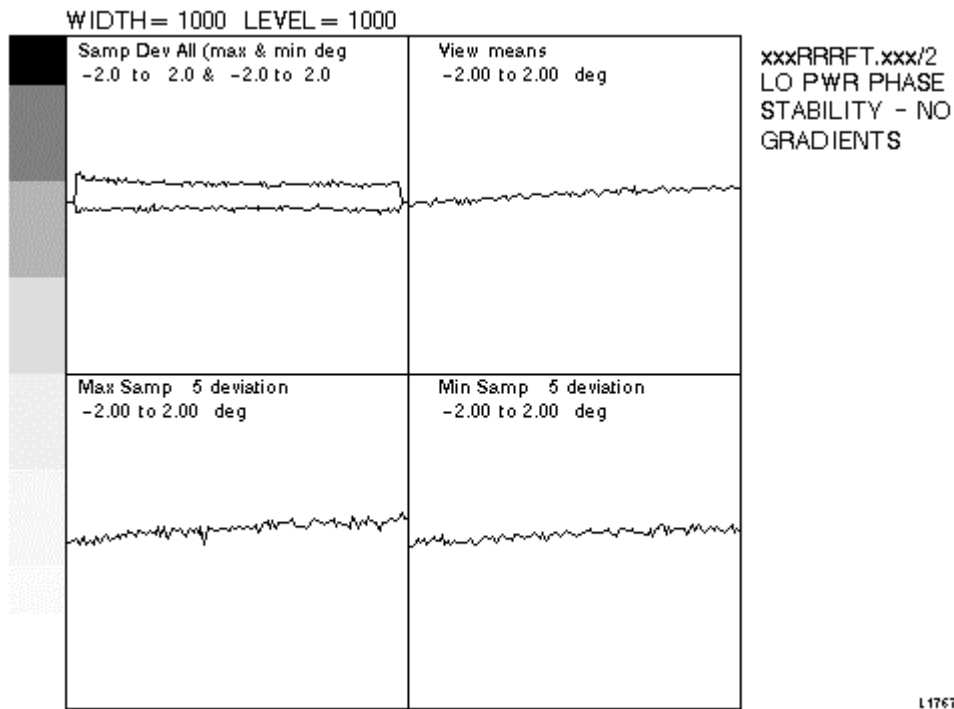
xxxxxxxx.RFT/2 LO PWR PHASE STABILITY - NO GRADIENTS

=====

All values below are phase deviation in degrees.

max-min(pp)	+ stdev	- stdev	+ area	- area
0.47	0.02	0.03	18.79	-28.01

		p-p	stdev	
max sample column	5	0.47	0.09	
min sample column	5	0.47	0.09	
view means		0.12	0.02	



RFT LOW POWER PHASE STABILITY (NO GRADIENTS) SAMPLE PLOTS
ILLUSTRATION 7-1

7-4-3 High Power Magnitude (No Gradients) Stability Data

xxxxxxxx.RFT/3 HI PWR MAGNITUDE STABILITY - NO GRADIENTS

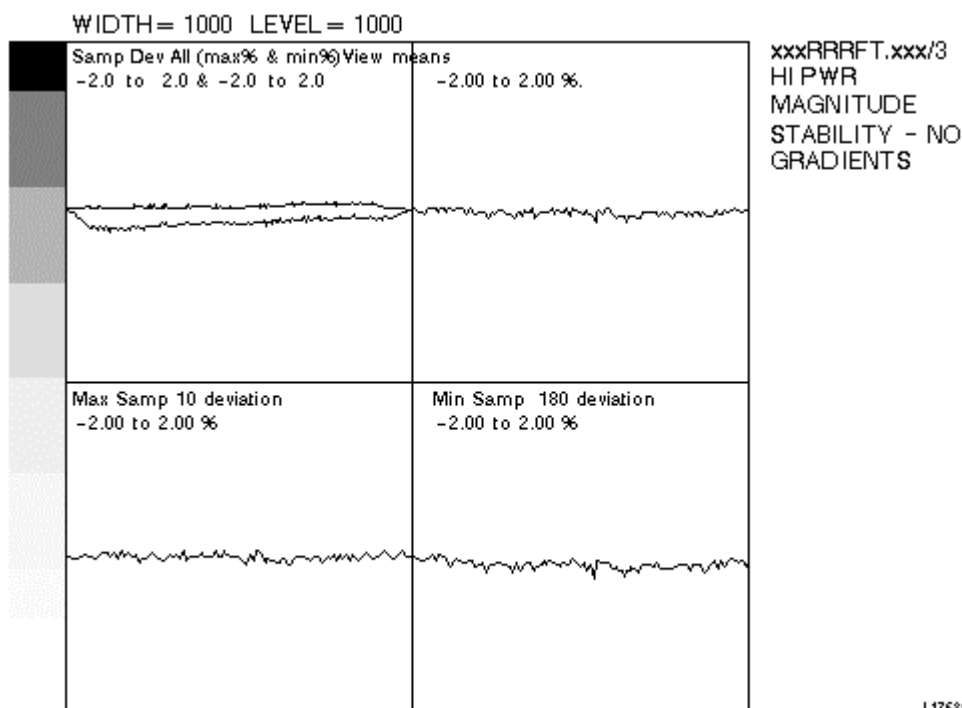
=====

All values below are percent of peak pulse amplitude.

max-min(pp)	+ stdev	- stdev	+ area	- area
0.28	0.01	0.03	10.26	-37.37

		p-p	stdev
max sample column 10		0.16	0.02
min sample column 180		0.21	0.04
view means		0.17	0.03

Plot below may be incorrect-refer to Illustration 7-1.



HIGH POWER MAGNITUDE STABILITY (NO GRADIENTS) SAMPLE PLOTS
ILLUSTRATION 7-2

7-4-4 High Power Phase (No Gradients) Stability Data

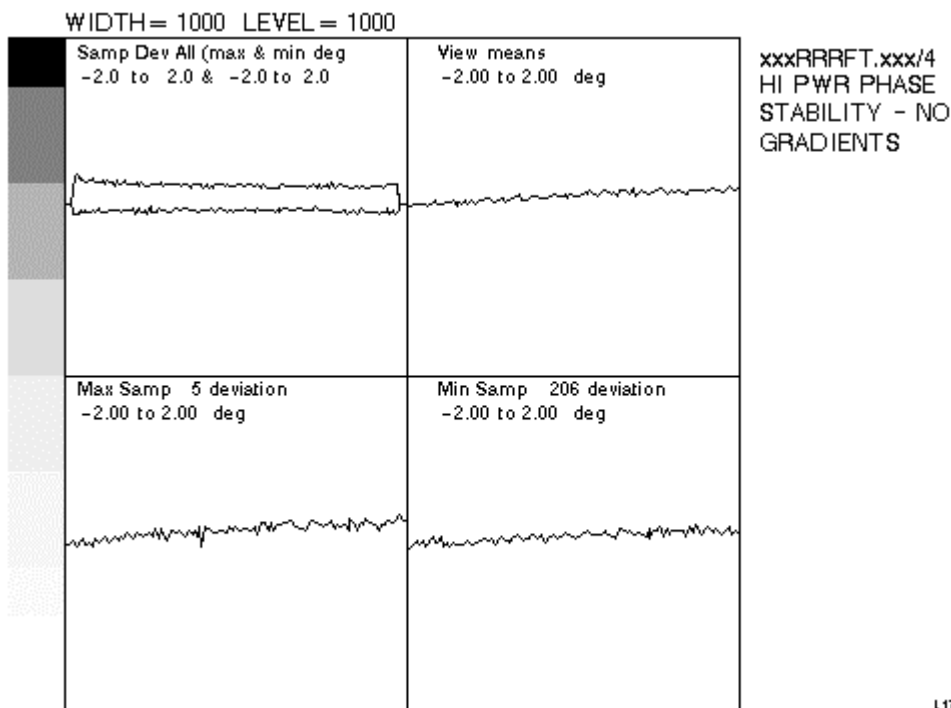
xxxxxxxx.RFT/4 HI PWR PHASE STABILITY - NO GRADIENTS

=====

All values below are phase deviation in degrees.

```
-----
max-min(pp)    + stdev    - stdev    + area    - area
      0.50         0.04         0.03        49.85    -32.21
-----
max sample column 5          p-p          stdev
min sample column 206        0.37          0.08
view means              0.40          0.05
                        0.29          0.04
-----
```

Plot below may be incorrect-refer to Illustration 7-1.



L17639

HIGH POWER PHASE STABILITY (NO GRADIENTS) SAMPLE PLOTS
ILLUSTRATION 7-3

7-4-5 Low Power Magnitude (With Gradients) Stability Data

xxxxxxxx.RFT/5 LO PWR MAGNITUDE STABILITY WITH GRADIENTS

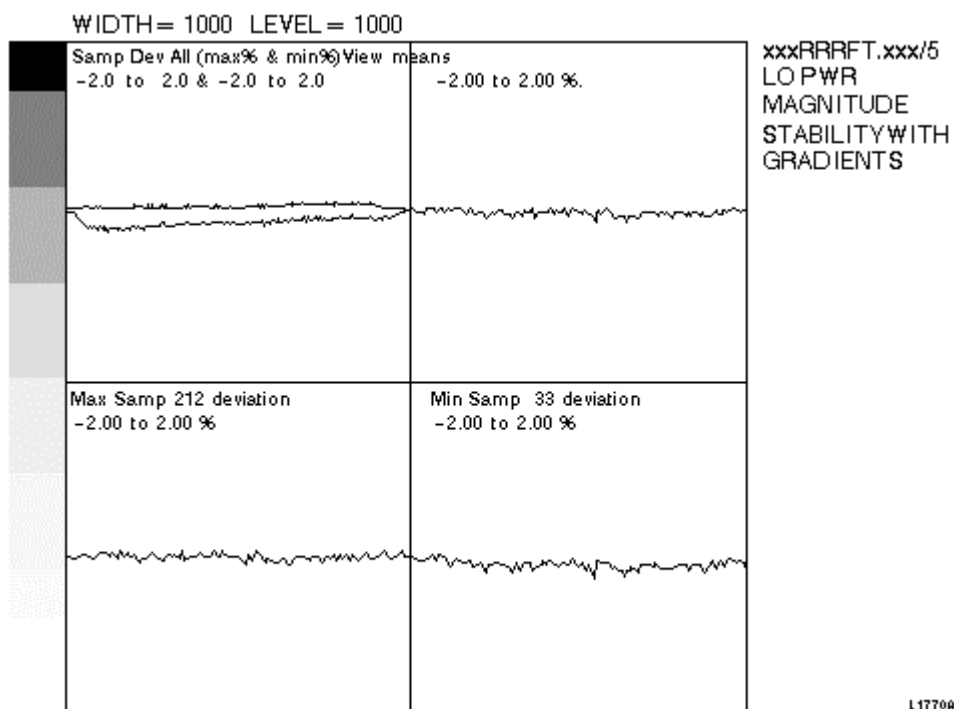
=====

All values below are percent of peak pulse amplitude.

max-min(pp) + stdev - stdev + area - area
 0.41 0.02 0.06 13.21 -32.16

	p-p	stdev
max sample column 212	0.15	0.03
min sample column 33	0.35	0.05
view means	0.10	0.02

Plot below may be incorrect-refer to Illustration 7-1.



LOW POWER MAGNITUDE STABILITY (WITH GRADIENTS) SAMPLE PLOTS
ILLUSTRATION 7-4

7-4-6 Low Power Phase (With Gradients) Stability Data

xxxxxxxx.RFT/6 LO PWR PHASE STABILITY WITH GRADIENTS

=====

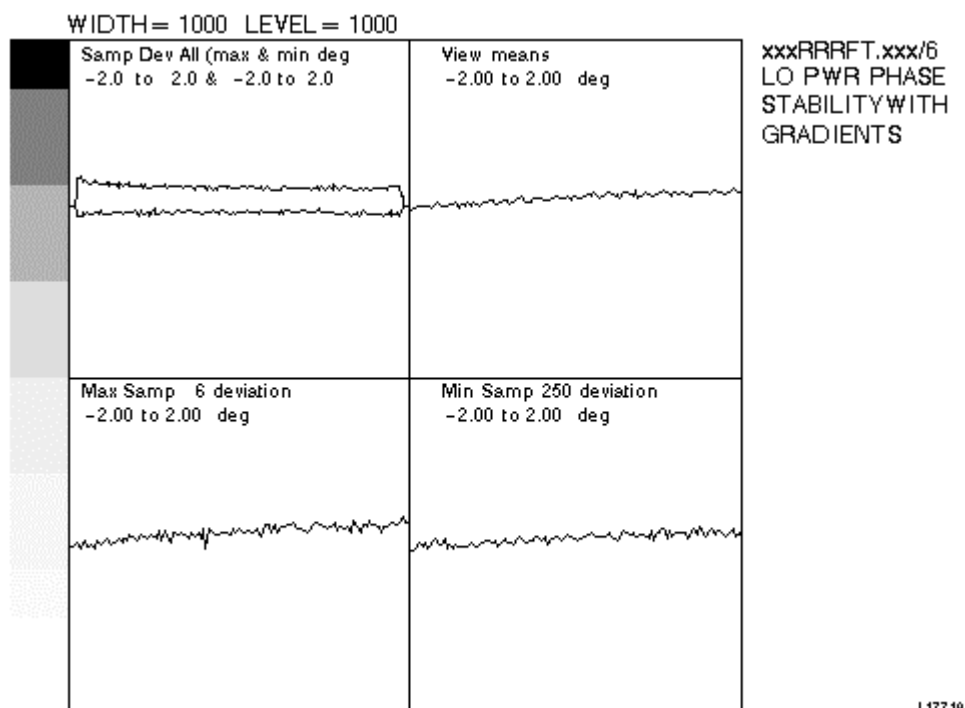
All values below are phase deviation in degrees.

```

-----
max-min(pp)    + stdev    - stdev    + area    - area
      0.42        0.02        0.03       15.76    -33.28
-----
max sample column 6          p-p          stdev
min sample column 250        0.41          0.06
view means                0.34          0.06
                        0.12          0.02
-----

```

Plot below may be incorrect-refer to Illustration 7-1.



LOW POWER PHASE STABILITY (WITH GRADIENTS) SAMPLE PLOTS
ILLUSTRATION 7-5

7-4-7 High Power Magnitude (With Gradients) Stability Data

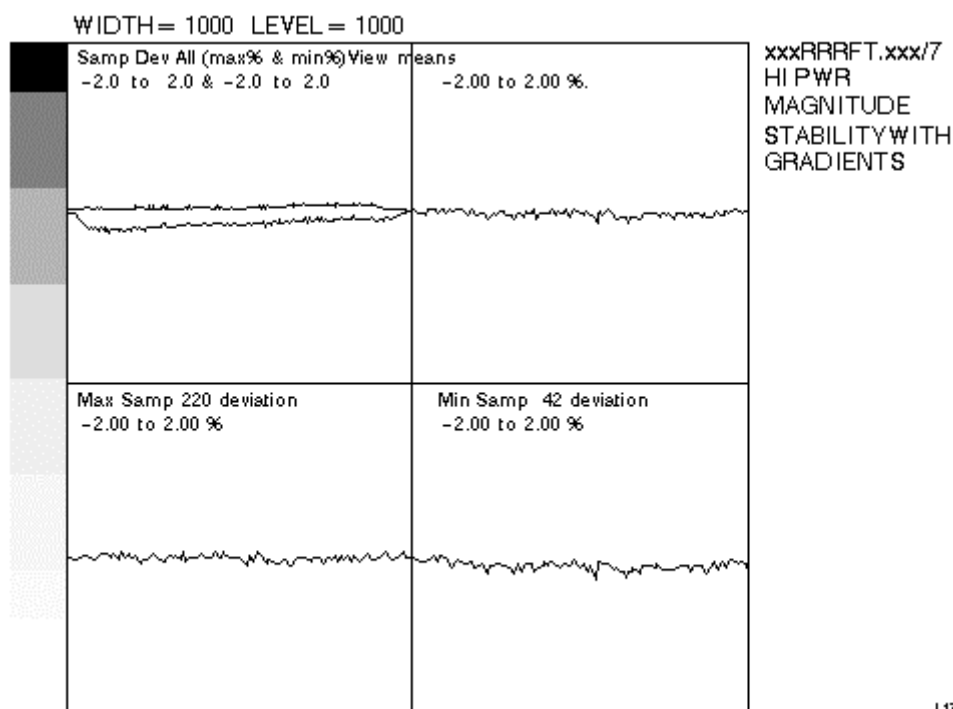
xxxxxxxx.RFT/7 HI PWR MAGNITUDE STABILITY WITH GRADIENTS

=====
All values below are percent of peak pulse amplitude.

```
-----
max-min(pp)  + stdev    - stdev    + area    - area
      0.31      0.01      0.05      10.68    -34.00
-----

max sample column 220      p-p      stdev
min sample column  42      0.15      0.03
view means              0.26      0.04
                        0.14      0.03
-----
```

Plot below may be incorrect-refer to Illustration 7-1.



L1772R

HIGH POWER MAGNITUDE STABILITY (WITH GRADIENTS) SAMPLE PLOTS
ILLUSTRATION 7-6

7-4-8 High Power Phase (With Gradients) Stability Data

xxxxxxxx.RFT/8 HI PWR PHASE STABILITY WITH GRADIENTS

=====

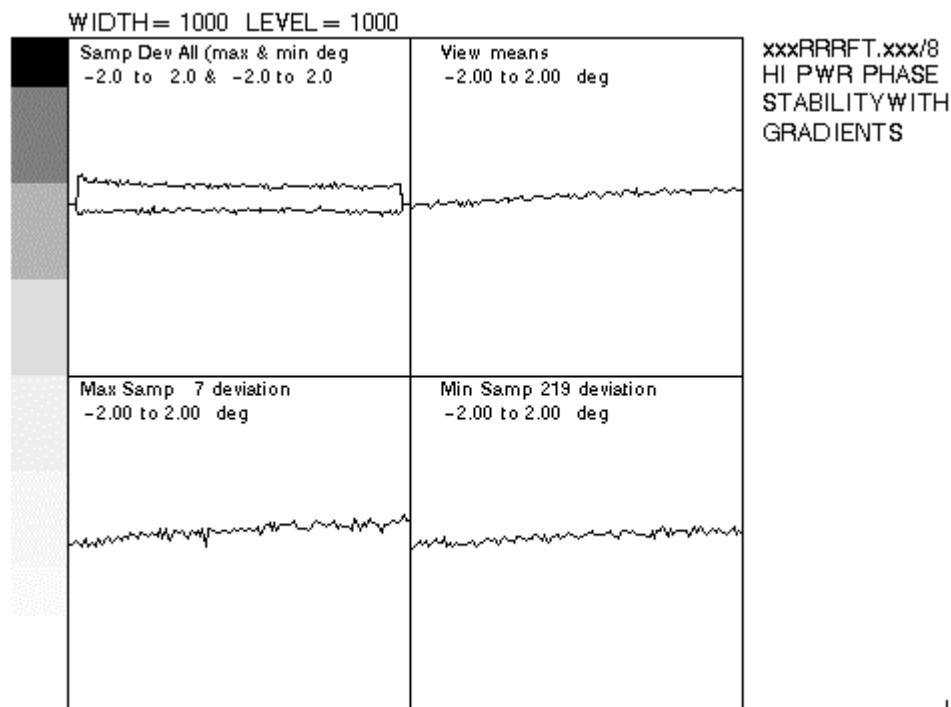
All values below are phase deviation in degrees.

```

-----
max-min(pp)  + stdev    - stdev    + area    - area
      0.45      0.05      0.01      57.23    -16.91
-----
max sample column 7      p-p      stdev
min sample column 219    0.42      0.08
view means              0.34      0.06
                        0.25      0.06
-----

```

Plot below may be incorrect-refer to Illustration 7-1.



HIGH POWER PHASE STABILITY (WITH GRADIENTS) SAMPLE PLOTS
ILLUSTRATION 7-7

L1773a

7-4-9 Linearity Data

xxxxxxxx.RFT/9 LINEARITY -61 to 0 dB

=====

Carrier Leakage (dB below full scale modulation)

N/A

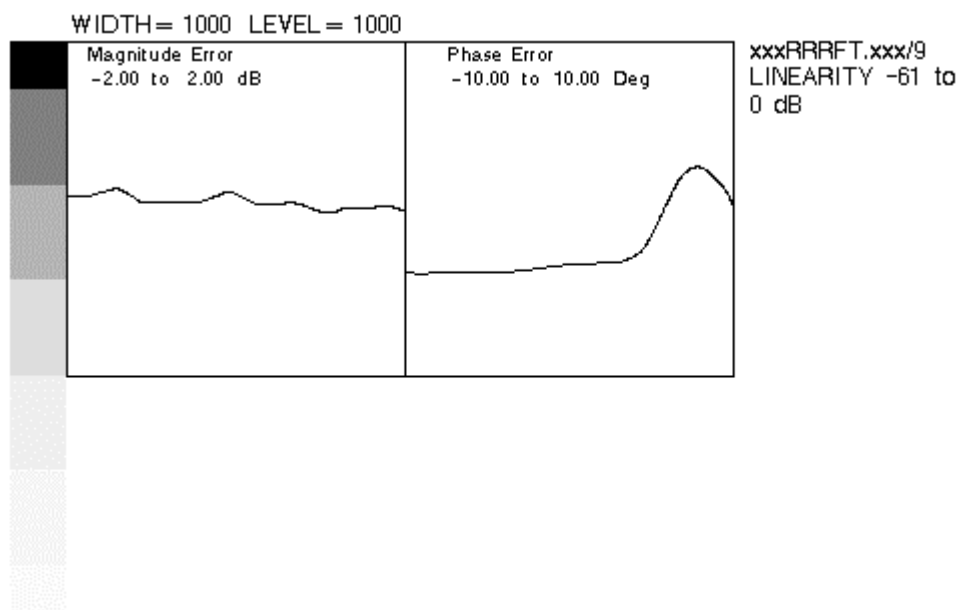
Magnitude Linearity Error (dB)

max	Min	P-P	Std Dev
0.23	-0.02	0.25	0.06

Phase Linearity Error (degrees)

max	Min	P-P	Std Dev
2.41	-3.75	6.17	2.09

Plot below may be incorrect for 8X.



LINEARITY SAMPLE PLOT
ILLUSTRATION 7-8

L1774R

7-4-10 Magnitude Droop Data

xxxxxxxx.RFT/10 MAGNITUDE DROOP

=====

Peak Amplitude 27260.2 counts

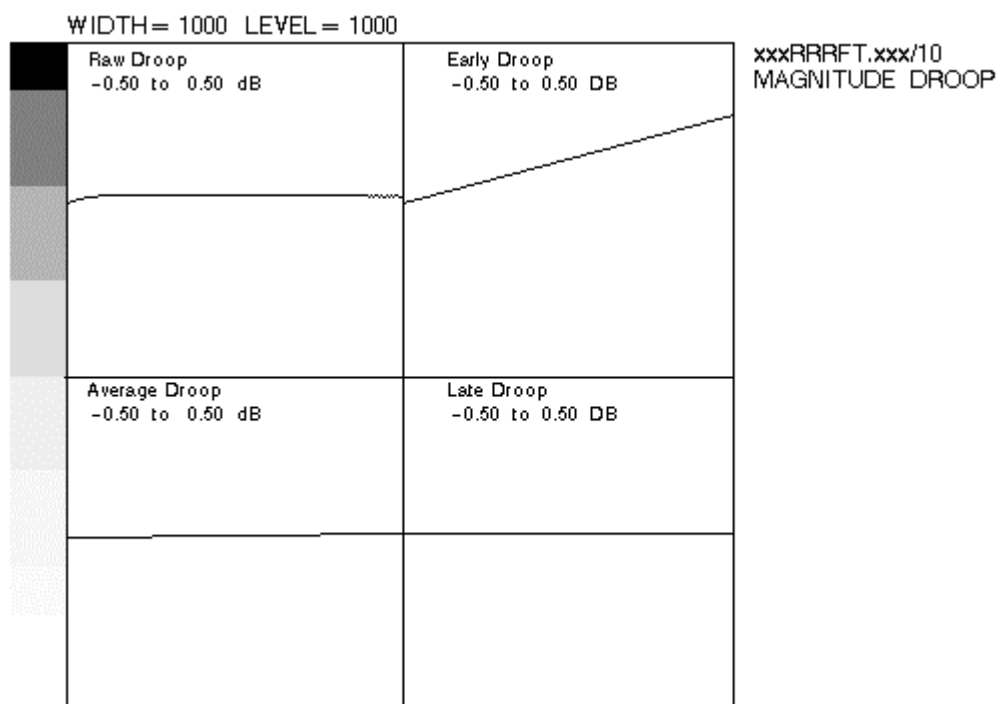
Droop Rates in dB/millisecond

Early 0.0820

Average 0.0030

Late -0.0000

Plot below may be incorrect for 8X.



L1775A

MAGNITUDE DROOP SAMPLE PLOT
ILLUSTRATION 7-9

7-4-11 Sinc 2 Pulse Fidelity Data

xxxxxxxx.RFT/11 SINC 2 PULSE FIDELITY

Positive Modulation

Full Scale

Slice Amplitude 121.71 % of ideal slice
Total Error 303.27 dB x samples > -45 dB
Pulse Amplitude 81.15 % of ideal pulse

Half Scale

Slice Amplitude 58.02 % of ideal slice
Total Error 290.59 dB x samples > -45 dB

Negative Modulation

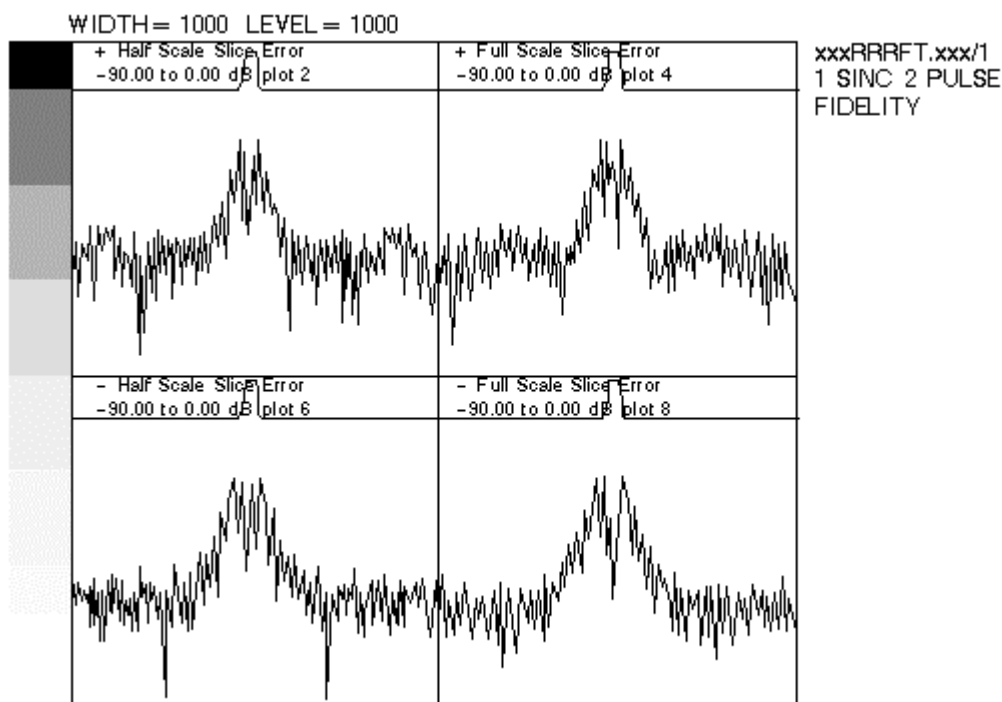
Full Scale

Slice Amplitude 121.04 % of ideal slice
Total Error 318.04 dB x samples > -45 dB
Pulse Amplitude 80.98 % of ideal pulse

Half Scale

Slice Amplitude 57.23 % of ideal slice
Total Error 347.59 dB x samples > -45 dB

Plot below may be incorrect for 8X.



SINC 2 PULSE FIDELITY SAMPLE PLOT
ILLUSTRATION 7-10

L1776A

7-4-12 Sinc 8 Pulse Fidelity Data

xxxxxxxx.RFT/12 SINC 8 PULSE FIDELITY

=====

Positive Modulation

Full Scale

Slice Amplitude 30.25 % of ideal slice
Total Error 1655.74 dB x samples > -45 dB
Pulse Amplitude 80.45 % of ideal pulse

Half Scale

Slice Amplitude 14.41 % of ideal slice
Total Error 2030.24 dB x samples > -45 dB

Negative Modulation

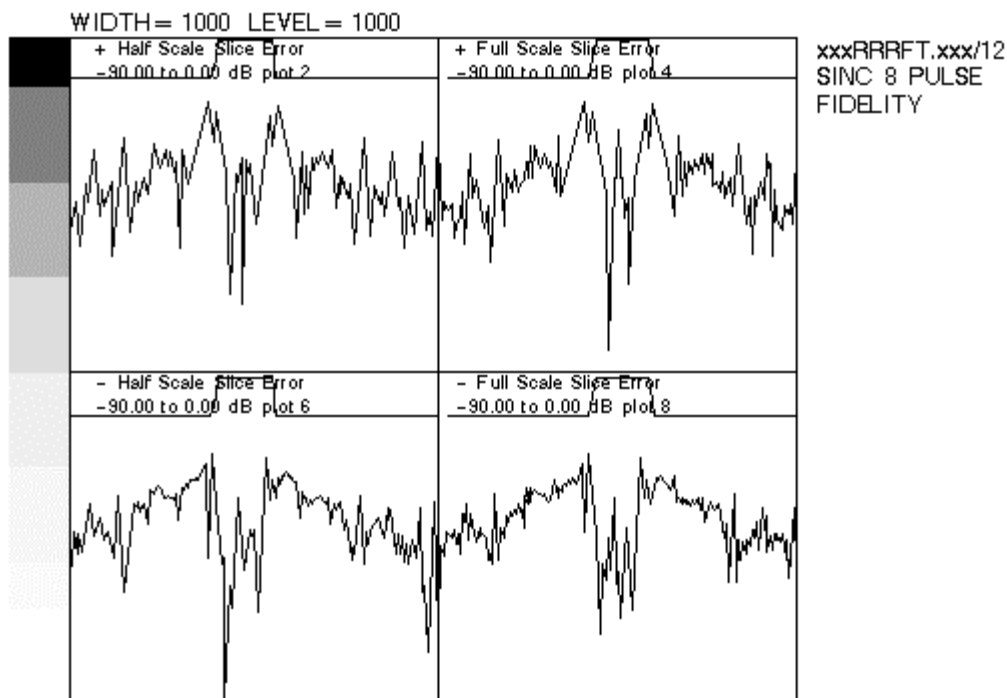
Full Scale

Slice Amplitude 30.19 % of ideal slice
Total Error 1683.67 dB x samples > -45 dB
Pulse Amplitude 80.40 % of ideal pulse

Half Scale

Slice Amplitude 14.35 % of ideal slice
Total Error 2127.08 dB x samples > -45 dB

Plot below may be incorrect for 8X.



SINC 8 PULSE FIDELITY SAMPLE PLOT
ILLUSTRATION 7-11

L1777A

7-4-13 Bandwidth Data

```

xxxxxxxx.RFT/13    BANDWIDTH
=====
Center Frequency          -0.25 dB Bandwidth

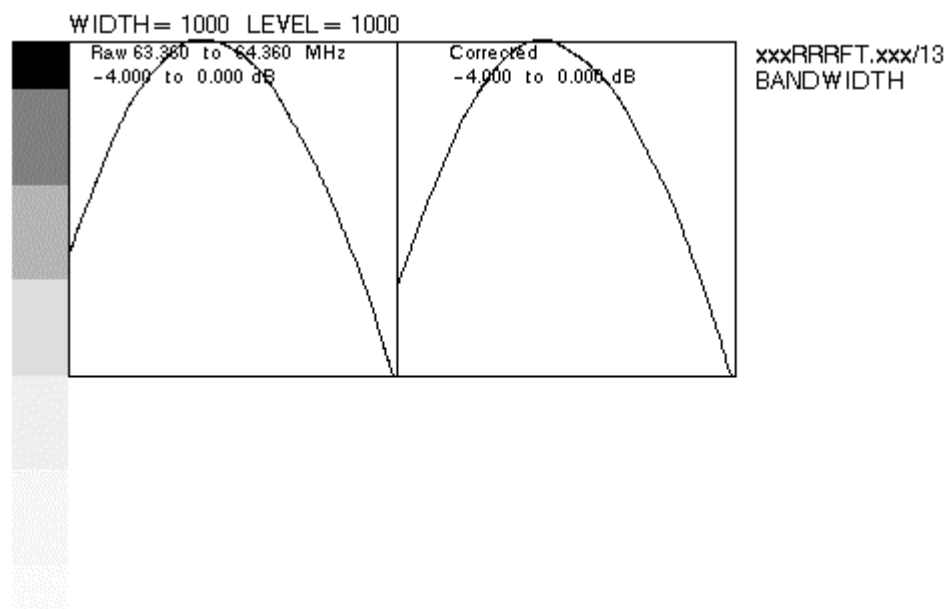
    63.777 MHz              0.346 MHz
-----
Low -0.25 dB freq        High -0.25 dB freq

    63.612 MHz              63.958 MHz
-----
Sweep start freq          Sweep end freq

    63.360 MHz              64.360 MHz
-----
Cal File Data Acquired
-----

```

Plot below may be incorrect for 8X.



RFT BANDWIDTH SAMPLE PLOT
ILLUSTRATION 7-12

L1778A

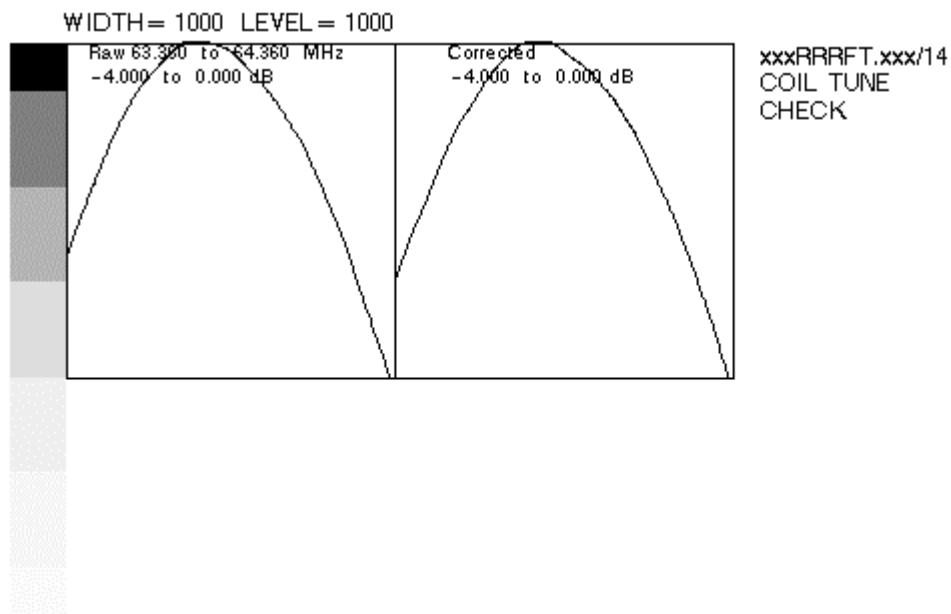
7-4-14 Coil Tune Check Data

```

xxxxxxxx.RFT/14    COIL TUNE CHECK
=====
Center Frequency          -3.0 dB Bandwidth
    63.777 MHz              0.346 MHz
-----
Low -3.0 dB freq          High -3.0 dB freq
    63.612 MHz              63.958 MHz
-----
Sweep start freq          Sweep end freq
    63.360 MHz              64.360 MHz
-----
Cal File Data Acquired
-----

```

Plot below may be incorrect for 8X.



COIL TUNE CHECK SAMPLE PLOT
ILLUSTRATION 7-13

AL1779A

7-4-15 Config File Data

xxxxxxx.RFT/15/15 SERVICE CONFIG FILE

=====

magnetSerNum = "R244"
magnetType = "GELCC-15"
fieldStrength = "1.5T"
room shielding (Yes/No) = "No"
magniShield = (Yes/No) = "No"
shieldCooler = type (0,1,2,3,4,5,6) = "4"
resistive shim power supply (0,1,2,3,4,5,6) = "0"
RFAmpType = (ErbtecTube, ErbtecSolidState, TELI Solid State) = "4"
spectroAmp = (none, ENI) = "None"
line CondType (0,1,2) = "2"
lineFreq (50, 60) = "60"
SRMode (120,77,20,17) = "120"
ThisLineNotUsed = ""
ThisLineNotUsed = ""
ThisLineNotUsed = ""

8- RFT THEORY

8-1 RFT Overview

RFT is an RF loop-back test that will quickly evaluate stability, magnitude and phase linearity, droop, pulse fidelity, and bandwidth of the RF transmit chain in a Signa system. This is accomplished by playing out four different RF pulses and collecting each one as a separate "echo". The first pulse has a wide flat top and is used for stability, droop and bandwidth testing. The second pulse has a narrower flat top and its amplitude increments during the scan for magnitude and phase linearity testing. The third pulse is a sinc 2 and the fourth pulse is a sinc 8. The amplitude of the third and fourth pulses is incremented like the second pulse for pulse fidelity analysis.

All tests are run at the following frequencies: (Actual magnet center frequency is not used.)

1.5T - 63.86 MHz

1.0T - 42.57 MHz

For the bandwidth test, provision is made to characterize the Exciter/Receiver combination so that when the RF amp is tested the results can be corrected for errors in the Exciter/Receiver. When testing bandwidth using a sense loop in one of the scanners transmit coils, the results will likewise be corrected for errors in the Exciter/RF Amplifier combination. User CVs are provided so the analysis tool can be told the exact status of the loop back configuration.

- **RFT Hardware**
A Switchable Attenuator controlled by the PSD through the DAB is required to extend the dynamic range of the linearity and stability tests. Linearity and stability data are collected twice, first with the transmit path attenuator set to 30 dB (IN) and again with it set to 0 dB (OUT). The receive path attenuator is 30 dB when the transmit path attenuator is 0 dB and vice versa. Stability data is collected twice more with a gradient pulse applied during the RF pulse. The Switchable Attenuator is used as before.
- **Testing Sequence**
This document is structured for four basic modes of running RFT: 1) baseline (reference) data collection, 2) coil tune check, 3) stability troubleshooting, and 4) pulse fidelity troubleshooting. Because the number of possible hardware and software configurations is large, **flowcharts** are provided for each mode. The **flowcharts** direct you to the appropriate tables and diagrams for software and hardware setup, prescan, and scan references.

8-2 Data Collection

First slice

The 128 views are collected with the Switchable Attenuator IN (i.e., transmit path set to 30 dB). Echo 2, 3, and 4 pulse amplitudes are incremented as specified in Illustration 8-1.

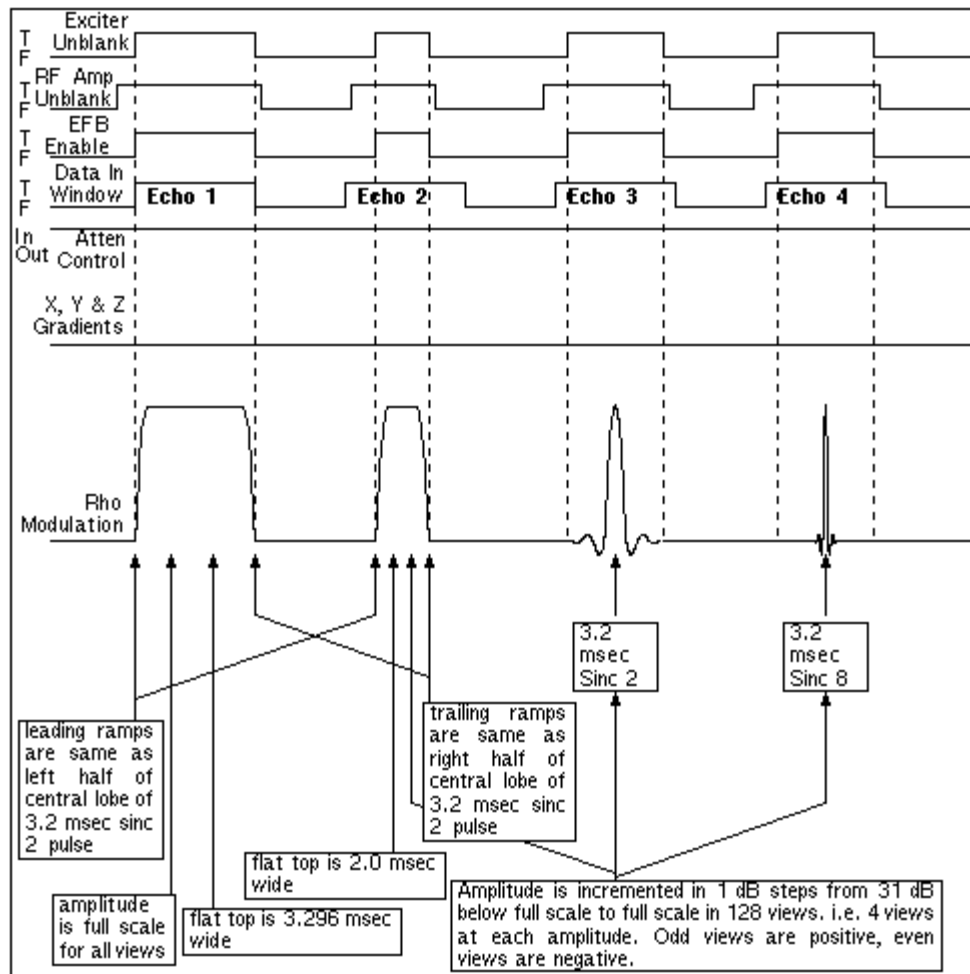


ILLUSTRATION 8-1

Second slice

The 128 views are collected with the Switchable Attenuator OUT (i.e., transmit path set to 0 dB). Echo 2, 3, and 4 pulse amplitudes are incremented as specified in Illustration 8-1.

Third slice

The 128 views are collected with the Switchable Attenuator IN (i.e., transmit path set to 30 dB). Echo 2, 3, and 4 pulse amplitudes are set to zero with modulator unblank, EFB enable and Data In Windows disabled as specified in Illustration 8-2.

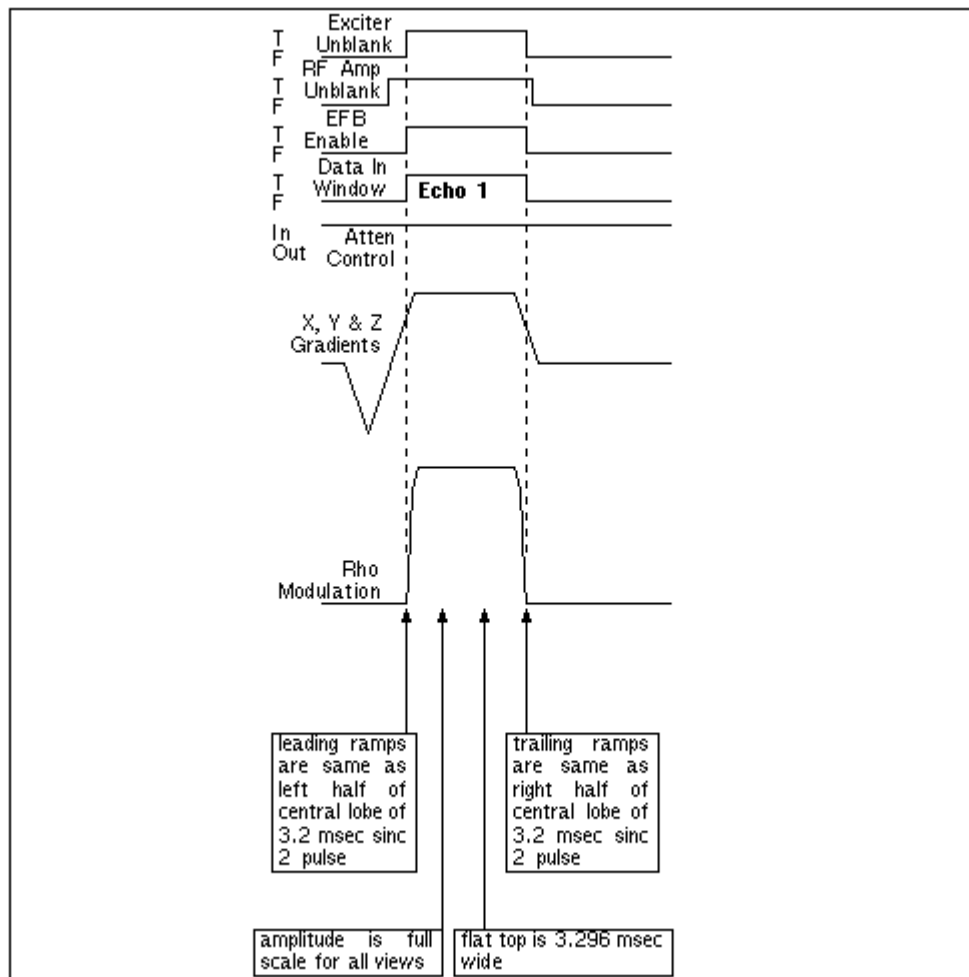


ILLUSTRATION 8-2

Fourth slice

The 128 views are collected with the Switchable Attenuator OUT (i.e., transmit path set to 0 dB). Echo 2, 3, and 4 pulse amplitudes are set to zero with modulator unblank, EFB enable and Data In Windows disabled as specified in Illustration 8-2.

Fifth slice

The Switchable Attenuator is OUT (i.e. transmit path set to 0 dB) and operating frequency is set to 500 kHz below 63.86 MHz (42.57 or 21.89 MHz). Transmit and receive frequency are incremented (1000/127) kHz per view while the echo 1 pulse is played out at full scale. The pulse amplitude for echoes 2, 3, and 4 is zero and the exciter and RF amplifier are not unblanked during the 128 views of the third slice. See Illustration 8-2.

8-3 Magnitude Stability Analysis (groups 1, 3, 5 and 7)

A magnitude matrix is calculated from echo 1 raw data. Each element in the normalization vector (**nvc**) contains the mean of the corresponding column in the magnitude matrix. Peak magnitude (**pkmag**) is the mean of elements 31 through 37 in the normalization vector. Each element in the sample max vector (**smaxc**) contains the maximum of the corresponding column in the normalized magnitude matrix. (**cmax**) is element number (**c**) for which smaxc is maximum. Each element in the sample min vector (**sminc**) contains the minimum of the corresponding column in the normalized magnitude matrix. (**cmin**) is element number (**c**) for which sminc is minimum. Each element in the view means vector (**vmeanr**) contains the mean of the corresponding row in the normalized magnitude matrix.

The test provides plots of "Sample Deviation All" (max% & min%), "View Means" (%), "Max Sample Deviation" (%), and "Min Sample Deviation" (%).

Report data file summary:

- Max-Min (p-p)
- +Stdev
- -Stdev
- +Area
- -Area
- Max Sample Column
- Min Sample Column
- View Means

8-4 Phase Stability Analysis (groups 2, 4, 6 & 8)

An unwrapped phase matrix is calculated from echo 1 raw data. Each element in the normalization vector (**nvc**) contains the mean of the corresponding column in the unwrapped phase matrix. Each element in the sample max vector (**smaxc**) contains the maximum of the corresponding column in the normalized phase matrix. (**cmax**) is element number (**c**) for which smaxc is maximum. Each element in the sample min vector (**sminc**) contains the minimum of the corresponding column in the normalized phase matrix. (**cmin**) is element number (**c**) for which sminc is minimum. Each element in the view means vector (**vmeanr**) contains the mean of the corresponding row in the normalized phase matrix.

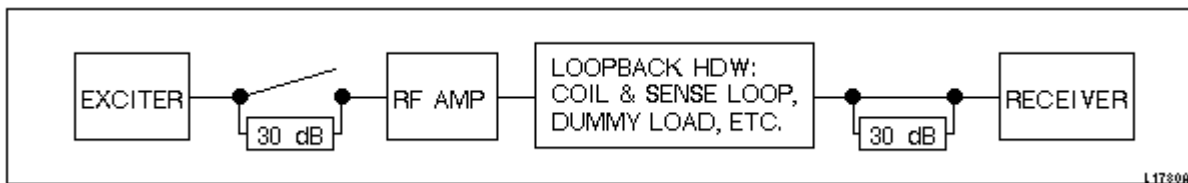
The test provides plots of "Sample Deviation All" (max% & min%), "View Means" (%), "Max Sample Deviation" (%), and "Min Sample Deviation" (%).

Report data file summary:

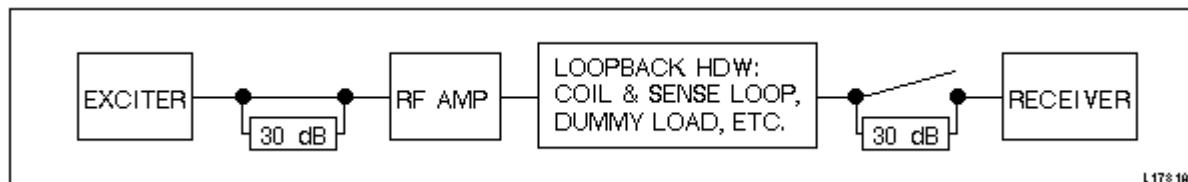
- Max-Min (p-p)
- +Stdev
- -Stdev
- +Area
- -Area
- Max Sample Column
- Min Sample Column
- View Means

8-5 Linearity And Carrier Leakage Analysis (group 9)

Data for Linearity and Carrier Leakage Analysis is collected in passes (slices) 1 and 2 of the scan. The second echo is analyzed for linearity. Polarity of the RF modulation is alternated on every other view and 4 views are taken at each magnitude. Therefore, there are 2 positive pulses and 2 negative pulses for each amplitude. Amplitude is incremented in 1.0 dB steps beginning at 31.0 dB below full scale and ending at full scale for a total of 32 different amplitudes in each pass. The Switchable Attenuator inserts 30 dB between output of the exciter and input of the RF amplifier or in the loopback path between output of the RF amplifier and input of the receiver. See Illustration 8-3 and Illustration 8-4. Since pulse amplitude is incremented through 32 dB in each pass and the attenuation change is 30 dB between passes, there is 2 dB of overlap in RF amplifier output between passes 1 and 2. This overlap is used to smoothly splice together linearity data for the two passes yielding a total dynamic range for the linearity test of 62 dB.



PASS 1 & 3 SWITCHABLE ATTENUATOR CONDITION
ILLUSTRATION 8-3



PASS 2, 4, & 5 SWITCHABLE ATTENUATOR CONDITION
ILLUSTRATION 8-4

The test provides plots of "Magnitude Error" (dB), and "Phase Error" (deg).

Report data file summary:

- Carrier Leakage
- Magnitude Linearity Error (Max)
- Magnitude Linearity Error (Min)
- Magnitude Linearity Error (p-p)
- Magnitude Linearity Error (Std Dev)
- Phase Linearity (Max)
- Phase Linearity (Min)
- Phase Linearity (p-p)
- Phase Linearity (Std Dev)

8-6 Magnitude Droop Analysis (group 10)

Data for this test is collected in pass (slice) 2 of the scan. The first echo is analyzed for magnitude droop.

The test provides plots of "Raw Droop", "Early Droop", "Average Droop" and "Late Droop". Report data file summary:

- Peak Amplitude
- Early Droop Rate
- Average Droop Rate
- Late Droop Rate

8-7 Sinc 2 Pulse Fidelity Analysis (group 11) (TPS Loopback only)

Data for this test is collected in pass (slice) 2 of the scan. The third echo is analyzed for pulse fidelity.

The test provides plots of "+Half Scale Slice Error" (dB), "+Full Scale Slice Error" (%), "-Half Scale Slice Error" (%) and "-Full Scale Slice Error" (dB).

Report data file summary:

- + Full Scale Slice Amplitude
- + Full Scale Total Error
- + Full Scale Pulse Amplitude
- + Half Scale Slice Amplitude
- + Half Scale Total Error
- - Full Scale Slice Amplitude
- - Full Scale Total Error
- - Full Scale Pulse Amplitude
- - Half Scale Slice Amplitude
- - Half Scale Total Error

8-8 Sinc 8 Pulse Fidelity Analysis (group 12) (TPS Loopback only)

Data for this test is collected in pass (slice) 2 of the scan. The fourth echo is analyzed for pulse fidelity.

The test provides plots of "+Half Scale Slice Error" (dB), "+Full Scale Slice Error" (%), "-Half Scale Slice Error" (%) and "-Full Scale Slice Error" (dB).

The Report data file summary:

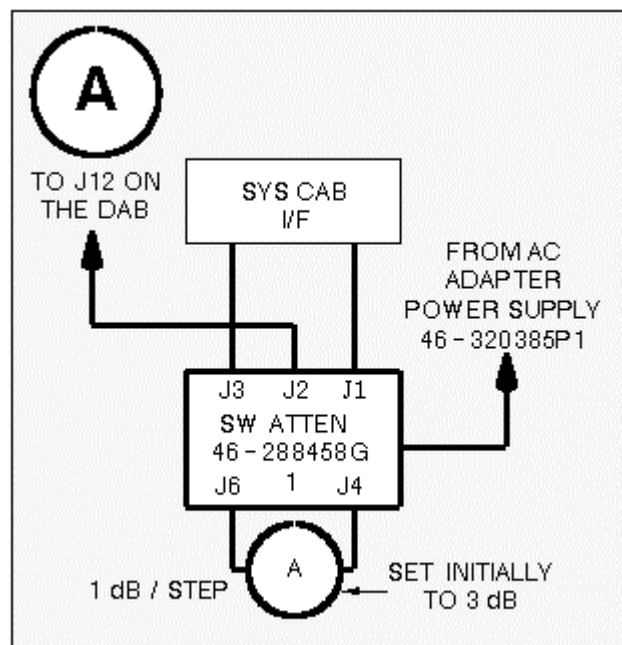
- + Full Scale Slice Amplitude
- + Full Scale Total Error
- + Full Scale Pulse Amplitude
- + Half Scale Slice Amplitude
- + Half Scale Total Error
- - Full Scale Slice Amplitude
- - Full Scale Total Error
- - Full Scale Pulse Amplitude
- - Half Scale Slice Amplitude
- - Half Scale Total Error

8-9 Magnitude Bandwidth Analysis (group 13 & 14)

Data for this test is collected in pass (slice) 5 of the scan. The first echo is analyzed for magnitude bandwidth.

Reference files are stored in /usr/g/service/data. They are required for the following conditions:

- TPS Reference File (Group 13)
The bandwidth test is run with output of the TPS looped back to the input of the receiver via the Switchable Attenuator as shown in Illustration 8-5. See Table 8-1 to determine when a copy of the filtered plot data will be written to the file /usr/g/service/data/tpsbw.cal.



L1725A

TPS LOOPBACK SETUP
ILLUSTRATION 8-5

TABLE 8-1
BW CAL FILE LOGIC

OPUSER 1	OPUSER 2	OPUSER 7	READ CAL FILE	WRITE CAL FILE	RESULT FILE NUMBER	DBDOWN
0	0	0	none	none	13	-.5
0	0	1	none	tpsbw.cal	13	-.5
1	0	0	tpsbw.cal	none	13	-.5
2	0	0	tpsbw.cal	none	13	-.5
2	0	1	tpsbw.cal	headbw.cal	13	-.5
2	1	0	headbw.cal	none	13	-.5
3	0	0	tpsbw.cal	none	13	-.5
4	0	0	tpsbw.cal	none	13	-.5
4	0	1	tpsbw.cal	bodybw.cal	13	-.5
4	1	0	bodybw.cal	none	13	-.5
7	0	0	tpsbw.cal	none	13	-.5
7	0	1	tpsbw.cal	preampbw.cal	13	-.5
7	1	0	preampbw.cal	none	14	-.3

- RF Amp Body Mode Reference File (Group 13)
The bandwidth test will be run with output of the RF amplifier looped back through suitable attenuation to the input of the receiver via the Switchable Attenuator as shown in Body tests (Body, Dummy Load, EFB Enabled) & (Body, Dummy Load, EFB Bypassed). See Table 8-1 to determine when a copy of the filtered plot data (prior to TPS bandwidth correction) will be written to the file /usr/g/service/data/bodybw.cal.
- RF Amp Head Mode Reference File (Group 13)
The bandwidth test will be run with output of the RF amplifier looped back through suitable attenuation to the input of the receiver via the Switchable Attenuator as shown in Section (Head, Dummy Load, EFB Enabled) and Section (Head, Dummy Load, EFB Bypassed). See Table 8-1 to determine when a copy of the filtered plot data (prior to TPS bandwidth correction) will be written to the file /usr/g/service/data/headbw.cal.
- Coil Tune Check Preamplifier Mode Reference File (Group 14)
The bandwidth test will be run with output of the body preamplifier looped back through suitable attenuation to the input of the receiver as shown in Body tests (Coil Tune Body Preamplifier) & (Body Coil Tune Sense Loop), and Head tests (Coil Tune Head Preamplifier) & (Head Coil Tune Sense Loop). See Table 8-1 to determine when a copy of the filtered plot data (prior to TPS bandwidth correction) will be written to the file /usr/g/service/data/preampbw.cal.

The test provides plots of "Raw" (MHz), and "Corrected" (MHz).

Report data file summary:

- Center Frequency
- -0.5dB Bandwidth
- Low -0.5dB Frequency
- High -0.5dB Frequency
- Sweep Start Frequency
- Sweep End Frequency
- Cal File Data Acquired

APPENDIX A - DATA SHEETS

1.5T RFT BODY/HEAD DATA

Site:		Name:		Date:
RFT File Number:				
Coil:	☐ Body ☐ Head	☐ Body ☐ Head	Body/Head Specs	
LO PWR MAGNITUDE STABILITY - NO GRADIENTS WITH EFB ENABLED			5.0%*(1.0% no EFB) (MAX - MIN (PP) %)	
LO PWR PHASE STABILITY - NO GRADIENTS WITH EFB ENABLED			10.0°* (1.5° no EFB) (MAX - MIN (PP) °)	
HI PWR MAGNITUDE STABILITY - NO GRADIENTS WITH EFB ENABLED			1.0% (MAX - MIN (PP) %)	
HI PWR PHASE STABILITY - NO GRADIENTS WITH EFB ENABLED			1.5° (MAX - MIN (PP) %)	
LO PWR MAGNITUDE STABILITY - WITH GRADIENTS WITH EFB ENABLED			1.0% (MAX - MIN (PP) %)	
LO PWR PHASE STABILITY - WITH GRADIENTS WITH EFB ENABLED			1.5° (MAX - MIN (PP) %)	
HI PWR MAGNITUDE STABILITY - WITH GRADIENTS WITH EFB ENABLED			1.0% (MAX - MIN (PP) %)	
HI PWR PHASE STABILITY - WITH GRADIENTS WITH EFB ENABLED			1.5° (MAX - MIN (PP) %)	
MAGNITUDE LINEARITY -61 to 0 dB with EFB ENABLED			2.0 dB p-p (MAX (dB))	
MAGNITUDE LINEARITY -61 to 0 dB with EFB BYPASSED			2.5 dB p-p (MAX (dB))	
PHASE LINEARITY -61 to 0 dB with EFB ENABLED			30 %* (MAX - MIN (PP) %)	
PHASE LINEARITY -61 to 0 dB with EFB BYPASSED			12 %* (MAX - MIN (PP) %)	
MAGNITUDE DROOP			Not Available	
BANDWIDTH			.5 dB down @ +/- 100 kHz from 63.86 MHz	
COIL TUNE CHECK			BODY: 63.86 MHz +/- 50 kHz HEAD: 63.11 MHz +/- 50 kHz	

Note 1: The RFT analysis program will interpret the plot, and report pass/fail if MRsrv.cfg entries are correct. See Group 9 in the Report.

Note 2: With the APB in the system low power stability and phase linearity (even with the correct entry in MRsrv.cfg) can significantly exceed the limits. When running Body Dummy load with EFB enabled, it is common to see low power magnitude stability numbers approaching 5 p-p, and low power phase close to 10 p-p. Magnitude linearity should pass with or without an APB in the system. Phase linearity with an APB in the system (EFB enabled) can be as high as 30, but usually drops to 12 or below when bypassed. MR Engineering is currently investigating ways to reduce these numbers. Note that the numbers for Head Dummy Load (less RF drive) will have phase linearity numbers about 40% lower than those in the Body mode.

1.0T RFT BODY/HEAD DATA

Site:		Name:		Date:
RFT File Number:				
Coil:	☐ Body ☐ Head	☐ Body ☐ Head	Body/Head Specs	
LO PWR MAGNITUDE STABILITY - NO GRADIENTS WITH EFB ENABLED			5.0%*(1.0% no EFB) (MAX - MIN (PP) %)	
LO PWR PHASE STABILITY - NO GRADIENTS WITH EFB ENABLED			10.0°* (1.5° no EFB) (MAX - MIN (PP) °)	
HI PWR MAGNITUDE STABILITY - NO GRADIENTS WITH EFB ENABLED			1.0% (MAX - MIN (PP) %)	
HI PWR PHASE STABILITY - NO GRADIENTS WITH EFB ENABLED			1.5° (MAX - MIN (PP) %)	
LO PWR MAGNITUDE STABILITY - WITH GRADIENTS WITH EFB ENABLED			1.0% (MAX - MIN (PP) %)	
LO PWR PHASE STABILITY - WITH GRADIENTS WITH EFB ENABLED			1.5° (MAX - MIN (PP) %)	
HI PWR MAGNITUDE STABILITY - WITH GRADIENTS WITH EFB ENABLED			1.0% (MAX - MIN (PP) %)	
HI PWR PHASE STABILITY - WITH GRADIENTS WITH EFB ENABLED			1.5° (MAX - MIN (PP) %)	
MAGNITUDE LINEARITY -61 to 0 dB with EFB ENABLED			2.0 dB p-p (MAX (dB))	
MAGNITUDE LINEARITY -61 to 0 dB with EFB BYPASSED			2.5 dB p-p (MAX (dB))	
PHASE LINEARITY -61 to 0 dB with EFB ENABLED			30 %* (MAX - MIN (PP) %)	
PHASE LINEARITY -61 to 0 dB with EFB BYPASSED			12 %* (MAX - MIN (PP) %)	
MAGNITUDE DROOP			Not Available	
BANDWIDTH			.5 dB down @ +/- 100 kHz from 42.68 MHz	
COIL TUNE CHECK			BODY: 42.68 MHz +/- 50 kHz HEAD: NO SPEC	

Note 1: The RFT analysis program will interpret the plot, and report pass/fail if MRconfig.cfg entries are correct. See Group 9 in the Report.

Note 2: With the APB in the system low power stability and phase linearity (even with the correct entry in MRconfig.cfg) can significantly exceed the limits. When running Body Dummy load with EFB enabled, it is common to see low power magnitude stability numbers approaching 5 p-p, and low power phase close to 10 p-p. Magnitude linearity should pass with or without an APB in the system. Phase linearity with an APB in the system (EFB enabled) can be as high as 30, but usually drops to 12 or below when bypassed. MR Engineering is currently investigating ways to reduce these numbers. Note that the numbers for Head Dummy Load (less RF drive) will have phase linearity numbers about 40% lower than those in the Body mode.

APPENDIX B - CV DESCRIPTIONS

Name	Meaning	Controls	Allowable Values	Conditions required for values other than zero
EFB Bypass (opuser0)	0= No 1= Yes	Sets allowable values for RFampMode (opuser1) & Bandwidth (opuser6)	[0.0 to 1.0]	None
RFampMode (opuser1)	0= Exciter 1= Head Dummy Load 2 = Head coil load 3= Body Dummy Load 4= Body coil load 5= Test 6= SSdriver 7= Coil tune check	RF Amp Operating Mode 0= Standby 1= Head Mode 2 = Head mode 3= Body Mode 4= Body mode 5= Test mode 6= Head Mode 7= Standby	[0.0 to 7.0]	If RFampMode = 5 or 6 then EFB bypass must = 1. If RFampMode = 1, 2 or 6, then op coil must = 1 (Head coil) If RFampMode = 3 or 4, then op coil must = 2 (Body coil) Note: opcoil is coil selection
Sense Loop (opuser2)	0= No 1= Yes	Sets allowable valuefor BW Cal (opuser7)	[0.0 to 1.0]	RFampMode = 2, 4 or 7 only
StabNoGrad (opuser3)	0 = Tests Disabled 1= Yes	0 = No Stability Analysis 1 = Do Stability Analysis Note: Slices 1 & 2, echo 1 will always be played out even if StabNoGrad = 0.	[0.0 to 1.0]	None
Lin&Fidlty (opuser4)	0 = Test Disabled 1 = Lin only 2 = Lin & Fidelity	0 = Slices 1 & 2, echo 1 only 1 = Slices 1 & 2, echoes 1 & 2 2 = Slices 1 & 2, 4 echoes	[0.0 to 2.0]	If Lin&Fidlty = 2 then RFampMode must = 0 (Exciter)

Stab/Grad (opuser5)	0 = Test Disabled 1 = X Gradient Only 2 = Y Gradient Only 3 = Z Gradient Only 4 = All Gradients	0 = Slices 3 & 4 not done 1 = do slices 3 & 4 w/ X grad. 2 = do slices 3 & 4 w/ Y grad 3 = do slices 3 & 4 w/ Z grad 4 = do slices 3 & 4 w/ all grad. Note: Slices 3 & 4 will be played out but not analyzed if Stab/Grad = 0 and Bancwidth = 1.	[0.0 to 4.0]	None
Bandwidth (opuser6)	0 = Test Disabled 1 = Test Enabled	0 = slice 5 not done 1 = Do slice 5 Sets allowable value for BW Cal	[0.0 to 1.0]	RFampMode = 0 or 7 OR RFampMode = 1, 2, 3 or 4 AND EFB Bypass = 1
BW Cal (opuser7)	0 = Disabled 1 = Enabled	Analysis Use Only	[0.0 to 1.0]	Bandwidth = 1 AND RFampMode = 0, 2, 4 or 7 AND Sense Loop = 0

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
A	Sep 17, 1998	K. L-P	Initial version in Word
0	Dec 9, 1998	K. L-P	Initial release
1	Jan 19, 1999	Bob Schmidt	Updated Illustration 2-1 per SPR MRIge49476.
2	May 7, 1999	Resa Lambert	Began correctional updates to clean up procedure.
3	July 21, 1999	Resa Lambert	Changed BRM/F measured value and starting position.